EvoST: Adapting Traffic Forecasters to Evolving Sensor Graphs

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Abstract

Web-based navigation and traffic information services depend on accurate forecasts as sensor networks evolve. Controlled comparisons show that updating only shared weights or adding node prompts leaves substantial degradation in old checkpoints. We identify outdated node identities as a major source of this degradation, while shared spatio-temporal weights remain largely reusable with calibration. We propose EvoST, a framework that maintains traffic forecasters by refreshing node identities while keeping shared weights fixed. EvoST updates existing identities using recently released labels and initializes new sensors with identities borrowed from behaviorally similar established sensors. Once sufficient delayed labels arrive, it jointly refits all sensor identities and recalibrates their outputs. A per-sensor recursive least-squares calibrator corrects forecasts as labels become available. Across twelve PeMS region-year settings, identity refresh reduces the relative MAE gap over the one-hour forecast horizon between five-year-old STAEFormer checkpoints and models trained in the target year from 17.7% to 2.5%, with all compared forecasters calibrated, and its MAE is 0.7% above that of full-parameter fine-tuning on the same data. In Graph WaveNet, whose adaptive adjacency is generated from node embeddings, identity refresh recovers 88% of the five-year MAE gap to calibrated target-year models. Similar recovery holds for STAEFormer checkpoints up to fifteen years old, while refreshing STID recovers 67% of its five-year MAE gap to calibrated target-year counterparts. On an annual benchmark covering nine EvoXXL-Traffic districts and Transport for NSW, EvoST has the lowest average MAE, RMSE, and MAPE among the reported non-foundation methods on all ten networks.

Our code and an extended version of this paper are available at https://anonymous.4open.science/r/EvoST-0551.

CCS Concepts

• **Information systems → Sensor networks**.

Keywords

Traffic Forecasting, Evolving Traffic Sensor, Test Time Adaptation

1 Introduction

Traffic forecasting is vital for building smart cities and intelligent transportation systems, helping alleviate congestion and improve mobility by supporting efficient traffic management and route planning [4, 16, 20, 22, 23, 33, 39–43, 46]. Usually, existing traffic forecasting studies are built upon fixed sensor graphs, where nodes represent traffic sensors deployed at specific locations and edges characterize the spatial connectivity of road networks. To model complex spatio-temporal traffic dynamics and accurately predict future traffic states, e.g., traffic flow, speed, and demand, various data-driven methods have been developed, typically following a workflow of traffic data collection, graph construction, spatio-temporal modeling, and future-state prediction. This pipeline has substantially broadened the range of downstream applications that traffic forecasting can support, such as congestion management, route planning, and traffic control, etc.

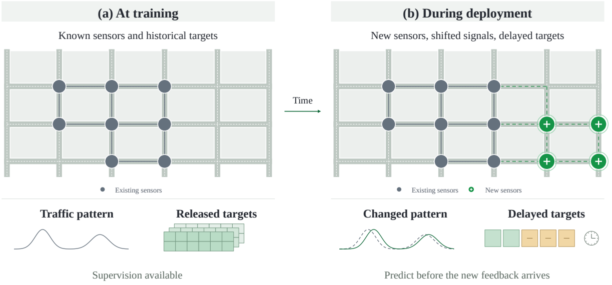

Figure 1: Traffic forecasting on evolving sensor networks. (a) At training time, historical observations and targets are available for known sensors. (b) During deployment, new sensors join the network and traffic patterns change. Forecasts must be issued before their delayed targets are released. The traffic curves are schematic.

Traditional methods generally assume a static traffic network, following a train-once-and-deploy paradigm where the trained model is expected to remain effective in a long period [9–11, 24]. Unfortunately, this assumption rarely holds in practice, as real-world traffic sensor networks continuously evolve over time. On the one hand, road networks undergo continuous construction and renewal, leading to the deployment of new sensors and the removal of outdated ones. Meanwhile, ongoing urbanization reshapes road layouts and connectivity, further altering the underlying sensor graph. On the other hand, human mobility patterns also evolve over time due to changes in commuting behavior, population distribution, and transportation policies, resulting in shifts in traffic dynamics even when the road topology remains unchanged. In these cases, conventional train-once-and-deploy forecasting models may suffer from substantial performance degradation, as both the underlying sensor graph and traffic dynamics progressively shift away from those observed during training. To handle the above dynamic changes, a straightforward solution is to regularly train the model. However, retraining can be expensive and may overwrite useful knowledge with short-lived patterns. Under this circumstance, a promising alternative is to adapt a pretrained model with data observed during deployment, as in test-time adaptation (TTA) [8, 29], rather than retraining it from scratch. Such adaptation reuses the knowledge encoded in the pretrained backbone, but it leaves open which of the backbone's parameters should be updated.

WWW '27, Dublin, Ireland
2027.

Adapting a checkpoint to an evolving sensor network raises three difficulties. First, it is unclear which parameters become outdated as the network evolves; existing methods update lightweight output modules [8], node prompts [4], or the whole model with replay [6]. Second, newly deployed sensors have short histories, which make reliable fitting of their sensor-specific parameters difficult at arrival. Third, as Figure 1 illustrates, a target becomes available only after its forecast horizon has elapsed, so any correction must rely on delayed feedback.

We therefore first ask which parameters need maintenance. We reuse STAEFormer checkpoints trained five years before the target year on twelve PeMS region-year settings; with all forecasters calibrated, their MAE over the full one-hour forecast horizon is 17.7% higher than that of checkpoints trained in the target year. Using the same recent windows, tuning only the shared spatio-temporal weights, the output head, or an EAC-style input prompt recovers much less of this gap than refreshing only the node identities, which recovers 86% of it; its MAE is 0.7% above that of full fine-tuning (Section 6.4). These results point to outdated node identities as a major source of degradation, while the shared weights remain largely reusable with calibration.

Building on this diagnosis, we propose EvoST, which maintains a deployed forecaster by refreshing node identities while keeping its shared spatio-temporal weights fixed (Figure 2). EvoST refits the identities of sensors with recent labeled history; for backbones that generate adaptive adjacency from node embeddings, such as Graph WaveNet [33], this also updates the learned graph. A new sensor starts from the mean identity of behaviorally similar established sensors and joins a refit of all identities once enough delayed labels have been released. Between fitting stages, a per-sensor recursive least-squares calibrator corrects forecasts using only targets whose complete horizon has elapsed.

Our main contributions are summarized as follows:

- We study how to maintain traffic forecasters as sensor networks evolve and labels arrive with delay, and we compare which parameters to update with the same recent data and calibration. The comparison points to outdated node identities as a major source of degradation, while shared spatio-temporal weights remain largely reusable.
- We propose EvoST, which refreshes node identities with the shared backbone frozen, lets new sensors borrow identities from behaviorally similar established sensors until a joint refit, and corrects forecasts with per-sensor recursive least squares.
- With all compared forecasters calibrated, identity refresh recovers 86% of the MAE gap over the one-hour horizon between five-year-old and target-year STAEFormer checkpoints on twelve PeMS region-year settings. Similar recovery holds for STAEFormer checkpoints up to fifteen years old; on sixteen settings, STID and Graph WaveNet recover 67% and 88% of their respective five-year gaps. On an annual benchmark covering nine EvoXXLTraffic districts and TFNSW with a separate checkpoint for each year, EvoST has the lowest average MAE, RMSE, and MAPE of the reported non-foundation methods. On average across these networks, calibration accounts for most of the Avg-MAE reduction from the matched frozen checkpoints.

2 Related Work

Traffic forecasting has developed from statistical time-series models [1, 31] to neural models of temporal and spatial dependencies. T-GCN, STGCN, and DCRNN combine temporal modeling with graph convolutions or diffusion over sensor graphs [13, 39, 44]. Graph WaveNet learns adaptive adjacency, while GMAN and STTN use attention to model spatial dependencies [33, 34, 45]. STID and STAEFormer use learnable node and spatio-temporal embeddings within MLP and Transformer backbones, respectively [15, 23]. These models are generally evaluated with the same sensors in training and testing. Wang et al. [30] show that adaptive embeddings can generalize poorly across years and evaluate both embedding fine-tuning and PCA-based representations.

Continual traffic forecasting methods address changing networks through replay, pattern memories, and selective updates [6, 27, 28]. EAC keeps the forecasting backbone frozen and adapts a pool of node prompts through expansion and low-rank compression [4]. STRAP retrieves historical spatio-temporal patterns to augment predictions [40], while ST-TTC updates a spectral output calibrator using delayed labels [5]. We examine which parameters require updating when reusing checkpoints trained years earlier on an evolving sensor network. We compare updates to node identities, shared weights, and the full model using the same recent observations and delayed-feedback protocol. EvoST builds on this diagnosis to refresh existing node representations, including those defining learned adjacency, and initialize identities for new sensors with the shared backbone frozen. Further related work is discussed in Appendix A.2.

3 Dataset and Empirical Motivation

3.1 EvoXXLTraffic Construction

EvoXXLTraffic organizes five-minute records from the California Department of Transportation Performance Measurement System (PeMS) [3, 38] into yearly graph snapshots for nine districts (D03–D08 and D10–D12). For each district, $\mathcal{V}_y$ contains sensors observed on the first available day in the initial year and on January 1 in subsequent years. Persistent IDs align the flow matrices and identify additions and removals between yearly snapshots. Sensor coordinates define a spatial proximity graph $\mathbf{A}_y$ through Haversine distances and a thresholded Gaussian kernel ($\epsilon = 0.1$). TFNSW is a long-span external test network.

3.2 Sensor Network Growth

The sensor cohorts extend and densify coverage along road corridors, with sensor counts increasing by up to approximately 95×. Appendix A.1 maps the cohorts (Figure 8), gives the district-level scale (Table 4), and shows non-monotonic changes in average degree and graph density, motivating adaptation to changing sensor sets and spatial connectivity.

4 Problem Formulation

Following EvoXXLTraffic [37, 38], we model each district as yearly graph snapshots whose sensor set, adjacency matrix, and traffic observations may change.

4.1 Evolving Sensor Graph

For district d and year y, let $\mathbf{X}^d_y$, $\mathbf{A}^d_y$, and $\mathcal{V}^d_y$ denote the traffic tensor, adjacency matrix, and active sensor set:

$$\mathcal{D}^d_{\text{evo}} = \left\{\left(\mathbf{X}^d_y, \mathbf{A}^d_y, \mathcal{V}^d_y\right)\right\}_{y=y_0}^{y_1}. \tag{1}$$

Here, $\mathbf{X}^d_y \in \mathbb{R}^{T^d_y \times N^d_y \times C}$, $N^d_y = |\mathcal{V}^d_y|$, and T^d_y and C are the numbers of time steps and channels. Both $\mathcal{V}^d_y$ and $\mathbf{A}^d_y$ may change between years. Sensors newly active relative to the preceding year form

$$\Delta\mathcal{V}^d_y = \mathcal{V}^d_y \setminus \mathcal{V}^d_{y-1}. \tag{2}$$

The intersection contains retained sensors, while $\mathcal{V}^d_{y-1} \setminus \mathcal{V}^d_y$ contains inactive ones. This annual expansion cohort appears in year y's training, validation, and test partitions.

We omit the district superscript below. Let $\mathcal{V}_{\text{ckpt}}$ contain the sensor IDs represented in the reused checkpoint. The active sensors with checkpoint identities are $\mathcal{K} = \mathcal{V}_y \cap \mathcal{V}_{\text{ckpt}}$; those absent from the checkpoint are $\mathcal{U} = \mathcal{V}_y \setminus \mathcal{K}$. Thus, $\mathcal{U}$ depends on the checkpoint year and can differ from the annual expansion cohort $\Delta\mathcal{V}_y$. In the arrival evaluation, each sensor in $\mathcal{U}$ has a specified arrival time. Its identity initialization, donor matching, and sensor-specific fitting use observations from that time onward.

4.2 Forecasting Task

We focus on univariate traffic flow ($C = 1$). Within snapshot y, let $\mathbf{s}_t \in \mathbb{R}^{N_y}$ denote the traffic state at time t. The input and target windows are

$$\begin{aligned} \mathbf{X}_t &= [\mathbf{s}_{t-L+1}, \dots, \mathbf{s}_t] \in \mathbb{R}^{N_y \times L}, \\ \mathbf{Y}_t &= [\mathbf{s}_{t+1}, \dots, \mathbf{s}_{t+H}] \in \mathbb{R}^{N_y \times H}. \end{aligned} \tag{3}$$

We process one chronological window at a time. Before predicting at t, record i can supply feedback only after its complete target window has elapsed, that is, $i + H \le t$.

Checkpoint maintenance. We reuse a checkpoint trained in an earlier year, keeping its shared spatio-temporal weights and normalization statistics fixed. Available labeled history supports node identity fitting and output calibration. Subsequent feedback updates the calibrators under the complete-horizon release rule. For incoming sensors with short histories, EvoST borrows identities before a scheduled refit. The task is to maintain forecasting accuracy on the current sensor network through these updates.

5 Methodology

5.1 Overview

Figure 2 illustrates EvoST. Given the current sensor network and input $\mathbf{X}_t$, the forecaster produces a base prediction using frozen shared weights and maintained node identities. When recent labeled history is available for all active sensors, EvoST fits their identities before deployment. With restricted arrival history, established sensors are first refreshed on their own network, while incoming sensors borrow identities from established sensors with similar observations. A later refit updates all active identities, followed by reinitialization of the output calibrators. Between fitting stages, per-sensor recursive least squares (RLS) updates the forecast correction using released feedback.

5.2 Node Identity Maintenance

Parameters. We separate the checkpoint into shared parameters θ and node identities $\mathbf{E}$:

$$\widehat{\mathbf{Y}}^{\text{base}}_t = f_\theta(\mathbf{X}_t, \mathbf{A}_y; \mathbf{E}). \tag{4}$$

Here, $\mathbf{E}$ contains the backbone's existing node-specific parameters. STAEFormer uses an adaptive spatio-temporal embedding block for each sensor [15]. STID uses spatial embeddings [23], and Graph WaveNet uses two node embedding tables [33]. We write $\mathbf{e}_n$ for all identity components of sensor n. Shared layers, output heads, shared temporal parameters, and normalization statistics remain fixed. Identity fitting keeps the backbone in evaluation mode. The input $\mathbf{A}_y$ is supplied when used by the backbone.

Identity refresh. Let $\mathcal{R}$ contain recent windows with released targets and $\mathcal{S}$ the sensors being fitted. We reserve the final chronological quarter of $\mathcal{R}$ for early stopping and optimize on the earlier windows, $\mathcal{R}_{\text{fit}}$:

$$\begin{aligned} \mathcal{L}_{\text{id}}(\mathbf{E}; \mathcal{R}, \mathcal{S}) = \frac{1}{Z} \sum_{i \in \mathcal{R}_{\text{fit}}} \sum_{n \in \mathcal{S}} \sum_{h=1}^{H} m_{i,n,h} \\ \cdot \left| \hat{y}^{\text{base}}_{i,n,h}(\mathbf{E}) - y_{i,n,h} \right|. \end{aligned} \tag{5}$$

The mask $m_{i,n,h}$ equals one for finite, nonzero targets; Z counts these entries over the fitted windows and sensors. When all active sensors have recent history, $\mathcal{S} = \mathcal{V}_y$: matched checkpoint identities initialize $\mathcal{K}$, and the componentwise mean checkpoint identity initializes $\mathcal{U}$. For arrival preparation, we first fit $\mathcal{K}$ using inputs and any adjacency restricted to that sensor set, then initialize $\mathcal{U}$ as described in Section 5.4.

For Graph WaveNet, the maintained tables also define its adaptive adjacency:

$$\mathbf{A}_{\text{adp}} = \text{softmax}_{\text{row}}\left(\text{ReLU}(\mathbf{E}_1 \mathbf{E}_2^\top)\right). \tag{6}$$

Refreshing $\mathbf{E}_1$ and $\mathbf{E}_2$ changes the learned adjacency. The static adjacency is retained alongside it, with shared propagation weights fixed.

5.3 Causal Delayed-Feedback Pool

A forecast issued at i covers $i+1 : i+H$. Its record enters a pending queue $\mathcal{Q}$ after prediction. Before predicting at t, eligible records move to the released pool:

$$\mathcal{P}_t = \mathcal{P}_{t-1} \cup \{q_i \in \mathcal{Q} \mid i + H \le t\}. \tag{7}$$

The pool starts with eligible history available before the stream. Each record stores its input, the base forecast computed at issue time, sensor IDs, and origin index; its target is attached upon release. Released records leave $\mathcal{Q}$. Identity fitting selects recent released windows. Ordinary RLS updates consume each new eligible record once, in time order, using its cached base forecast. Feedback awaiting an RLS update is tracked separately from the history retained for fitting.

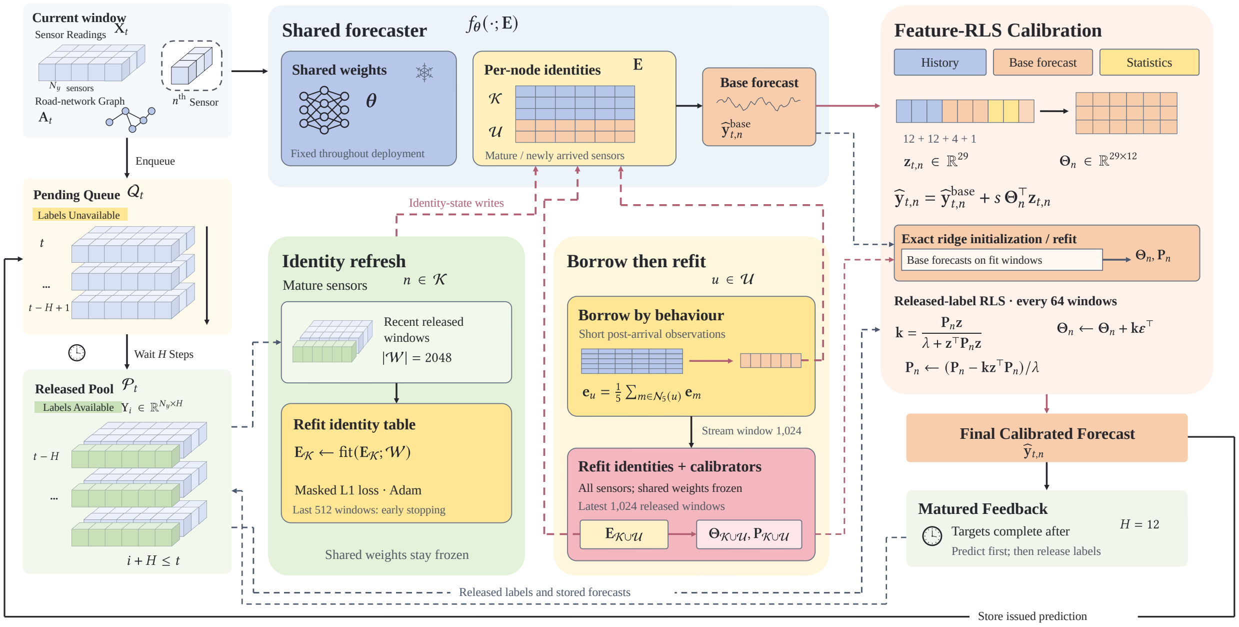
Current window X_t
Sensor Readings
N_s sensors
n^{th} Sensor
Road-network Graph
A_t
Enqueue
Pending Queue Q_t
Labels Unavailable
t
...
$t-H+1$
Wait H Steps
Released Pool $\mathcal{P}_t$
Labels Available $Y_i \in \mathbb{R}^{N_s \times H}$
$t-H$
...
$i+H \le t$
Shared forecaster $f_\theta(\cdot;\mathrm{E})$
Shared weights
θ
Fixed throughout deployment
Per-node identities
E
$\mathcal{K}$
$\mathcal{U}$
Mature / newly arrived sensors
Base forecast
$\hat{y}^{\text{base}}_{t,n}$
Identity-state writes
Identity refresh
Mature sensors $n \in \mathcal{K}$
Recent released windows
$|\mathcal{W}| = 2048$
Refit identity table
$\mathrm{E}_\mathcal{K} \leftarrow \mathrm{fit}(\mathrm{E}_\mathcal{K};\mathcal{W})$
Masked L1 loss · Adam
Last 512 windows; early stopping
Shared weights stay frozen
Borrow then refit
$u \in \mathcal{U}$
Borrow by behaviour
Short post-arrival observations
$\mathbf{e}_u = \frac{1}{5}\sum_{m\in\mathcal{N}_5(u)} \mathbf{e}_m$
Stream window 1,024
Refit identities + calibrators
All sensors; shared weights frozen
Latest 1,024 released windows
$\mathrm{E}_{\mathcal{K}\cup\mathcal{U}}$
$\Theta_{\mathcal{K}\cup\mathcal{U}}, \mathrm{P}_{\mathcal{K}\cup\mathcal{U}}$
Feature-RLS Calibration
History
Base forecast
Statistics
12 + 12 + 4 + 1
$\mathbf{z}_{t,n} \in \mathbb{R}^{29}$
$\Theta_n \in \mathbb{R}^{29\times 12}$
$\hat{y}_{t,n} = \hat{y}^{\text{base}}_{t,n} + s\,\Theta_n^\top \mathbf{z}_{t,n}$
Exact ridge initialization / refit
Base forecasts on fit windows
Θ_n, P_n
Released-label RLS · every 64 windows
$\mathrm{k} = \frac{\mathrm{P}_n\mathrm{z}}{\lambda + \mathrm{z}^\top \mathrm{P}_n \mathrm{z}}$
$\Theta_n \leftarrow \Theta_n + \mathrm{k}\varepsilon^\top$
$\mathrm{P}_n \leftarrow (\mathrm{P}_n - \mathrm{k}\mathrm{z}^\top\mathrm{P}_n)/\lambda$
Final Calibrated Forecast
$\hat{y}_{t,n}$
Matured Feedback
Targets complete after
$H = 12$
Predict first; then release labels
Released labels and stored forecasts
Store issued prediction

Figure 2: Overview of EvoST. Shared spatio-temporal weights θ remain frozen while node identities E are maintained. Sensors with recent labeled history undergo identity refresh. The borrowing and scheduled-refit branches handle incoming sensors with limited history: they start from identities of behaviorally similar established sensors, then join a refit of all active identities. Base forecasts on the refitting history are recomputed to rebuild every sensor's calibration state. Feature-RLS corrects outputs using released feedback, with $i + H \le t$. Adjacency inputs depend on the backbone; identity donors are selected from observed traffic behavior.

5.4 New-Sensor Initialization and Joint Refitting

Borrowing an identity. For $u \in \mathcal{U}$ and $|\mathcal{K}| > 0$, we compare its short post-arrival history with contemporaneous observations from $v \in \mathcal{K}$. Let $\mathcal{O}_{uv}$ contain the times available before initialization when both sensors have finite readings. We use

$$\begin{aligned} d(u,v) &= \frac{1}{|\mathcal{O}_{uv}|}\sum_{r\in\mathcal{O}_{uv}} (s_{r,u}-s_{r,v})^2, \\ \mathbf{e}_u^{(0)} &= \frac{1}{k}\sum_{v\in\mathcal{N}_k(u)} \mathbf{e}_v, \end{aligned} \tag{8}$$

We set $k = \min(5, |\mathcal{K}|)$ and use the nearest eligible donors as $\mathcal{N}_k(u)$. Averaging covers all identity components. If fewer than k donors meet the observation requirements, we draw the entire donor set from $\mathcal{K}$ with a fixed random seed. Borrowed identities remain fixed until the joint refit. Appendix A.5 gives the eligibility rules.

Refitting the expanded network. At the scheduled time $t_\star$, we collect the latest W released windows, including eligible prestream history when needed. Starting from the current identity table, we optimize eq. (5) over all active sensors, $\mathcal{S} = \mathcal{K} \cup \mathcal{U}$. We then recompute base forecasts on those windows with the updated identities and rebuild every calibrator's coefficients and inverse covariance state by ridge regression. The selected ridge penalty, forgetting factor, and output scale are retained. Pending records keep their original cached forecasts. After rebuilding, recursive updates resume with newly released feedback. The refitting history is not repeatedly consumed by RLS.

5.5 Per-Sensor Recursive Calibration

For window t and sensor n, let $\mathbf{b}_{t,n} = \hat{\mathbf{y}}^{\text{base}}_{t,n}$. We stack the scaled base forecast, checkpoint-normalized history $\bar{\mathbf{x}}_{t,n}$, four temporal summaries, and an intercept into a column vector:

$$\mathbf{z}_{t,n} = \mathrm{col}\big(\mathbf{b}_{t,n}/s, \bar{\mathbf{x}}_{t,n}, \mathrm{Stat}(\bar{\mathbf{x}}_{t,n}), 1\big). \tag{9}$$

The summaries are the last value, mean, population standard deviation, and least-squares slope of the history. The global output scale s is fixed from valid targets in the initial training tail; Appendix A.5 specifies its scope in the arrival evaluation. With $D = H + L + 5$, a separate matrix $\Theta_n \in \mathbb{R}^{D\times H}$ corrects each sensor:

$$\hat{\mathbf{y}}_{t,n} = \mathbf{b}_{t,n} + s\Theta_n^\top \mathbf{z}_{t,n}. \tag{10}$$

Let $\mathcal{J}_n$ contain fitting records whose H targets are all finite and nonzero and which satisfy the sensor's arrival restriction. With

Algorithm 1 EvoST with delayed feedback

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Require: Frozen $\theta$; prepared $\mathbf{E}$ and $\{(\Theta_n, \mathbf{P}_n)\}$; eligible history $\mathcal{H}_{\text{pre}}$; fixed $\rho, \lambda, s$; delay $H$; cadence $\delta$; optional $t_\star$ with $t_\star \bmod \delta = 0$; refit length $W$
Ensure: Forecasts $\{\widehat{\mathbf{Y}}_t\}$ using released feedback
$\mathcal{Q} \leftarrow \emptyset$; $\mathcal{P} \leftarrow \mathcal{H}_{\text{pre}}$; $\mathcal{I} \leftarrow \emptyset$ {feedback awaiting RLS}
for each chronological window $t = 0, 1, \ldots$ do
    Move records with $i + H \leq t$ from $\mathcal{Q}$ to $\mathcal{P}$ and $\mathcal{I}$, attaching their targets
    if $t \bmod \delta = 0$ then
        Update RLS on valid records in $\mathcal{I}$ in time order using cached base forecasts and eq. (12)
        $\mathcal{I} \leftarrow \emptyset$
    end if
    if refitting is enabled and $t = t_\star$ then
        $\mathcal{R} \leftarrow \text{Tail}_W(\mathcal{P})$
        Refit all active identities from the current $\mathbf{E}$ using eq. (5)
        Recompute base forecasts, features, and residual targets on $\mathcal{R}$
        Rebuild all $(\Theta_n, \mathbf{P}_n)$ using eq. (11), retaining $\rho, \lambda, s$
    end if
    $\widehat{\mathbf{Y}}_t^{\text{base}} \leftarrow f_\theta(\mathbf{X}_t, \mathbf{A}_y; \mathbf{E})$
    Correct each sensor's forecast using eq. (10)
    Emit $\widehat{\mathbf{Y}}_t$; enqueue the record and its base forecast in $\mathcal{Q}$ with release time $t + H$
end for
```

$\mathbf{r}_{i,n} = (\mathbf{y}_{i,n} - \mathbf{b}_{i,n})/s$ and ridge penalty $\rho > 0$, we initialize

$$\begin{aligned} \mathbf{P}_n^{(0)} &= \left(\rho \mathbf{I}_D + \sum_{i \in \mathcal{J}_n} \mathbf{z}_{i,n} \mathbf{z}_{i,n}^\top \right)^{-1}, \\ \Theta_n^{(0)} &= \mathbf{P}_n^{(0)} \sum_{i \in \mathcal{J}_n} \mathbf{z}_{i,n} \mathbf{r}_{i,n}^\top. \end{aligned} \tag{11}$$

Calibration uses the full eligible fitting history, including the quarter reserved for identity early stopping. For a newly eligible record, RLS updates

$$\begin{aligned} \mathbf{k}_n &= \frac{\mathbf{P}_n \mathbf{z}_{i,n}}{\lambda + \mathbf{z}_{i,n}^\top \mathbf{P}_n \mathbf{z}_{i,n}}, \\ \epsilon_{i,n} &= \mathbf{r}_{i,n} - \Theta_n^\top \mathbf{z}_{i,n}, \\ \Theta_n^+ &= \Theta_n + \mathbf{k}_n \epsilon_{i,n}^\top, \\ \mathbf{P}_n^+ &= \lambda^{-1} \left(\mathbf{P}_n - \mathbf{k}_n \mathbf{z}_{i,n}^\top \mathbf{P}_n \right), \end{aligned} \tag{12}$$

where $\lambda \in (0, 1]$ and the right-hand sides use the state before the update. The same complete-target validity rule applies to initialization and recursion. These online updates change only the calibration state; identities remain fixed between fitting stages.

5.6 Unified Online Procedure

Algorithm 1 summarizes the feedback order after identity preparation, calibration initialization, and validation-based hyperparameter selection. The arrival refit coincides with an RLS update, $t_\star \bmod \delta = 0$, so all feedback released by then is processed before refitting. Test predictions follow the complete-horizon release rule.

6 Experiments

6.1 Experimental Settings

PeMS experiments use the first 31 days of each selected year; Table 4 summarizes the full data archive. Each year uses a chronological 60%/20%/20% train/validation/test split. PEMS windows contain 12 five-minute steps and TFNSW windows contain 12 hourly steps. In the annual benchmark, conventional forecasters share graph snapshots, splits, raw input and target windows, and metric definitions. Conventional baseline training uses the benchmark's cached inputs, which are standardized within each split using that split's statistics, following TrafficStream [6]. The EvoST pipeline rebuilds validation and test inputs from raw windows and standardizes them with the statistics of the model's training split. Residual non-finite inputs are set to zero after normalization, and invalid targets are masked. Foundation models receive raw-scale contexts and retain their model-specific input normalization.

Annual benchmark. For each year, EvoST starts from the benchmark's STAEFormer checkpoint of that year, so every sensor of the year already has an identity. It refreshes the identity table on up to the latest 2,048 training windows, using the final 512 window starts for early stopping and for selecting the learning rate from $\{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$, and fits each sensor's calibrator on the training tail. Calibration settings are selected on up to the first 1,024 validation windows. Validation labels are used in this selection; the resulting RLS states are discarded, and new calibrators fitted on the training tail run over the test portion only, using each target once it is released 12 steps later and updating every 64 windows. Static forecasters do not adapt to test labels. Appendix A.4 gives the remaining settings and matched controls. We report cumulative mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE) at horizons 3, 6, and 12. Avg averages the cumulative scores from horizons 1 to 12, with Avg-MAE as the primary metric. Results use five seeds unless noted otherwise.

Checkpoint maintenance. These experiments reuse the annual benchmark's per-year STAEFormer checkpoints (checkpoint seeds 42–44) on later target years, with the statistics of each checkpoint's training split. The main panel applies five-year-old checkpoints to PEMS03, PEMS06, PEMS08, PEMS10, PEMS11, and PEMS12 in 2020 and 2025, giving twelve region-year settings. Parameter updates, including the prompt and scratch controls, draw from the latest 2,048 training windows of the target year, with the final 512 reserved for early stopping and learning-rate selection. Both the fitting and holdout subsets are sampled at a stride of two (four above 1,600 sensors). Each variant that updates parameters in the main comparison selects its learning rate on this holdout; no validation or test value enters these initial fits. The scratch control uses target-year training statistics. All variants and the reference then use the same per-sensor RLS procedure, with labels released after 12 steps and updates every 64 windows. Its penalty and forgetting factor are selected separately for each variant on the first 1,024 validation windows; RLS is then reinitialized from the training tail, validation and test are replayed chronologically, and only test errors are reported. The reference is the target-year STAEFormer checkpoint with this calibration. We report MAE over the one-hour horizon, accumulated over steps 1–12 (H12). Errors are averaged over evaluation seeds 51–53 within each setting, divided by the reference, and combined by the geometric mean across settings; 95% intervals use 10,000 bootstrap draws over districts, keeping both years of a district together. Extensions to other checkpoint ages, backbones, sensor arrivals, and label budgets state their panels and any fixed learning rates with their results; Appendix A.5 provides implementation details.

Table 1: Average results on PEMS03, PEMS04, and TFNSW. Each row is one baseline, with MAE, RMSE, and MAPE reported for each dataset. Avg is the mean cumulative-horizon score over prediction lengths from 1 to 12. Chronos-2 uses multivariate graph-grouped inference, and its fine-tuned variant uses per-year low-rank adaptation (LoRA). Values are mean ± standard deviation when available (for EvoST, the sample standard deviation over five seeds); deterministic zero-shot runs are shown as ±0.00. MAPE is in percent. Bold and underlining mark the best and second-best non-foundation methods; foundation-model rows are not ranked.

Category	Baseline	PEMS03			PEMS04			TFNSW		
		MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
Static STGNN Backbones	DCRNN	$14.99_{\pm0.18}$	$24.90_{\pm0.19}$	$28.69_{\pm0.86}$	$22.22_{\pm0.27}$	$34.58_{\pm0.32}$	$20.53_{\pm0.40}$	$179.87_{\pm14.50}$	$303.16_{\pm19.20}$	$211.72_{\pm42.72}$
	ASTGNN	$15.15_{\pm0.10}$	$25.15_{\pm0.17}$	$26.56_{\pm0.43}$	$22.45_{\pm0.09}$	$34.96_{\pm0.12}$	$19.29_{\pm0.42}$	$140.15_{\pm0.82}$	$247.22_{\pm1.02}$	$125.28_{\pm3.02}$
	TGCN	$15.88_{\pm0.10}$	$30.12_{\pm0.40}$	$28.95_{\pm0.89}$	$23.60_{\pm0.37}$	$38.82_{\pm0.55}$	$20.15_{\pm0.28}$	$208.33_{\pm1.48}$	$390.75_{\pm3.25}$	$177.95_{\pm3.89}$
	GWN	$15.33_{\pm0.26}$	$25.15_{\pm0.39}$	$28.01_{\pm0.71}$	$23.41_{\pm0.27}$	$35.69_{\pm0.22}$	$19.94_{\pm0.55}$	$170.89_{\pm3.29}$	$313.61_{\pm5.86}$	$138.49_{\pm10.57}$
Naïve Schemes	Pretrain	$31.37_{\pm1.11}$	$41.64_{\pm0.99}$	$50.04_{\pm2.80}$	$153.33_{\pm6.62}$	$186.37_{\pm7.58}$	$82.25_{\pm2.99}$	$262.63_{\pm10.87}$	$383.38_{\pm12.29}$	$430.83_{\pm35.95}$
	Retrain	$14.16_{\pm0.15}$	$23.46_{\pm0.14}$	$26.29_{\pm0.58}$	$21.64_{\pm0.20}$	$33.58_{\pm0.22}$	$19.15_{\pm0.30}$	$138.04_{\pm0.62}$	$247.58_{\pm1.00}$	$104.00_{\pm3.59}$
	Online-NN	$17.47_{\pm0.16}$	$28.29_{\pm0.08}$	$29.62_{\pm1.04}$	$38.46_{\pm0.27}$	$51.85_{\pm0.46}$	$26.93_{\pm0.71}$	$174.36_{\pm3.31}$	$318.17_{\pm5.87}$	$129.82_{\pm8.24}$
	Online-AN	$13.80_{\pm0.05}$	$22.89_{\pm0.11}$	$24.96_{\pm0.21}$	$21.01_{\pm0.26}$	$32.73_{\pm0.31}$	$17.82_{\pm0.31}$	$130.61_{\pm0.96}$	$235.35_{\pm1.95}$	$99.93_{\pm3.54}$
Evolving Graph	TrafficStream	$14.32_{\pm0.20}$	$23.60_{\pm0.28}$	$27.63_{\pm1.67}$	$21.43_{\pm0.53}$	$33.35_{\pm0.60}$	$18.83_{\pm0.59}$	$145.42_{\pm2.41}$	$259.40_{\pm4.67}$	$106.92_{\pm8.01}$
	PECPM	$14.90_{\pm0.81}$	$24.32_{\pm0.98}$	$27.78_{\pm2.30}$	$25.16_{\pm3.18}$	$39.17_{\pm4.12}$	$22.29_{\pm3.69}$	$174.19_{\pm22.07}$	$318.61_{\pm45.67}$	$106.24_{\pm4.44}$
	STKEC	$14.47_{\pm0.22}$	$23.89_{\pm0.34}$	$27.58_{\pm0.64}$	$22.78_{\pm1.14}$	$34.78_{\pm1.18}$	$21.54_{\pm1.53}$	$147.86_{\pm1.05}$	$259.82_{\pm2.21}$	$148.01_{\pm7.69}$
	EAC	$15.91_{\pm0.48}$	$26.40_{\pm0.88}$	$30.12_{\pm0.81}$	$34.02_{\pm9.03}$	$48.20_{\pm9.87}$	$39.19_{\pm17.54}$	$137.90_{\pm1.42}$	$245.35_{\pm1.93}$	$119.37_{\pm14.85}$
Retrieval TTC	STRAP	$14.15_{\pm0.13}$	$23.37_{\pm0.22}$	$26.12_{\pm0.31}$	$21.93_{\pm0.67}$	$33.78_{\pm0.64}$	$19.49_{\pm0.57}$	$135.57_{\pm1.80}$	$243.47_{\pm4.15}$	$104.21_{\pm3.09}$
	ST-TTC	$13.98_{\pm0.14}$	$23.18_{\pm0.13}$	$26.15_{\pm0.57}$	$21.33_{\pm0.20}$	$33.21_{\pm0.22}$	$19.04_{\pm0.29}$	$144.41_{\pm0.72}$	$257.96_{\pm1.08}$	$106.33_{\pm3.01}$
Static Forecasting Backbones	STID	$15.14_{\pm0.14}$	$27.10_{\pm0.38}$	$26.44_{\pm0.62}$	$22.43_{\pm0.18}$	$35.63_{\pm0.31}$	$18.51_{\pm0.26}$	$153.63_{\pm6.20}$	$307.19_{\pm10.11}$	$112.25_{\pm11.68}$
	STNorm	$18.20_{\pm0.47}$	$28.58_{\pm0.65}$	$52.94_{\pm2.57}$	$23.99_{\pm0.62}$	$36.65_{\pm0.80}$	$27.98_{\pm1.51}$	$148.78_{\pm2.33}$	$275.25_{\pm3.16}$	$182.00_{\pm6.10}$
	iTrans	$14.21_{\pm0.13}$	$23.42_{\pm0.14}$	$26.79_{\pm0.71}$	$21.22_{\pm0.11}$	$33.04_{\pm0.17}$	$19.36_{\pm0.25}$	$144.04_{\pm0.82}$	$257.27_{\pm2.16}$	$113.97_{\pm6.78}$
	DLinear	$15.73_{\pm0.06}$	$26.23_{\pm0.04}$	$28.55_{\pm0.98}$	$23.58_{\pm0.18}$	$36.54_{\pm0.22}$	$19.60_{\pm0.33}$	$182.20_{\pm1.03}$	$318.13_{\pm1.10}$	$187.17_{\pm5.55}$
	STAEFormer	$12.26_{\pm0.11}$	$20.58_{\pm0.12}$	$23.06_{\pm0.64}$	$19.38_{\pm0.94}$	$30.64_{\pm1.20}$	$16.97_{\pm0.95}$	$96.12_{\pm1.67}$	$190.33_{\pm2.83}$	$66.43_{\pm2.35}$
Times-series Forecasting Foundation Model	Chronos-2 (ZS)	$15.27_{\pm0.00}$	$26.98_{\pm0.00}$	$25.83_{\pm0.00}$	$21.58_{\pm0.00}$	$35.22_{\pm0.00}$	$17.62_{\pm0.00}$	$227.74_{\pm0.00}$	$440.18_{\pm0.00}$	$120.67_{\pm0.00}$
	Chronos-2 (FT)	$13.31_{\pm0.01}$	$23.66_{\pm0.03}$	$22.80_{\pm0.03}$	$18.94_{\pm0.04}$	$31.89_{\pm0.12}$	$15.25_{\pm0.01}$	$160.20_{\pm0.68}$	$323.47_{\pm2.06}$	$78.26_{\pm0.28}$
	TimesFM2.5 (ZS)	$15.91_{\pm0.00}$	$27.86_{\pm0.00}$	$26.71_{\pm0.00}$	$23.49_{\pm0.00}$	$37.69_{\pm0.00}$	$19.32_{\pm0.00}$	$266.77_{\pm0.00}$	$487.06_{\pm0.00}$	$160.45_{\pm0.00}$
	TimesFM2.5 (FT)	$14.16_{\pm0.01}$	$25.36_{\pm0.04}$	$23.79_{\pm0.01}$	$20.92_{\pm0.02}$	$35.20_{\pm0.02}$	$16.65_{\pm0.02}$	$215.22_{\pm0.33}$	$420.65_{\pm0.52}$	$124.25_{\pm0.83}$
	Moirai-2.0 (ZS)	$16.19_{\pm0.00}$	$28.00_{\pm0.00}$	$26.15_{\pm0.00}$	$24.52_{\pm0.00}$	$38.87_{\pm0.00}$	$19.11_{\pm0.00}$	$294.05_{\pm0.00}$	$545.09_{\pm0.00}$	$137.97_{\pm0.00}$
	Moirai-2.0 (FT)	$13.95_{\pm0.01}$	$24.51_{\pm0.03}$	$24.18_{\pm0.01}$	$20.68_{\pm0.03}$	$34.42_{\pm0.09}$	$16.78_{\pm0.02}$	$142.36_{\pm0.28}$	$303.78_{\pm0.69}$	$62.69_{\pm0.22}$
Ours	EvoST	$\mathbf{10.92}_{\pm0.05}$	$\mathbf{18.47}_{\pm0.07}$	$\mathbf{22.52}_{\pm0.21}$	$\mathbf{16.58}_{\pm0.21}$	$\mathbf{27.14}_{\pm0.26}$	$\mathbf{15.52}_{\pm0.23}$	$\mathbf{65.16}_{\pm0.34}$	$\mathbf{137.77}_{\pm0.59}$	$\mathbf{52.89}_{\pm0.29}$

6.2 Baselines

We compare nine static forecasters (DCRNN, ASTGNN, TGCN, GWN, STID, STNorm, iTrans, DLinear, and STAEFormer); Pretrain, Retrain, Online-NN, and Online-AN; four evolving-graph methods (TrafficStream, PECPM, STKEC, and EAC); STRAP and ST-TTC; and zero-shot (ZS) or yearly fine-tuned (FT) versions of Chronos-2, TimesFM 2.5, and Moirai-2.0.

EvoST starts from stored per-year STAEFormer checkpoints of the benchmark. The Online-AN and STAEFormer rows report the benchmark's test errors recorded at the end of training on split-specific cached inputs. The STAEFormer row includes later single-year retraining for PEMS04/2002 and TFNSW/2006. The EvoST row rebuilds inputs with training-split statistics, and its selected STAEFormer checkpoints differ in some cases from those behind the STAEFormer row.

Maintenance controls. On five-year-old STAEFormer checkpoints, we compare output-head tuning (Head), shared-weight tuning (Shared), full-parameter fine-tuning (Full FT), an EAC-style low-rank input prompt [4] (Prompt), node-identity refresh (EvoST), and joint identity and prompt tuning (EvoST+Prompt). Old keeps the forecaster parameters fixed, while Scratch trains a randomly initialized model on the same target-year fitting windows. All variants and the target-year checkpoint reference use the per-sensor RLS procedure in Section 5.5.

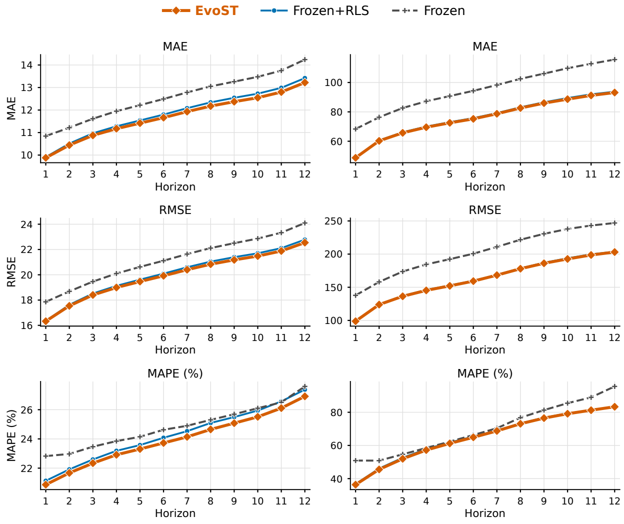
6.3 Main Results

Overall performance. Table 1 reports three representative networks. EvoST has the lowest MAE, RMSE, and MAPE among the non-foundation methods on all three, and also on the other seven PEMS districts (Tables 5 and 6 in Appendix A.3). The annual benchmark uses checkpoints trained for the same year. With these checkpoints and training-split input normalization, EvoST reduces Avg-MAE relative to the frozen forecaster by 7.2%, 10.7%, and 22.0% on PEMS03, PEMS04, and TFNSW (Figure 11), and the matched frozen-plus-RLS control already achieves most of this reduction. On all ten networks, EvoST's Avg-MAE is below that of Frozen+RLS, by 0.8%

on average; on the three networks the reductions are 0.8%, 0.3%, and 0.6%, with lower cross-year means in five, three, and five of the five paired evaluation seeds. On PEMS03, calibrating the refreshed model raises MAPE from 21.66 to 22.52.

Comparison with time-series foundation models. TSFMs are reference rows outside the non-foundation ranking. EvoST has lower MAE and RMSE than every zero-shot and fine-tuned TSFM cell in Table 1, and lower MAPE except on PEMS04, where Chronos-2 (FT) reaches 15.25 against 15.52. At horizons 3, 6, and 12 on all ten networks (Tables 7 and 8), EvoST also has lower MAE and RMSE than every TSFM reference; Appendix A.3 lists the MAPE exceptions.

Forecast horizons. On PEMS03 and TFNSW, at each of the 12 steps (Figure 3), the calibrator gives most of the MAE and RMSE reduction relative to the frozen checkpoints, and identity refresh lowers both further; on PEMS03, its MAE reduction over Frozen+RLS grows from 0.3% at the first step to 1.5% at the last. Over all ten networks, EvoST is below Frozen in all 360 combinations of network, metric, and step and below Frozen+RLS in 343 (Figure 10); 14 of the 17 exceptions are MAPE steps on PEMS04 and PEMS05.



(a) PEMS03

(b) TFNSW

Figure 3: All-sensor per-step errors of EvoST and its matched controls on PEMS03 and TFNSW: the same checkpoints with the calibrator only (Frozen+RLS) and without adaptation (Frozen), over the same 25 and 21 years and five seeds. Lower is better.

6.4 Which Parameters Need Maintenance?

Figure 4 shows the maintenance controls on the main panel. The five-year-old checkpoint (Old) is 17.7% worse than the target-year checkpoint. Refreshing only the node identities (EvoST) lowers the ratio to 1.025 and recovers 86% of this gap. Full fine-tuning recovers slightly more (1.018): identity refresh is 0.7% worse than full fine-tuning (95% interval 0.3–1.3%). Tuning only the shared spatio-temporal weights (1.078) or the output head (1.148) recovers much less, as do a rank-6 EAC-style input prompt (1.130) and a model trained from scratch on the same target-year windows (1.123). Refitting the prompt jointly with the identities reduces MAE by a further 0.25% relative to EvoST. On the H100 runs, the median ratio of epoch time for identity refresh to Full FT is 0.82; fitting still backpropagates through the frozen backbone. Identity refresh changes only per-node parameters and leaves the shared model unchanged. These comparisons use the specified fitting budgets and learning-rate grids; Appendix A.5 reports the settings and selection counts.

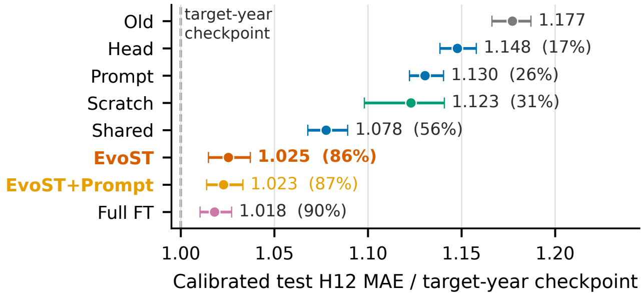


Figure 4: Parameter maintenance of five-year-old STAEFormer checkpoints (Scratch: trained anew). Calibrated test H12 MAE relative to the calibrated target-year checkpoint (dashed); geometric means over twelve settings, after averaging three seeds, with 95% district-bootstrap intervals; percentages give the share of the Old gap recovered.

Table 2: Five-year reuse on sixteen region-year settings (eight districts, 2020 and 2025). Calibrated test H12 MAE relative to each backbone's calibrated target-year checkpoint; geometric means after averaging three seeds per setting. STAEFormer and STID use a fixed refresh learning rate of 3×10^{-3}. Recovery is the fraction of the old-to-reference error gap closed.

Backbone	Settings	Old	Refresh	Recovery
STAEFormer	16	1.180	1.033	82%
STID	16	1.045	1.015	67%
Graph WaveNet	16	1.091	1.011	88%

6.5 Other Backbones and Time Inputs

The finding is not specific to STAEFormer (Table 2). On sixteen settings from eight districts, five-year-old STID and Graph WaveNet checkpoints are 4.5% and 9.1% worse than the target-year ones, against 18.0% for STAEFormer. Refreshing the node embeddings of STID and the node vectors that generate the adaptive adjacency of Graph WaveNet recovers 67% and 88% of these gaps. The benchmark's STAEFormer has no time-of-day or day-of-week inputs, so its identities could absorb daily profiles; retrained per year with these inputs, a five-year-old checkpoint on the main panel is still 14.0% worse than the target-year one (20.5% for matched retraining without these inputs), and identity refresh recovers 90% of the gap (Appendix A.5).

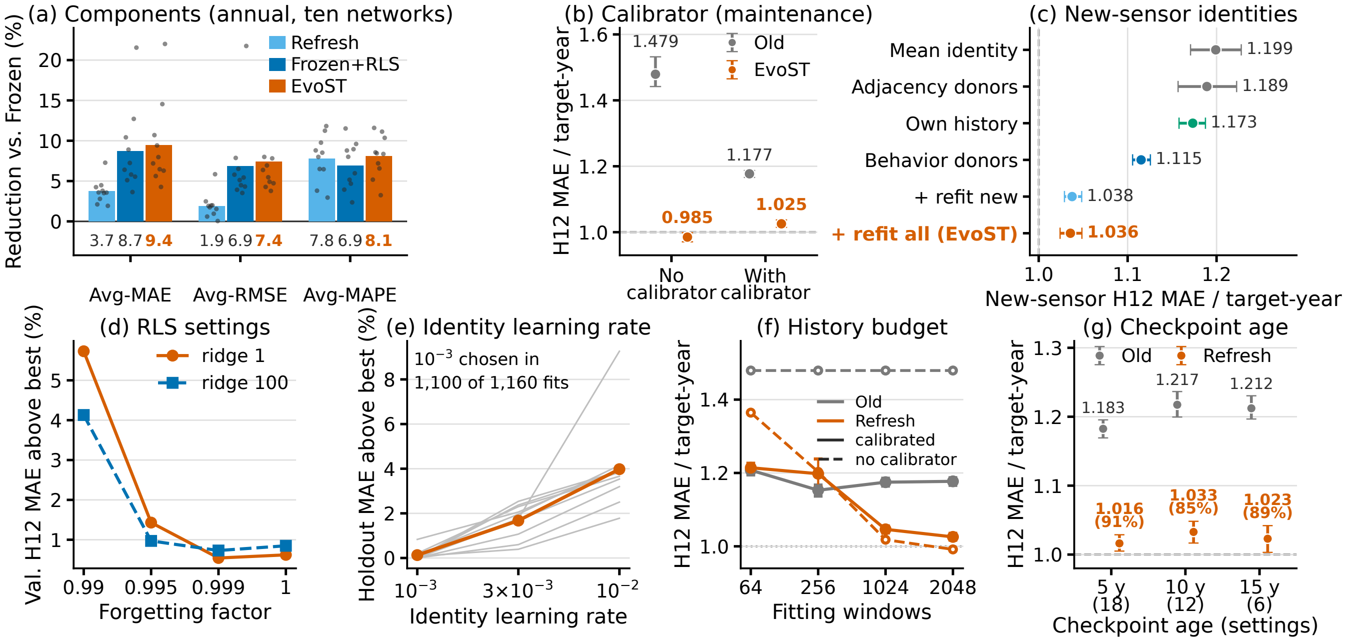


Figure 5: Ablations (a–c) and sensitivity (d–g). (a) Ten networks: reduction relative to the frozen per-year checkpoints; bars: equal-weight means; dots: networks. (b) Old checkpoint and EvoST, all uncalibrated or all calibrated. (c) New sensors after the arrival refit, calibrated. (d, e) Validation (RLS) and holdout (learning rate) MAE above each run's best (1,165 runs; 1,160 fits). (f) History budget, calibrated (solid) or uncalibrated (dashed). (g) Checkpoint ages on different target years. Details: Appendix A.5.

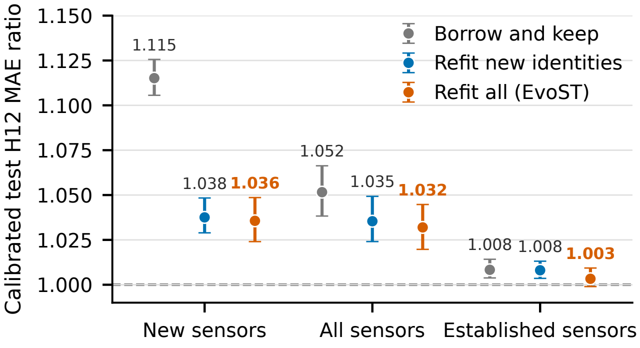


Figure 6: Sensor arrivals on twelve region-year settings (six districts, 2020 and 2025). Calibrated test H12 MAE ratios, geometrically averaged after averaging three seeds per setting, with 95% district-bootstrap intervals. New and All use the target-year checkpoint as reference; Established uses the reference whose backbone runs on established sensors alone (Appendix A.5).

6.6 Newly Added Sensors

Figure 6 tracks sensors that the five-year-old checkpoint has never seen. Each arrives with 64 windows of history before the stream and borrows the mean identity of its five behaviorally nearest established sensors; at stream window 1,024, EvoST refits all identities and rebuilds every calibrator (Section 5.4). On the test portion, which lies entirely after this refit, keeping the borrowed identities leaves new sensors 11.5% worse than the target-year checkpoint. Refitting only the new identities and rebuilding their calibrators lowers this to 3.8% but leaves established sensors 0.8% worse than the calibrated reference whose backbone runs on established sensors alone, with their identities refreshed before the stream. Refitting all identities, as EvoST does, gives 3.6% and 0.3%. The borrowed start matters: kept unchanged, it gives 5.0% lower MAE than independently fitting new identities from the established-sensor mean on the same short histories and 7.0% lower MAE than the mean identity, whereas donors chosen by road adjacency improve on the mean identity by only 0.8% (Figure 5(c)). Appendix A.5 describes how the all-identity refit was specified.

6.7 Ablation Study

Figure 5(a–c) separates the parts of EvoST. On the annual benchmark (a), identity refresh alone lowers ten-network Avg-MAE relative to the frozen checkpoints by 3.7%, the calibrator alone by 8.7%, and both by 9.4%; for Avg-MAPE, refresh alone gives the larger reduction (7.8% against 6.9%). Without any calibrator (b), the old checkpoint's H12 MAE is 47.9% above that of the uncalibrated target-year checkpoint and the refreshed checkpoint's is 1.5% below it; with all three models calibrated, the ratios are 1.177 and 1.025.

6.8 Sensitivity Analysis

Figure 5(d–g) varies the settings of EvoST. On the tested annual validation-selection grid (d), the ten-network mean error varies more across forgetting factors than across ridge penalties: at $\lambda =$

Table 3: History budgets on twelve region-year settings (six districts, 2020 and 2025). Calibrated test H12 MAE relative to the calibrated target-year checkpoint at the same budget, geometrically averaged after averaging three seeds per setting. Fit windows include the identity holdout; selection windows are an additional validation prefix for calibration. The identity learning rate is fixed at 3×10^{-3}.

Fit windows	Selection	Old	Refresh
64	32	1.208	1.214
256	128	1.153	1.198
1,024	512	1.175	1.046
2,048	1,024	1.177	1.026

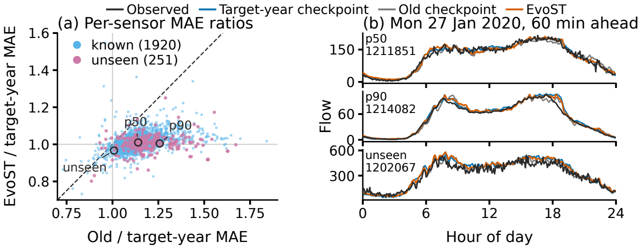


Figure 7: Case study on PEMS12/2020 (calibrated): (a) per-sensor H12 MAE ratios to the target-year checkpoint; (b) 60-minute-ahead forecasts; see Appendix A.5.

0.99, the mean excess H12 MAE above each run’s best setting is 5.7% for $\rho = 1$ and 4.1% for $\rho = 100$, and at $\lambda \in \{0.999, 1\}$ both means stay below 1%. The lowest identity learning rate, 10^{-3}, was selected in 1,100 of the 1,160 annual fits (e), and 10^{-2} has a ten-network mean excess holdout MAE of 4.0% above each fit’s best candidate. Refresh needs enough recent history (f, Table 3; Appendix A.5); the budget sets both the fitting windows and the calibrator’s selection prefix. With 1,024 fitting windows, about 3.6 days of 5-minute data, refresh lowers the calibrated ratio from 1.175 to 1.046, and with 2,048 to 1.026. Without calibration, it lowers MAE at every tested budget, by 8% with 64 windows and by 33% with 2,048; Section 7 covers the calibrated small budgets. Refresh also stays effective for older checkpoints (g): at ages of five, ten, and fifteen years, it recovers 91%, 85%, and 89% of gaps of 18.3%, 21.7%, and 21.2% (ratios 1.016, 1.033, and 1.023).

6.9 Case Study

Figure 7 examines PEMS12/2020. For sensors with valid test targets, the median ratio of calibrated test H12 MAE to the target-year checkpoint falls from 1.13 to 1.01 for the 1,920 the old checkpoint has seen and from 1.18 to 1.00 for the 251 it has not, and EvoST beats the old checkpoint on 97% of them (a). On 27 January (b), EvoST’s mean absolute differences from the target-year forecasts are 5.4, 2.5, and 14.1, versus 8.3, 3.9, and 25.8 for the old checkpoint.

7 Limitations

EvoST relies on labels. Identity refresh uses recent windows whose targets have been released, and the calibrator learns from delayed feedback. The method therefore does not cover label-free deployment. With short histories, EvoST can be worse than calibrating the old checkpoint alone: at the two smallest budgets in Figure 5(f), the calibrated old checkpoint has lower MAE than the calibrated refreshed model. Identities are carried across snapshots by sensor ID. The arrival study provides 64 post-arrival history windows before identity borrowing and does not evaluate commissioning without such history. EvoST applies to backbones whose node-specific parameters can be separated from the shared ones, and we evaluate three such backbones. Identity fitting still backpropagates through the frozen backbone, so its time and memory depend on the backbone and the graph size. The evaluation covers univariate traffic flow with yearly snapshots. In the annual benchmark, where each year has its own checkpoint, most of the Avg-MAE reduction relative to the frozen checkpoint comes from the calibrator.

8 Conclusion

We studied how to maintain a deployed traffic forecaster while its sensor graph evolves. EvoST keeps the shared spatio-temporal weights of a checkpoint fixed and refreshes node identities from released labels. New sensors borrow identities from behaviorally similar established sensors until a joint refit, and per-sensor recursive least squares corrects forecasts as feedback arrives. With all compared forecasters calibrated, identity refresh recovers 86% of the test H12 MAE gap between five-year-old and target-year STAEFormer checkpoints on twelve PeMS region-year settings. Its MAE is 0.7% above full fine-tuning on the same recent data. Similar recovery holds for checkpoints up to fifteen years old, and on sixteen settings the refresh recovers 67% and 88% of the five-year gaps for STID and Graph WaveNet. On the annual benchmark of ten networks, EvoST achieves the lowest Avg-MAE, Avg-RMSE, and Avg-MAPE among the reported non-foundation methods. These results point to outdated node identities as a major source of degradation in the evaluated forecasters, while shared weights remain largely reusable with calibration.

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Table 4: EvoXXLTraffic scale and sensor growth.

District	Span	#Yr	$N_{\text{first}} \to N_{\text{last}}$	Timesteps	Obs.	Size
D03	2001–2025	25	174→1859	2,611,872	2.56B	10.2 GB
D04	2001–2025	25	1266→4110	2,559,744	6.63B	26.5 GB
D05	2005–2025	21	6→572	2,185,344	0.50B	2.0 GB
D06	2005–2025	21	13→746	2,138,976	0.84B	3.4 GB
D07	2001–2025	25	3215→4888	2,629,728	11.28B	45.1 GB
D08	2001–2025	25	262→2074	2,612,736	3.49B	14.0 GB
D10	2006–2025	20	45→1396	2,051,712	1.70B	6.8 GB
D11	1999–2025	27	424→1440	2,788,128	3.23B	12.9 GB
D12	2002–2025	24	1783→2587	2,524,608	5.73B	22.9 GB
Total	1999–2025	–	–	22,102,848	**35.97B**	**144 GB**

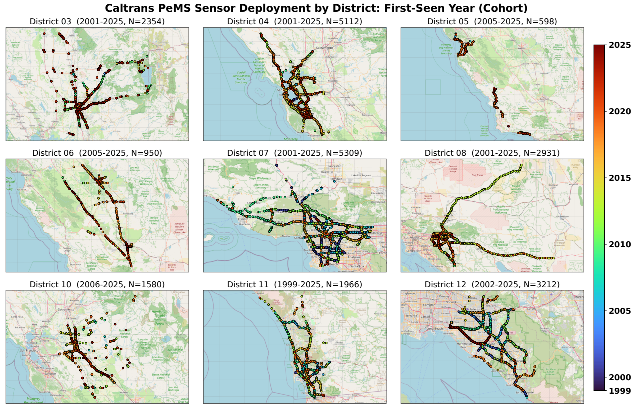
Caltrans PeMS Sensor Deployment by District: First-Seen Year (Cohort)
District 03 (2001-2025, N=2354)
District 04 (2001-2025, N=5112)
District 05 (2005-2025, N=598)
District 06 (2005-2025, N=950)
District 07 (2001-2025, N=5309)
District 08 (2001-2025, N=2931)
District 10 (2006-2025, N=1580)
District 11 (1999-2025, N=1966)
District 12 (2002-2025, N=3212)
2025
2020
2015
2010
2005
2000
1999

Figure 8: Sensor coverage across nine districts. Colors indicate first appearance in the yearly dataset snapshots; N counts all sensors with valid coordinates observed across the displayed years.

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A Appendix

A.1 Structural Evolution Details

Figure 9 tracks average node degree and graph density for the yearly proximity graphs. With N nodes and E undirected edges counted once, these quantities are $2E/N$ and $2E/[N(N-1)]$, respectively. Both change non-monotonically as sensor additions and removals alter local neighborhoods and spatial connections. Successive snapshots therefore differ in both dimension and connectivity.

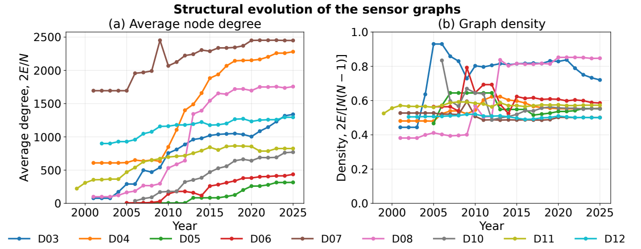


Figure 9: Yearly evolution of the sensor proximity graphs. Average node degree and graph density both vary non-monotonically.

A.2 Extended Related Work

A.2.1 Spatio-Temporal Traffic Forecasting. Traffic forecasting, including traffic flow, speed, and congestion status, has always been one of the most important issues in computational urban and intelligent transportation systems. Early traffic forecasting methods primarily relied on statistical modeling and traditional machine learning methods, such as Historical Average (HA), Autoregressive Integrated Moving Average (ARIMA) [1], Seasonal ARIMA [31], Kalman filtering [12], Vector Autoregression (VAR) [21], Support Vector Regression (SVR) [32], and k-Nearest Neighbors (kNN) [2]. These methods typically treat traffic sequences as one-dimensional or low-dimensional time series and make predictions by modeling trends, periodicity, and short-term autocorrelation in historical observations. Among them, HA uses historical averages over the same period as predictions, making it simple and efficient but difficult to adapt to sudden changes. ARIMA and its variants can characterize linear temporal dependence and periodic patterns but usually rely on stationarity assumptions. Kalman filtering is suitable for online state estimation and short-term forecasting, while SVR and kNN improve predictive capability through nonlinear mapping or similar-pattern matching. Nevertheless, these methods mostly rely on manual feature design and struggle to characterize complex nonlinear temporal variations and spatial dependencies within road networks simultaneously.

Deep traffic forecasting has used convolutional neural networks (CNNs), recurrent neural networks (RNNs), graph neural networks (GNNs), Transformers, and, more recently, state-space models. Early temporal models used long short-term memory (LSTM) or gated recurrent unit (GRU) networks to capture long-range dependencies. CNNs [25] and temporal convolutional networks (TCNs) [26] later improved parallelism while retaining local temporal structure. Graph models then made road-network dependencies explicit. Representative methods include T-GCN [44], STGCN [39], and DCRNN [13]. T-GCN places graph convolution inside GRU units, STGCN combines graph and temporal convolutions, and DCRNN models directed propagation with diffusion convolution and an encoder-decoder architecture. GNNs have also been applied to regional multimodal demand forecasting: Ma et al. [18] release an NYC taxi-bike dataset and study multi-source, multi-graph, and meta-information-enhanced STGNNs. Our setting instead concerns sensor graphs whose nodes and connectivity change over time.

Traffic forecasting models have since developed toward adaptive graph learning, attention mechanisms, and large-scale sequence modeling. Graph WaveNet [33], for example, reduces reliance on predefined adjacency matrices by learning adaptive spatial dependencies. ASTGCN [9], GMAN [45], and STTN [34] introduce spatio-temporal attention to dynamically model relationships among sensors and time steps. Transformer-based models are now widely used because of their long-sequence modeling capability, while state-space models offer linear-complexity alternatives for large-scale forecasting. This development has produced many strong models, including MTMGNN [36], STQCL [35], STAEFormer [15], STID [23], HIMNet [7], PDFormer [10], and AutoSTF [17].

Despite their progress on static benchmarks, most existing methods assume that the sensor set and road-network topology remain unchanged between training and testing. In real deployments, however, sensor networks expand over time, new sensors are commissioned, and connectivity and traffic patterns drift. A deployed forecaster therefore faces not merely a longer time series but a continuously evolving traffic graph. The resulting node growth, topology change, and temporal distribution shift challenge models built under a fixed-graph assumption.

A.2.2 *Online and Continual Traffic Forecasting.*

Long-term traffic-system deployment has consequently motivated online and continual forecasting. New sensors are repeatedly added, while traffic patterns at existing nodes evolve over time. An effective model must therefore acquire new spatio-temporal patterns without catastrophically forgetting previously learned knowledge.

TrafficStream [6] was among the first methods to formalize network expansion and pattern evolution as a streaming continual-learning problem. It uses local subgraphs to integrate neighborhood information for new sensors, Jensen–Shannon divergence to detect drift at existing nodes, and historical replay with parameter smoothing to mitigate catastrophic forgetting. Following an explicit-memory approach, PECPM [28] maintains a library of representative spatio-temporal patterns and updates only new or conflicting nodes when a new graph arrives. STKEC [27] selects influential old nodes to construct local subgraphs, combines them with a memory bank for long-term periodic patterns, and preserves prior knowledge through parameter-distance regularization. To reduce update cost, EAC [4] uses a continual prompt pool to represent knowledge from different stages and follows an expansion–compression strategy that avoids large-scale updates of the whole model. STBP [14] freezes a general spatio-temporal backbone and expands only a scalable contextual pattern library, balancing stability and plasticity. STEV [19] treats new sensors as a variable-expansion problem and uses a flattening scheme to improve robustness to varying input scales. STRAP [40] constructs an external pattern library and retrieves similar patterns as prompts at inference time. ST-TTC [5] performs test-time calibration with frequency-domain alignment and streaming memory queues, but primarily addresses temporal drift within a fixed node set.

Unlike methods that update substantial model components or depend on large replay memories, EvoST keeps the shared spatio-temporal weights of a deployed checkpoint fixed. It refreshes only the node identities from released labels, lets new sensors borrow identities from behaviorally similar established sensors until a joint refit, and corrects the forecasts of each sensor by recursive least squares as delayed feedback arrives.

A.3 Annual Benchmark Results on All Ten Networks

Average performance on the remaining districts. Tables 5 and 6 report PEMS05 to PEMS08 and PEMS10 to PEMS12. Together with Table 1, EvoST has the lowest MAE, RMSE, and MAPE among the non-foundation methods in all 30 Avg dataset-metric comparisons, each strictly below every other non-foundation method. Averaged equally across the seven networks of these two tables, its MAE, RMSE, and MAPE are 10.5%, 8.1%, and 5.3% lower than those of the STAEFormer row (13.1%, 10.6%, and 6.8% across all ten networks). The STAEFormer row reports the benchmark's results, which differ from EvoST in input standardization and in some checkpoint selections (Section 6); Figure 11 isolates the effect of maintaining the same checkpoints.

Table conventions. In Tables 5 and 6, each row is one baseline and each dataset reports MAE, RMSE, and MAPE in percent. Avg denotes the mean of cumulative-horizon scores over prediction lengths 1–12. Entries are mean±standard deviation (SD) where available; for EvoST, the SD is the sample SD over five seeds. Deterministic zero-shot references are single-run and shown with ±0.00. Chronos-2 uses multivariate graph-grouped inference, with per-year low-rank adaptation (LoRA) for its FT variant. Bold and underlined entries mark the best and second-best non-foundation methods, respectively. Foundation-model rows are excluded from highlighting.

Per-horizon performance and TSFM comparison. For compactness, the preceding tables average cumulative errors over prediction lengths 1 to 12. Tables 7 and 8 report horizons 3, 6, and 12. EvoST has the lowest error among the non-foundation methods in all 120 reported cells (three metrics at horizons 3, 6, and 12 and Avg on ten networks). Averaged equally over the ten networks, its MAE is 14.0%, 12.8%, and 12.0% lower than that of the STAEFormer column at horizons 3, 6, and 12, with the same caveat on this reported baseline as above.

Across the three horizons and Avg rows, EvoST also has lower error than the best TSFM column in 102 of the 120 cells, including all 80 MAE and RMSE cells and 27 of the 30 Avg cells. The 18 exceptions are MAPE cells: horizon 3 on all ten networks, horizon 6 on PEMS04, PEMS05, PEMS07, PEMS11, and PEMS12, and Avg on PEMS04, PEMS05, and PEMS11. Fine-tuning remains important for the foundation models: Chronos-2, TimesFM 2.5, and Moirai-2.0 improve over their zero-shot counterparts in all 30 Avg dataset-metric comparisons, with mean relative reductions of 13.5%, 10.9%, and 15.4%, respectively.

Per-horizon table conventions. In Tables 7 and 8, EvoST is evaluated over five seeds with the sample SD, and MAPE is reported in percent. Compact headers pair zero-shot and fine-tuned variants: Chro2-Z/F, TimesF-Z/T, Moira1-Z/T, and Moira2-Z/T. Deterministic zero-shot references are single-run and shown with ±0.00, whereas fine-tuned references report mean±SD over three seeds. Chronos-2 uses multivariate graph-grouped inference and per-year LoRA fine-tuning for its FT variant. TSFM columns are excluded

Table 5: Average results on PEMS05–08 (mean±standard deviation; for EvoST, the sample standard deviation over five seeds).

Category	Baseline	PEMS05			PEMS06			PEMS07			PEMS08		
		MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
Static STGNN Backbones	DCRNN	11.69$_{\pm0.07}$	18.34$_{\pm0.09}$	26.98$_{\pm0.74}$	14.87$_{\pm0.31}$	23.29$_{\pm0.43}$	24.66$_{\pm0.77}$	19.39$_{\pm0.28}$	32.62$_{\pm0.30}$	31.09$_{\pm1.27}$	15.50$_{\pm0.13}$	25.20$_{\pm0.19}$	23.31$_{\pm0.41}$
	ASTGNN	11.69$_{\pm0.04}$	18.52$_{\pm0.08}$	25.31$_{\pm0.60}$	15.52$_{\pm0.16}$	24.04$_{\pm0.28}$	23.41$_{\pm0.46}$	19.56$_{\pm0.05}$	33.05$_{\pm0.13}$	29.95$_{\pm0.26}$	15.22$_{\pm0.05}$	25.23$_{\pm0.10}$	20.85$_{\pm0.39}$
	TGCN	11.94$_{\pm0.10}$	19.43$_{\pm0.16}$	26.83$_{\pm0.59}$	15.52$_{\pm0.46}$	25.38$_{\pm0.56}$	24.06$_{\pm0.40}$	20.62$_{\pm0.41}$	37.75$_{\pm0.75}$	32.43$_{\pm1.02}$	15.86$_{\pm0.17}$	27.85$_{\pm0.40}$	22.13$_{\pm0.71}$
	GWN	12.57$_{\pm0.13}$	19.49$_{\pm0.18}$	27.19$_{\pm0.86}$	16.16$_{\pm0.54}$	24.84$_{\pm0.65}$	24.76$_{\pm0.81}$	19.51$_{\pm0.99}$	32.22$_{\pm1.35}$	31.81$_{\pm2.36}$	15.69$_{\pm0.16}$	25.27$_{\pm0.29}$	21.90$_{\pm0.74}$
Naïve Schemes	Pretrain	24.94$_{\pm0.99}$	34.77$_{\pm1.06}$	116.10$_{\pm7.29}$	78.66$_{\pm1.39}$	97.96$_{\pm2.03}$	79.40$_{\pm1.33}$	40.30$_{\pm2.05}$	54.10$_{\pm1.93}$	49.24$_{\pm2.54}$	52.25$_{\pm2.71}$	62.21$_{\pm3.13}$	57.09$_{\pm1.58}$
	Retrain	11.31$_{\pm0.04}$	18.03$_{\pm0.07}$	24.42$_{\pm0.41}$	14.37$_{\pm0.34}$	22.39$_{\pm0.39}$	22.54$_{\pm0.33}$	18.02$_{\pm0.27}$	30.73$_{\pm0.23}$	27.64$_{\pm1.35}$	14.34$_{\pm0.07}$	23.85$_{\pm0.09}$	20.58$_{\pm0.25}$
	Online-NN	15.98$_{\pm0.52}$	26.35$_{\pm0.67}$	29.06$_{\pm0.76}$	27.64$_{\pm0.65}$	38.29$_{\pm1.20}$	32.16$_{\pm0.38}$	22.72$_{\pm0.59}$	35.82$_{\pm0.68}$	35.08$_{\pm1.56}$	17.30$_{\pm0.05}$	26.86$_{\pm0.13}$	24.09$_{\pm0.35}$
	Online-AN	11.10$_{\pm0.07}$	17.63$_{\pm0.13}$	23.49$_{\pm0.44}$	14.02$_{\pm0.21}$	21.68$_{\pm0.28}$	21.93$_{\pm0.52}$	17.67$_{\pm0.31}$	30.13$_{\pm0.32}$	26.68$_{\pm1.27}$	13.92$_{\pm0.10}$	23.20$_{\pm0.15}$	19.28$_{\pm0.22}$
Evolving Graph	TrafficStream	11.32$_{\pm0.13}$	18.04$_{\pm0.14}$	24.56$_{\pm0.93}$	14.46$_{\pm0.34}$	22.41$_{\pm0.47}$	22.43$_{\pm0.49}$	17.54$_{\pm0.31}$	30.04$_{\pm0.29}$	26.67$_{\pm2.19}$	14.04$_{\pm0.13}$	23.35$_{\pm0.18}$	20.96$_{\pm1.33}$
	PECPM	11.67$_{\pm0.31}$	18.50$_{\pm0.44}$	25.13$_{\pm0.85}$	18.02$_{\pm3.19}$	29.03$_{\pm6.07}$	26.88$_{\pm5.50}$	19.23$_{\pm1.47}$	32.52$_{\pm2.46}$	27.85$_{\pm1.72}$	15.58$_{\pm0.77}$	25.25$_{\pm1.04}$	22.26$_{\pm1.64}$
	STKEC	11.48$_{\pm0.16}$	18.26$_{\pm0.21}$	25.11$_{\pm0.78}$	14.90$_{\pm0.24}$	23.19$_{\pm0.34}$	24.14$_{\pm0.58}$	17.07$_{\pm1.26}$	29.33$_{\pm1.84}$	25.66$_{\pm3.04}$	14.47$_{\pm0.38}$	23.89$_{\pm0.50}$	21.55$_{\pm1.34}$
	EAC	16.35$_{\pm3.53}$	23.77$_{\pm3.62}$	61.09$_{\pm37.33}$	21.10$_{\pm4.19}$	31.23$_{\pm4.99}$	36.63$_{\pm16.01}$	18.26$_{\pm0.27}$	30.41$_{\pm0.37}$	28.73$_{\pm0.39}$	14.76$_{\pm0.28}$	23.76$_{\pm0.37}$	26.61$_{\pm1.73}$
Retrieval TTC	STRAP	11.33$_{\pm0.04}$	17.97$_{\pm0.04}$	25.25$_{\pm0.88}$	14.44$_{\pm0.46}$	22.53$_{\pm0.49}$	22.60$_{\pm0.66}$	17.81$_{\pm0.16}$	30.37$_{\pm0.18}$	26.88$_{\pm0.66}$	14.15$_{\pm0.03}$	23.54$_{\pm0.04}$	19.96$_{\pm0.41}$
	ST-TTC	11.22$_{\pm0.04}$	17.89$_{\pm0.07}$	24.33$_{\pm0.41}$	14.25$_{\pm0.34}$	22.21$_{\pm0.38}$	22.42$_{\pm0.33}$	17.74$_{\pm0.27}$	30.37$_{\pm0.23}$	27.40$_{\pm1.33}$	14.16$_{\pm0.07}$	23.63$_{\pm0.08}$	20.45$_{\pm0.25}$
Static Forecasting Backbones	STID	11.64$_{\pm0.06}$	18.69$_{\pm0.08}$	24.51$_{\pm0.52}$	15.39$_{\pm0.23}$	24.34$_{\pm0.40}$	22.72$_{\pm0.38}$	19.27$_{\pm0.19}$	32.22$_{\pm0.21}$	34.13$_{\pm0.84}$	15.12$_{\pm0.07}$	25.69$_{\pm0.17}$	20.33$_{\pm0.76}$
	STNorm	12.57$_{\pm0.10}$	19.00$_{\pm0.17}$	35.42$_{\pm0.64}$	16.70$_{\pm0.35}$	24.97$_{\pm0.45}$	32.58$_{\pm1.07}$	25.17$_{\pm1.15}$	39.57$_{\pm1.56}$	58.70$_{\pm3.03}$	18.33$_{\pm0.31}$	28.00$_{\pm0.33}$	43.33$_{\pm2.01}$
	iTrans	11.42$_{\pm0.06}$	18.04$_{\pm0.07}$	24.92$_{\pm0.37}$	15.18$_{\pm0.41}$	23.39$_{\pm0.46}$	24.00$_{\pm0.96}$	18.64$_{\pm0.09}$	31.19$_{\pm0.17}$	30.76$_{\pm1.13}$	14.60$_{\pm0.08}$	23.99$_{\pm0.13}$	22.07$_{\pm0.99}$
	DLinear	12.16$_{\pm0.05}$	19.40$_{\pm0.06}$	26.15$_{\pm0.70}$	16.40$_{\pm0.13}$	25.55$_{\pm0.15}$	24.49$_{\pm0.37}$	19.85$_{\pm0.01}$	33.90$_{\pm0.07}$	29.94$_{\pm0.50}$	15.58$_{\pm0.04}$	26.07$_{\pm0.04}$	21.16$_{\pm0.42}$
	STAEFormer	10.16$_{\pm0.07}$	16.07$_{\pm0.10}$	22.83$_{\pm0.44}$	13.13$_{\pm0.60}$	19.88$_{\pm0.68}$	20.28$_{\pm0.85}$	15.32$_{\pm0.28}$	26.93$_{\pm0.26}$	21.39$_{\pm0.65}$	12.52$_{\pm0.03}$	20.88$_{\pm0.09}$	17.66$_{\pm0.23}$
Times-series Forecasting Foundation Model	Chronos-2 (ZS)	12.67$_{\pm0.00}$	20.54$_{\pm0.00}$	25.64$_{\pm0.00}$	15.59$_{\pm0.00}$	25.65$_{\pm0.00}$	22.27$_{\pm0.00}$	18.67$_{\pm0.00}$	34.18$_{\pm0.00}$	22.66$_{\pm0.00}$	16.19$_{\pm0.00}$	27.58$_{\pm0.00}$	20.06$_{\pm0.00}$
	Chronos-2 (FT)	11.12$_{\pm0.01}$	18.61$_{\pm0.02}$	21.67$_{\pm0.02}$	13.66$_{\pm0.01}$	23.12$_{\pm0.04}$	19.42$_{\pm0.04}$	16.63$_{\pm0.04}$	31.15$_{\pm0.13}$	20.04$_{\pm0.03}$	14.22$_{\pm0.02}$	25.08$_{\pm0.08}$	17.49$_{\pm0.03}$
	TimesFM2.5 (ZS)	13.11$_{\pm0.00}$	21.07$_{\pm0.00}$	26.75$_{\pm0.00}$	16.08$_{\pm0.00}$	26.22$_{\pm0.00}$	22.96$_{\pm0.00}$	19.55$_{\pm0.00}$	35.34$_{\pm0.00}$	23.68$_{\pm0.00}$	16.87$_{\pm0.00}$	28.38$_{\pm0.00}$	20.96$_{\pm0.00}$
	TimesFM2.5 (FT)	11.85$_{\pm0.01}$	19.92$_{\pm0.01}$	22.50$_{\pm0.02}$	14.28$_{\pm0.01}$	24.21$_{\pm0.01}$	20.06$_{\pm0.01}$	17.86$_{\pm0.02}$	33.57$_{\pm0.03}$	21.24$_{\pm0.02}$	14.90$_{\pm0.01}$	26.18$_{\pm0.02}$	18.26$_{\pm0.00}$
	Moirai-2.0 (ZS)	13.26$_{\pm0.00}$	21.14$_{\pm0.00}$	25.93$_{\pm0.00}$	16.33$_{\pm0.00}$	26.23$_{\pm0.00}$	22.46$_{\pm0.00}$	20.58$_{\pm0.00}$	36.42$_{\pm0.00}$	23.57$_{\pm0.00}$	17.64$_{\pm0.00}$	29.31$_{\pm0.00}$	20.70$_{\pm0.00}$
	Moirai-2.0 (FT)	11.66$_{\pm0.00}$	19.31$_{\pm0.00}$	23.30$_{\pm0.01}$	14.28$_{\pm0.01}$	23.75$_{\pm0.04}$	20.71$_{\pm0.01}$	17.53$_{\pm0.01}$	32.42$_{\pm0.05}$	21.17$_{\pm0.03}$	14.99$_{\pm0.02}$	25.97$_{\pm0.06}$	18.63$_{\pm0.04}$
Ours	EvoST	**9.48**$_{\pm0.03}$	**15.12**$_{\pm0.04}$	**21.77**$_{\pm0.17}$	**11.51**$_{\pm0.20}$	**17.87**$_{\pm0.29}$	**19.18**$_{\pm0.35}$	**13.45**$_{\pm0.06}$	**24.56**$_{\pm0.09}$	**19.85**$_{\pm0.19}$	**11.55**$_{\pm0.04}$	**19.57**$_{\pm0.09}$	**16.89**$_{\pm0.07}$

Table 6: Average results on PEMS10–12 (mean±standard deviation; for EvoST, the sample standard deviation over five seeds).

Category	Baseline	PEMS10			PEMS11			PEMS12		
		MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)	MAE	RMSE	MAPE (%)
Static STGNN Backbones	DCRNN	12.26$_{\pm0.16}$	20.13$_{\pm0.26}$	31.65$_{\pm0.35}$	19.69$_{\pm0.30}$	33.66$_{\pm0.34}$	31.28$_{\pm0.86}$	16.16$_{\pm0.17}$	28.46$_{\pm0.25}$	30.90$_{\pm0.78}$
	ASTGNN	12.45$_{\pm0.05}$	20.80$_{\pm0.06}$	30.22$_{\pm0.32}$	20.00$_{\pm0.16}$	34.32$_{\pm0.27}$	26.69$_{\pm0.39}$	15.98$_{\pm0.10}$	28.49$_{\pm0.18}$	27.32$_{\pm0.53}$
	TGCN	12.43$_{\pm0.08}$	21.49$_{\pm0.14}$	30.86$_{\pm0.22}$	21.35$_{\pm0.31}$	40.11$_{\pm0.53}$	30.37$_{\pm0.32}$	17.13$_{\pm0.08}$	33.43$_{\pm0.19}$	30.68$_{\pm0.38}$
	GWN	12.54$_{\pm0.11}$	20.45$_{\pm0.13}$	30.65$_{\pm0.69}$	20.29$_{\pm0.44}$	34.35$_{\pm0.80}$	27.55$_{\pm0.81}$	16.38$_{\pm0.23}$	28.47$_{\pm0.35}$	27.32$_{\pm0.41}$
Naïve Schemes	Pretrain	81.39$_{\pm3.16}$	84.52$_{\pm3.24}$	755.06$_{\pm28.01}$	85.43$_{\pm3.52}$	116.95$_{\pm3.38}$	82.12$_{\pm18.72}$	16.41$_{\pm0.21}$	27.96$_{\pm0.24}$	28.48$_{\pm0.81}$
	Retrain	11.85$_{\pm0.04}$	19.66$_{\pm0.03}$	30.30$_{\pm0.22}$	18.77$_{\pm0.13}$	32.21$_{\pm0.15}$	27.67$_{\pm0.81}$	14.91$_{\pm0.05}$	26.54$_{\pm0.06}$	26.88$_{\pm0.52}$
	Online-NN	12.35$_{\pm0.23}$	20.58$_{\pm0.45}$	31.79$_{\pm1.11}$	21.54$_{\pm1.59}$	36.45$_{\pm2.86}$	30.83$_{\pm1.25}$	18.04$_{\pm0.57}$	32.01$_{\pm1.14}$	29.42$_{\pm0.55}$
	Online-AN	11.60$_{\pm0.09}$	19.22$_{\pm0.17}$	29.80$_{\pm0.34}$	18.29$_{\pm0.13}$	31.40$_{\pm0.13}$	25.39$_{\pm0.75}$	14.54$_{\pm0.06}$	25.97$_{\pm0.07}$	25.48$_{\pm0.28}$
Evolving Graph	TrafficStream	11.93$_{\pm0.08}$	19.77$_{\pm0.14}$	32.18$_{\pm0.70}$	18.67$_{\pm0.35}$	31.90$_{\pm0.47}$	28.29$_{\pm1.63}$	14.74$_{\pm0.07}$	26.32$_{\pm0.10}$	26.88$_{\pm0.62}$
	PECPM	12.61$_{\pm1.16}$	21.29$_{\pm2.68}$	30.41$_{\pm1.69}$	25.89$_{\pm2.53}$	44.81$_{\pm4.91}$	36.90$_{\pm6.66}$	16.18$_{\pm1.49}$	28.57$_{\pm3.19}$	29.30$_{\pm1.38}$
	STKEC	11.92$_{\pm0.07}$	19.80$_{\pm0.11}$	31.06$_{\pm0.42}$	19.02$_{\pm0.23}$	32.38$_{\pm0.23}$	30.78$_{\pm1.62}$	14.71$_{\pm0.08}$	26.28$_{\pm0.12}$	26.66$_{\pm0.60}$
	EAC	13.60$_{\pm1.76}$	21.08$_{\pm1.43}$	59.52$_{\pm40.27}$	25.76$_{\pm4.14}$	41.08$_{\pm5.52}$	47.34$_{\pm12.72}$	14.98$_{\pm0.30}$	26.18$_{\pm0.40}$	29.71$_{\pm2.65}$
Retrieval TTC	STRAP	11.79$_{\pm0.09}$	19.58$_{\pm0.10}$	30.34$_{\pm0.58}$	18.60$_{\pm0.15}$	31.88$_{\pm0.20}$	28.85$_{\pm1.00}$	14.88$_{\pm0.14}$	26.48$_{\pm0.17}$	27.09$_{\pm1.22}$
	ST-TTC	11.76$_{\pm0.04}$	19.53$_{\pm0.04}$	30.14$_{\pm0.22}$	18.52$_{\pm0.13}$	31.83$_{\pm0.15}$	27.53$_{\pm0.80}$	14.74$_{\pm0.05}$	26.28$_{\pm0.05}$	26.77$_{\pm0.51}$
Static Forecasting Backbones	STID	12.53$_{\pm0.08}$	20.54$_{\pm0.09}$	33.11$_{\pm0.25}$	20.02$_{\pm0.32}$	33.89$_{\pm0.36}$	32.17$_{\pm1.75}$	16.07$_{\pm0.07}$	28.14$_{\pm0.19}$	33.77$_{\pm0.88}$
	STNorm	13.75$_{\pm0.32}$	21.81$_{\pm0.40}$	41.57$_{\pm1.79}$	21.77$_{\pm0.34}$	35.62$_{\pm0.53}$	48.00$_{\pm2.68}$	19.49$_{\pm0.39}$	32.19$_{\pm0.51}$	58.93$_{\pm2.28}$
	iTrans	11.83$_{\pm0.07}$	19.63$_{\pm0.10}$	30.28$_{\pm0.74}$	19.24$_{\pm0.28}$	32.79$_{\pm0.31}$	30.95$_{\pm2.77}$	15.27$_{\pm0.11}$	26.92$_{\pm0.15}$	29.32$_{\pm1.41}$
	DLinear	12.86$_{\pm0.04}$	21.36$_{\pm0.05}$	31.07$_{\pm0.38}$	20.62$_{\pm0.12}$	35.52$_{\pm0.13}$	27.66$_{\pm0.61}$	16.23$_{\pm0.02}$	29.24$_{\pm0.05}$	27.74$_{\pm0.47}$
	STAEFormer	10.34$_{\pm0.03}$	17.20$_{\pm0.05}$	27.83$_{\pm0.45}$	17.05$_{\pm0.12}$	28.82$_{\pm0.20}$	24.54$_{\pm0.69}$	12.91$_{\pm0.11}$	23.13$_{\pm0.12}$	23.94$_{\pm0.74}$
Times-series Forecasting Foundation Model	Chronos-2 (ZS)	13.30$_{\pm0.00}$	22.54$_{\pm0.00}$	30.51$_{\pm0.00}$	20.04$_{\pm0.00}$	37.05$_{\pm0.00}$	25.36$_{\pm0.00}$	16.67$_{\pm0.00}$	30.85$_{\pm0.00}$	26.63$_{\pm0.00}$
	Chronos-2 (FT)	11.85$_{\pm0.01}$	20.64$_{\pm0.04}$	27.84$_{\pm0.07}$	17.19$_{\pm0.02}$	32.41$_{\pm0.08}$	21.93$_{\pm0.04}$	14.66$_{\pm0.01}$	27.68$_{\pm0.07}$	22.84$_{\pm0.04}$
	TimesFM2.5 (ZS)	13.63$_{\pm0.00}$	22.92$_{\pm0.00}$	31.47$_{\pm0.00}$	20.91$_{\pm0.00}$	38.14$_{\pm0.00}$	26.41$_{\pm0.00}$	17.48$_{\pm0.00}$	31.92$_{\pm0.00}$	27.93$_{\pm0.00}$
	TimesFM2.5 (FT)	12.27$_{\pm0.01}$	21.11$_{\pm0.01}$	28.62$_{\pm0.03}$	18.32$_{\pm0.02}$	34.43$_{\pm0.07}$	23.11$_{\pm0.03}$	15.69$_{\pm0.02}$	29.74$_{\pm0.05}$	24.31$_{\pm0.02}$
	Moirai-2.0 (ZS)	14.18$_{\pm0.00}$	23.64$_{\pm0.00}$	30.73$_{\pm0.00}$	21.24$_{\pm0.00}$	38.56$_{\pm0.00}$	25.78$_{\pm0.00}$	17.90$_{\pm0.00}$	32.32$_{\pm0.00}$	27.26$_{\pm0.00}$
	Moirai-2.0 (FT)	12.43$_{\pm0.01}$	21.32$_{\pm0.03}$	29.22$_{\pm0.02}$	17.71$_{\pm0.01}$	32.78$_{\pm0.04}$	23.23$_{\pm0.03}$	15.32$_{\pm0.01}$	28.62$_{\pm0.02}$	24.38$_{\pm0.02}$
Ours	EvoST	**9.76**$_{\pm0.03}$	**16.30**$_{\pm0.04}$	**27.56**$_{\pm0.12}$	**13.70**$_{\pm0.09}$	**25.24**$_{\pm0.12}$	**22.27**$_{\pm0.17}$	**11.76**$_{\pm0.06}$	**21.36**$_{\pm0.10}$	**22.73**$_{\pm0.21}$

from best/second-best highlighting. Pretrain fixes the first-year model, Retrain starts from scratch each year, Online-NN updates new sensors, and Online-AN updates all active sensors.

A.3.1 Per-Step Results on All Ten Networks. Figure 10 extends Figure 3 to all ten networks. The curves use per-step rather than cumulative errors and compare EvoST with the same checkpoints without adaptation (Frozen) and with the calibrator only (Frozen+RLS), over the same years and five seeds on each network.

Table 7: Per-horizon results, part 1/2 (PEMS03, PEMS04, PEMS05, PEMS06, PEMS07; mean±std over seeds per dataset). Bold: best, $\underline{\text{underline}}$: second best. EvoST is measured over five seeds (sample standard deviation). MAPE values are reported in percent. Chronos-2 (ZS) and Chronos-2 (FT) use multivariate graph-grouped inference; FT uses per-year LoRA fine-tuning. TSFM columns are excluded from highlighting. Compact headers denote zero-shot/fine-tuned pairs: Chro2-Z/F for Chronos-2, TimesF-Z/T for TimesFM2.5, Moira1-Z/T for Moirai-MoE, and Moira2-Z/T for Moirai-2.0. Their deterministic zero-shot (ZS) references are single-run and shown with ±0.00 for visual consistency; fine-tuned (FT) references are reported as mean±std over 3 seeds.

		Static STGNN Backbones				Naïve Schemes				Evolving Graph				Retrieval TTC		Static Forecasting Backbones					TSFMs								Ours
Metric	**Len**	**DCRNN**	**ASTGNN**	**TGCN**	**GWN**	**Pretrain**	**Retrain**	**OL-NN**	**OL-AN**	**TrafStm**	**PECPM**	**STKEC**	**EAC**	**STRAP**	**ST-TTC**	**STID**	**STNorm**	**iTrans**	**DLinear**	**STAE**	**Chro2-Z**	**Chro2-F**	**TimesF-Z**	**TimesF-T**	**Moira1-Z**	**Moira1-T**	**Moira2-Z**	**Moira2-T**	**EvoST**
PEMS03																													
MAE	3	14.42$_{\pm0.21}$	13.70$_{\pm0.12}$	15.14$_{\pm0.10}$	14.22$_{\pm0.29}$	30.61$_{\pm1.22}$	13.09$_{\pm0.16}$	16.25$_{\pm0.19}$	12.76$_{\pm0.05}$	13.22$_{\pm0.11}$	13.88$_{\pm0.85}$	13.42$_{\pm0.27}$	14.18$_{\pm0.31}$	13.10$_{\pm0.12}$	12.89$_{\pm0.15}$	14.28$_{\pm0.15}$	16.46$_{\pm0.42}$	13.16$_{\pm0.13}$	14.17$_{\pm0.07}$	$\underline{11.81_{\pm0.11}}$	12.53$_{\pm0.00}$	11.73$_{\pm0.01}$	13.20$_{\pm0.00}$	12.13$_{\pm0.01}$	14.50$_{\pm0.00}$	16.03$_{\pm0.64}$	13.66$_{\pm0.00}$	12.16$_{\pm0.00}$	**10.40$_{\pm0.04}$**
	6	14.67$_{\pm0.18}$	14.89$_{\pm0.10}$	15.63$_{\pm0.10}$	15.08$_{\pm0.23}$	31.21$_{\pm1.19}$	14.01$_{\pm0.14}$	17.32$_{\pm0.16}$	13.68$_{\pm0.05}$	14.17$_{\pm0.17}$	14.76$_{\pm0.81}$	14.28$_{\pm0.20}$	15.64$_{\pm0.48}$	14.00$_{\pm0.12}$	13.82$_{\pm0.13}$	14.89$_{\pm0.14}$	17.86$_{\pm0.45}$	14.05$_{\pm0.14}$	15.45$_{\pm0.06}$	$\underline{12.22_{\pm0.10}}$	14.74$_{\pm0.00}$	13.09$_{\pm0.01}$	15.55$_{\pm0.00}$	13.87$_{\pm0.01}$	17.22$_{\pm0.00}$	18.39$_{\pm0.59}$	15.87$_{\pm0.00}$	13.70$_{\pm0.01}$	**10.91$_{\pm0.04}$**
	12	16.27$_{\pm0.18}$	17.57$_{\pm0.08}$	17.27$_{\pm0.11}$	17.24$_{\pm0.30}$	32.66$_{\pm0.99}$	15.92$_{\pm0.14}$	19.41$_{\pm0.20}$	15.46$_{\pm0.09}$	16.10$_{\pm0.35}$	16.57$_{\pm0.78}$	16.24$_{\pm0.18}$	18.65$_{\pm0.79}$	15.86$_{\pm0.15}$	15.74$_{\pm0.13}$	16.71$_{\pm0.13}$	21.02$_{\pm0.59}$	15.95$_{\pm0.13}$	18.34$_{\pm0.04}$	$\underline{12.98_{\pm0.13}}$	19.88$_{\pm0.00}$	15.90$_{\pm0.02}$	20.21$_{\pm0.00}$	17.38$_{\pm0.02}$	22.10$_{\pm0.00}$	22.80$_{\pm0.59}$	20.35$_{\pm0.00}$	16.86$_{\pm0.01}$	**11.71$_{\pm0.06}$**
	Avg	14.99$_{\pm0.18}$	15.15$_{\pm0.10}$	15.88$_{\pm0.10}$	15.33$_{\pm0.26}$	31.37$_{\pm1.11}$	14.16$_{\pm0.15}$	17.47$_{\pm0.16}$	13.80$_{\pm0.05}$	14.32$_{\pm0.20}$	14.90$_{\pm0.81}$	14.47$_{\pm0.22}$	15.91$_{\pm0.48}$	14.15$_{\pm0.13}$	13.98$_{\pm0.14}$	15.14$_{\pm0.14}$	18.20$_{\pm0.47}$	14.21$_{\pm0.13}$	15.73$_{\pm0.06}$	$\underline{12.26_{\pm0.11}}$	15.27$_{\pm0.00}$	13.31$_{\pm0.01}$	15.91$_{\pm0.00}$	14.16$_{\pm0.01}$	17.59$_{\pm0.00}$	18.68$_{\pm0.59}$	16.19$_{\pm0.00}$	13.95$_{\pm0.01}$	**10.92$_{\pm0.05}$**
RMSE	3	23.64$_{\pm0.22}$	22.30$_{\pm0.20}$	28.88$_{\pm0.42}$	23.01$_{\pm0.41}$	39.61$_{\pm1.17}$	21.27$_{\pm0.15}$	25.71$_{\pm0.14}$	20.83$_{\pm0.06}$	21.39$_{\pm0.10}$	22.19$_{\pm1.02}$	21.71$_{\pm0.38}$	22.90$_{\pm0.58}$	21.23$_{\pm0.22}$	20.98$_{\pm0.13}$	25.67$_{\pm0.42}$	25.72$_{\pm0.59}$	21.37$_{\pm0.14}$	23.03$_{\pm0.04}$	$\underline{19.61_{\pm0.15}}$	21.37$_{\pm0.00}$	20.23$_{\pm0.02}$	22.46$_{\pm0.00}$	20.93$_{\pm0.03}$	25.10$_{\pm0.00}$	27.89$_{\pm1.02}$	23.21$_{\pm0.00}$	20.77$_{\pm0.02}$	**17.45$_{\pm0.06}$**
	6	24.33$_{\pm0.17}$	24.71$_{\pm0.16}$	29.72$_{\pm0.40}$	24.70$_{\pm0.33}$	41.30$_{\pm1.09}$	23.20$_{\pm0.12}$	28.00$_{\pm0.10}$	22.70$_{\pm0.10}$	23.35$_{\pm0.22}$	24.08$_{\pm0.95}$	23.56$_{\pm0.28}$	25.90$_{\pm0.88}$	23.13$_{\pm0.21}$	22.93$_{\pm0.11}$	26.68$_{\pm0.38}$	27.99$_{\pm0.61}$	23.14$_{\pm0.15}$	25.74$_{\pm0.04}$	$\underline{20.52_{\pm0.12}}$	25.87$_{\pm0.00}$	23.16$_{\pm0.04}$	27.14$_{\pm0.00}$	24.74$_{\pm0.04}$	30.33$_{\pm0.00}$	32.52$_{\pm0.89}$	27.41$_{\pm0.00}$	23.96$_{\pm0.03}$	**18.48$_{\pm0.07}$**
	12	27.56$_{\pm0.21}$	29.82$_{\pm0.14}$	32.46$_{\pm0.37}$	28.79$_{\pm0.51}$	44.99$_{\pm0.98}$	26.94$_{\pm0.16}$	32.37$_{\pm0.35}$	26.14$_{\pm0.20}$	27.11$_{\pm0.58}$	27.75$_{\pm0.94}$	27.50$_{\pm0.37}$	31.87$_{\pm1.46}$	26.81$_{\pm0.26}$	26.70$_{\pm0.15}$	29.78$_{\pm0.35}$	33.28$_{\pm0.82}$	26.74$_{\pm0.15}$	31.45$_{\pm0.04}$	$\underline{22.05_{\pm0.13}}$	36.36$_{\pm0.00}$	29.28$_{\pm0.07}$	36.42$_{\pm0.00}$	32.31$_{\pm0.09}$	39.29$_{\pm0.00}$	40.97$_{\pm0.93}$	35.98$_{\pm0.00}$	30.56$_{\pm0.04}$	**20.00$_{\pm0.08}$**
	Avg	24.90$_{\pm0.19}$	25.15$_{\pm0.17}$	30.12$_{\pm0.40}$	25.15$_{\pm0.39}$	41.64$_{\pm0.99}$	23.46$_{\pm0.14}$	28.29$_{\pm0.08}$	22.89$_{\pm0.11}$	23.60$_{\pm0.28}$	24.32$_{\pm0.98}$	23.89$_{\pm0.34}$	26.40$_{\pm0.88}$	23.37$_{\pm0.22}$	23.18$_{\pm0.13}$	27.10$_{\pm0.38}$	28.58$_{\pm0.65}$	23.42$_{\pm0.14}$	26.23$_{\pm0.04}$	$\underline{20.58_{\pm0.12}}$	26.98$_{\pm0.00}$	23.66$_{\pm0.03}$	27.86$_{\pm0.00}$	25.36$_{\pm0.04}$	30.92$_{\pm0.00}$	32.99$_{\pm0.89}$	28.00$_{\pm0.00}$	24.51$_{\pm0.03}$	**18.47$_{\pm0.07}$**
MAPE (%)	3	28.08$_{\pm0.88}$	24.72$_{\pm0.40}$	28.13$_{\pm0.82}$	26.65$_{\pm0.67}$	50.48$_{\pm2.93}$	25.11$_{\pm0.64}$	28.43$_{\pm0.78}$	24.05$_{\pm0.22}$	26.49$_{\pm1.13}$	26.87$_{\pm2.27}$	26.48$_{\pm0.70}$	27.89$_{\pm1.36}$	25.03$_{\pm0.27}$	24.96$_{\pm0.64}$	25.31$_{\pm0.62}$	49.62$_{\pm2.31}$	25.56$_{\pm0.78}$	26.28$_{\pm0.91}$	$\underline{22.52_{\pm0.64}}$	22.59$_{\pm0.00}$	21.07$_{\pm0.03}$	23.53$_{\pm0.00}$	21.74$_{\pm0.02}$	23.11$_{\pm0.00}$	23.82$_{\pm0.36}$	23.23$_{\pm0.00}$	22.00$_{\pm0.01}$	**21.63$_{\pm0.17}$**
	6	28.30$_{\pm0.86}$	26.15$_{\pm0.42}$	28.46$_{\pm0.91}$	27.65$_{\pm0.69}$	49.81$_{\pm2.94}$	26.03$_{\pm0.57}$	29.35$_{\pm0.93}$	24.75$_{\pm0.17}$	27.29$_{\pm1.48}$	27.58$_{\pm2.27}$	27.28$_{\pm0.67}$	29.56$_{\pm0.89}$	25.82$_{\pm0.29}$	25.89$_{\pm0.57}$	26.15$_{\pm0.61}$	52.24$_{\pm2.52}$	26.53$_{\pm0.74}$	27.99$_{\pm0.96}$	$\underline{22.95_{\pm0.64}}$	25.06$_{\pm0.00}$	22.52$_{\pm0.03}$	26.19$_{\pm0.00}$	23.46$_{\pm0.02}$	25.83$_{\pm0.00}$	26.96$_{\pm0.32}$	25.68$_{\pm0.00}$	23.83$_{\pm0.01}$	**22.47$_{\pm0.20}$**
	12	30.21$_{\pm0.81}$	29.78$_{\pm0.50}$	30.89$_{\pm0.96}$	30.44$_{\pm0.82}$	49.61$_{\pm2.26}$	28.34$_{\pm0.49}$	31.64$_{\pm1.66}$	26.62$_{\pm0.25}$	29.70$_{\pm2.58}$	29.44$_{\pm2.39}$	29.57$_{\pm0.77}$	34.17$_{\pm1.00}$	28.11$_{\pm0.57}$	28.20$_{\pm0.48}$	28.48$_{\pm0.63}$	58.49$_{\pm3.04}$	28.91$_{\pm0.60}$	32.53$_{\pm1.08}$	$\underline{24.00_{\pm0.63}}$	31.43$_{\pm0.00}$	25.67$_{\pm0.02}$	31.92$_{\pm0.00}$	27.12$_{\pm0.01}$	31.32$_{\pm0.00}$	32.00$_{\pm0.24}$	30.84$_{\pm0.00}$	27.77$_{\pm0.03}$	**23.93$_{\pm0.29}$**
	Avg	28.69$_{\pm0.86}$	26.56$_{\pm0.43}$	28.95$_{\pm0.89}$	28.01$_{\pm0.71}$	50.04$_{\pm2.80}$	26.29$_{\pm0.58}$	29.62$_{\pm1.04}$	24.96$_{\pm0.21}$	27.63$_{\pm1.67}$	27.78$_{\pm2.30}$	27.58$_{\pm0.64}$	30.12$_{\pm0.81}$	26.12$_{\pm0.31}$	26.15$_{\pm0.57}$	26.44$_{\pm0.62}$	52.94$_{\pm2.57}$	26.79$_{\pm0.71}$	28.55$_{\pm0.98}$	$\underline{23.06_{\pm0.64}}$	25.83$_{\pm0.00}$	22.80$_{\pm0.03}$	26.71$_{\pm0.00}$	23.79$_{\pm0.01}$	26.24$_{\pm0.00}$	27.23$_{\pm0.31}$	26.15$_{\pm0.00}$	24.18$_{\pm0.01}$	**22.52$_{\pm0.21}$**
PEMS04																													
MAE	3	21.15$_{\pm0.32}$	20.23$_{\pm0.12}$	22.27$_{\pm0.38}$	22.00$_{\pm0.32}$	153.31$_{\pm6.85}$	19.92$_{\pm0.23}$	36.81$_{\pm0.33}$	19.34$_{\pm0.22}$	19.74$_{\pm0.48}$	23.62$_{\pm3.33}$	21.20$_{\pm1.34}$	31.75$_{\pm9.85}$	20.31$_{\pm0.70}$	19.59$_{\pm0.23}$	21.19$_{\pm0.21}$	22.37$_{\pm0.62}$	19.59$_{\pm0.15}$	21.27$_{\pm0.23}$	$\underline{18.62_{\pm0.99}}$	17.60$_{\pm0.00}$	16.52$_{\pm0.03}$	19.32$_{\pm0.00}$	17.78$_{\pm0.02}$	21.81$_{\pm0.00}$	26.08$_{\pm0.33}$	20.63$_{\pm0.00}$	17.86$_{\pm0.03}$	**15.68$_{\pm0.13}$**
	6	21.75$_{\pm0.26}$	22.06$_{\pm0.11}$	23.22$_{\pm0.37}$	22.97$_{\pm0.28}$	153.21$_{\pm6.57}$	21.35$_{\pm0.19}$	38.21$_{\pm0.32}$	20.78$_{\pm0.26}$	21.18$_{\pm0.50}$	24.90$_{\pm3.16}$	22.41$_{\pm1.06}$	33.61$_{\pm9.31}$	21.63$_{\pm0.64}$	21.04$_{\pm0.19}$	21.96$_{\pm0.17}$	23.63$_{\pm0.63}$	20.94$_{\pm0.11}$	23.12$_{\pm0.17}$	$\underline{19.33_{\pm0.95}}$	20.75$_{\pm0.00}$	18.59$_{\pm0.04}$	22.92$_{\pm0.00}$	20.46$_{\pm0.02}$	26.14$_{\pm0.00}$	30.01$_{\pm0.17}$	24.04$_{\pm0.00}$	20.27$_{\pm0.03}$	**16.55$_{\pm0.19}$**
	12	24.38$_{\pm0.25}$	26.11$_{\pm0.06}$	25.97$_{\pm0.36}$	26.06$_{\pm0.26}$	153.51$_{\pm6.60}$	24.45$_{\pm0.19}$	41.12$_{\pm0.17}$	23.68$_{\pm0.33}$	24.16$_{\pm0.63}$	27.74$_{\pm2.99}$	25.55$_{\pm1.01}$	37.67$_{\pm7.70}$	24.64$_{\pm0.65}$	24.17$_{\pm0.18}$	24.86$_{\pm0.16}$	26.74$_{\pm0.68}$	23.91$_{\pm0.11}$	27.45$_{\pm0.13}$	$\underline{20.61_{\pm0.94}}$	28.27$_{\pm0.00}$	22.84$_{\pm0.07}$	30.08$_{\pm0.00}$	25.86$_{\pm0.01}$	33.86$_{\pm0.00}$	39.47$_{\pm0.65}$	30.91$_{\pm0.00}$	25.23$_{\pm0.02}$	**17.95$_{\pm0.32}$**
	Avg	22.22$_{\pm0.27}$	22.45$_{\pm0.09}$	23.60$_{\pm0.37}$	23.41$_{\pm0.27}$	153.33$_{\pm6.62}$	21.64$_{\pm0.20}$	38.46$_{\pm0.27}$	21.01$_{\pm0.26}$	21.43$_{\pm0.53}$	25.16$_{\pm3.18}$	22.78$_{\pm1.14}$	34.02$_{\pm9.03}$	21.93$_{\pm0.67}$	21.33$_{\pm0.20}$	22.43$_{\pm0.18}$	23.99$_{\pm0.62}$	21.22$_{\pm0.11}$	23.58$_{\pm0.18}$	$\underline{19.38_{\pm0.94}}$	21.58$_{\pm0.00}$	18.94$_{\pm0.04}$	23.49$_{\pm0.00}$	20.92$_{\pm0.02}$	26.72$_{\pm0.00}$	31.16$_{\pm0.33}$	24.52$_{\pm0.00}$	20.68$_{\pm0.03}$	**16.58$_{\pm0.21}$**
RMSE	3	32.49$_{\pm0.37}$	31.23$_{\pm0.11}$	36.57$_{\pm0.57}$	33.27$_{\pm0.26}$	185.75$_{\pm7.85}$	30.57$_{\pm0.22}$	48.99$_{\pm0.57}$	29.87$_{\pm0.23}$	30.41$_{\pm0.49}$	36.55$_{\pm4.40}$	31.87$_{\pm1.33}$	44.57$_{\pm11.20}$	30.86$_{\pm0.67}$	30.17$_{\pm0.22}$	33.49$_{\pm0.37}$	34.23$_{\pm0.76}$	30.24$_{\pm0.20}$	32.51$_{\pm0.25}$	$\underline{29.26_{\pm1.26}}$	28.79$_{\pm0.00}$	27.45$_{\pm0.07}$	30.99$_{\pm0.00}$	29.58$_{\pm0.02}$	34.99$_{\pm0.00}$	41.11$_{\pm0.57}$	32.99$_{\pm0.00}$	29.40$_{\pm0.11}$	**25.64$_{\pm0.15}$**
	6	33.93$_{\pm0.28}$	34.42$_{\pm0.10}$	38.28$_{\pm0.55}$	35.06$_{\pm0.24}$	186.05$_{\pm7.49}$	33.18$_{\pm0.20}$	51.49$_{\pm0.51}$	32.43$_{\pm0.30}$	33.00$_{\pm0.54}$	38.80$_{\pm4.09}$	34.23$_{\pm1.05}$	47.58$_{\pm10.31}$	33.36$_{\pm0.61}$	32.80$_{\pm0.20}$	34.98$_{\pm0.29}$	36.16$_{\pm0.79}$	32.65$_{\pm0.16}$	35.90$_{\pm0.21}$	$\underline{30.61_{\pm1.20}}$	33.78$_{\pm0.00}$	31.20$_{\pm0.10}$	36.72$_{\pm0.00}$	34.39$_{\pm0.01}$	41.80$_{\pm0.00}$	47.11$_{\pm0.31}$	38.04$_{\pm0.00}$	33.63$_{\pm0.09}$	**27.13$_{\pm0.24}$**
	12	38.46$_{\pm0.30}$	40.98$_{\pm0.17}$	42.75$_{\pm0.53}$	40.07$_{\pm0.30}$	187.60$_{\pm7.68}$	38.43$_{\pm0.24}$	56.38$_{\pm0.30}$	37.23$_{\pm0.45}$	37.99$_{\pm0.82}$	43.47$_{\pm3.76}$	39.70$_{\pm1.19}$	53.93$_{\pm7.70}$	38.50$_{\pm0.67}$	38.08$_{\pm0.25}$	39.67$_{\pm0.26}$	40.75$_{\pm0.93}$	37.57$_{\pm0.19}$	43.10$_{\pm0.21}$	$\underline{32.76_{\pm1.20}}$	46.16$_{\pm0.00}$	39.07$_{\pm0.21}$	48.32$_{\pm0.00}$	43.96$_{\pm0.03}$	53.81$_{\pm0.00}$	64.62$_{\pm2.50}$	48.84$_{\pm0.00}$	42.50$_{\pm0.06}$	**29.37$_{\pm0.42}$**
	Avg	34.58$_{\pm0.32}$	34.96$_{\pm0.12}$	38.82$_{\pm0.55}$	35.69$_{\pm0.22}$	186.37$_{\pm7.58}$	33.58$_{\pm0.22}$	51.85$_{\pm0.46}$	32.73$_{\pm0.31}$	33.35$_{\pm0.60}$	39.17$_{\pm4.12}$	34.78$_{\pm1.18}$	48.20$_{\pm9.87}$	33.78$_{\pm0.64}$	33.21$_{\pm0.22}$	35.63$_{\pm0.31}$	36.65$_{\pm0.80}$	33.04$_{\pm0.17}$	36.54$_{\pm0.22}$	$\underline{30.64_{\pm1.20}}$	35.22$_{\pm0.00}$	31.89$_{\pm0.12}$	37.69$_{\pm0.00}$	35.20$_{\pm0.02}$	42.74$_{\pm0.00}$	49.27$_{\pm0.58}$	38.87$_{\pm0.00}$	34.42$_{\pm0.09}$	**27.14$_{\pm0.26}$**
MAPE (%)	3	19.95$_{\pm0.41}$	17.57$_{\pm0.43}$	19.35$_{\pm0.26}$	18.96$_{\pm0.52}$	82.38$_{\pm3.00}$	17.87$_{\pm0.35}$	25.59$_{\pm0.52}$	16.67$_{\pm0.26}$	17.63$_{\pm0.50}$	21.16$_{\pm3.83}$	20.61$_{\pm1.82}$	37.87$_{\pm18.81}$	18.37$_{\pm0.71}$	17.74$_{\pm0.35}$	17.69$_{\pm0.30}$	26.18$_{\pm1.42}$	18.27$_{\pm0.21}$	17.68$_{\pm0.31}$	$\underline{16.44_{\pm0.94}}$	14.71$_{\pm0.00}$	13.71$_{\pm0.01}$	16.27$_{\pm0.00}$	14.69$_{\pm0.02}$	16.45$_{\pm0.00}$	17.91$_{\pm0.07}$	16.23$_{\pm0.00}$	14.90$_{\pm0.01}$	**14.71$_{\pm0.17}$**
	6	20.17$_{\pm0.42}$	18.89$_{\pm0.42}$	19.76$_{\pm0.28}$	19.60$_{\pm0.54}$	82.26$_{\pm3.00}$	18.86$_{\pm0.28}$	26.67$_{\pm0.66}$	17.58$_{\pm0.32}$	18.53$_{\pm0.54}$	22.08$_{\pm3.77}$	21.21$_{\pm1.50}$	38.74$_{\pm17.46}$	19.14$_{\pm0.60}$	18.75$_{\pm0.28}$	18.18$_{\pm0.26}$	27.56$_{\pm1.51}$	19.11$_{\pm0.24}$	19.09$_{\pm0.32}$	$\underline{16.86_{\pm0.94}}$	16.93$_{\pm0.00}$	15.00$_{\pm0.01}$	18.84$_{\pm0.00}$	16.33$_{\pm0.02}$	19.03$_{\pm0.00}$	20.80$_{\pm0.03}$	18.75$_{\pm0.00}$	16.48$_{\pm0.02}$	**15.47$_{\pm0.22}$**
	12	21.91$_{\pm0.40}$	22.29$_{\pm0.42}$	21.86$_{\pm0.33}$	21.79$_{\pm0.59}$	82.01$_{\pm3.01}$	21.34$_{\pm0.25}$	29.13$_{\pm1.08}$	19.79$_{\pm0.43}$	20.92$_{\pm0.83}$	24.23$_{\pm3.51}$	23.44$_{\pm1.32}$	42.12$_{\pm15.90}$	21.54$_{\pm0.39}$	21.24$_{\pm0.24}$	20.14$_{\pm0.21}$	31.05$_{\pm1.68}$	21.23$_{\pm0.34}$	22.99$_{\pm0.46}$	$\underline{17.90_{\pm1.02}}$	22.62$_{\pm0.00}$	17.78$_{\pm0.01}$	24.24$_{\pm0.00}$	19.78$_{\pm0.02}$	23.86$_{\pm0.00}$	26.17$_{\pm0.16}$	23.60$_{\pm0.00}$	19.89$_{\pm0.03}$	**16.76$_{\pm0.36}$**
	Avg	20.53$_{\pm0.40}$	19.29$_{\pm0.42}$	20.15$_{\pm0.28}$	19.94$_{\pm0.55}$	82.25$_{\pm2.99}$	19.15$_{\pm0.30}$	26.93$_{\pm0.71}$	17.82$_{\pm0.31}$	18.83$_{\pm0.59}$	22.29$_{\pm3.69}$	21.54$_{\pm1.53}$	39.19$_{\pm17.54}$	19.49$_{\pm0.57}$	19.04$_{\pm0.29}$	18.51$_{\pm0.26}$	27.98$_{\pm1.51}$	19.36$_{\pm0.25}$	19.60$_{\pm0.33}$	$\underline{16.97_{\pm0.95}}$	17.62$_{\pm0.00}$	15.25$_{\pm0.01}$	19.32$_{\pm0.00}$	16.65$_{\pm0.02}$	19.38$_{\pm0.00}$	21.25$_{\pm0.07}$	19.11$_{\pm0.00}$	16.78$_{\pm0.02}$	**15.52$_{\pm0.23}$**
PEMS05																													
MAE	3	11.22$_{\pm0.09}$	10.77$_{\pm0.03}$	11.31$_{\pm0.11}$	11.71$_{\pm0.12}$	24.31$_{\pm1.08}$	10.54$_{\pm0.04}$	14.56$_{\pm0.51}$	10.39$_{\pm0.07}$	10.54$_{\pm0.12}$	10.93$_{\pm0.34}$	10.73$_{\pm0.17}$	15.58$_{\pm3.35}$	10.58$_{\pm0.05}$	10.45$_{\pm0.04}$	11.03$_{\pm0.07}$	11.98$_{\pm0.14}$	10.69$_{\pm0.07}$	11.00$_{\pm0.06}$	$\underline{9.88_{\pm0.06}}$	10.72$_{\pm0.00}$	10.01$_{\pm0.01}$	11.20$_{\pm0.00}$	10.44$_{\pm0.01}$	12.13$_{\pm0.00}$	13.76$_{\pm0.38}$	11.56$_{\pm0.00}$	10.39$_{\pm0.00}$	**9.17$_{\pm0.02}$**
	6	11.49$_{\pm0.06}$	11.55$_{\pm0.04}$	11.79$_{\pm0.09}$	12.34$_{\pm0.13}$	24.77$_{\pm0.99}$	11.20$_{\pm0.04}$	15.76$_{\pm0.54}$	11.00$_{\pm0.07}$	11.21$_{\pm0.12}$	11.55$_{\pm0.31}$	11.36$_{\pm0.14}$	16.21$_{\pm3.54}$	11.21$_{\pm0.03}$	11.11$_{\pm0.04}$	11.48$_{\pm0.05}$	12.44$_{\pm0.11}$	11.30$_{\pm0.06}$	11.96$_{\pm0.05}$	$\underline{10.12_{\pm0.07}}$	12.23$_{\pm0.00}$	10.94$_{\pm0.01}$	12.84$_{\pm0.00}$	11.62$_{\pm0.01}$	14.04$_{\pm0.00}$	15.48$_{\pm0.25}$	12.97$_{\pm0.00}$	11.46$_{\pm0.00}$	**9.47$_{\pm0.03}$**
	12	12.63$_{\pm0.05}$	13.21$_{\pm0.05}$	13.04$_{\pm0.08}$	14.10$_{\pm0.18}$	26.01$_{\pm1.05}$	12.57$_{\pm0.06}$	18.26$_{\pm0.54}$	12.24$_{\pm0.11}$	12.57$_{\pm0.16}$	12.87$_{\pm0.27}$	12.72$_{\pm0.17}$	17.64$_{\pm3.79}$	12.55$_{\pm0.05}$	12.48$_{\pm0.06}$	12.72$_{\pm0.05}$	13.56$_{\pm0.08}$	12.62$_{\pm0.05}$	14.06$_{\pm0.06}$	$\underline{10.60_{\pm0.08}}$	15.95$_{\pm0.00}$	12.93$_{\pm0.02}$	16.14$_{\pm0.00}$	14.09$_{\pm0.01}$	17.55$_{\pm0.00}$	18.96$_{\pm0.26}$	16.18$_{\pm0.00}$	13.73$_{\pm0.00}$	**9.96$_{\pm0.05}$**
	Avg	11.69$_{\pm0.07}$	11.69$_{\pm0.04}$	11.94$_{\pm0.10}$	12.57$_{\pm0.13}$	24.94$_{\pm0.99}$	11.31$_{\pm0.04}$	15.98$_{\pm0.52}$	11.10$_{\pm0.07}$	11.32$_{\pm0.13}$	11.67$_{\pm0.31}$	11.48$_{\pm0.16}$	16.35$_{\pm3.53}$	11.33$_{\pm0.04}$	11.22$_{\pm0.04}$	11.64$_{\pm0.06}$	12.57$_{\pm0.10}$	11.42$_{\pm0.06}$	12.16$_{\pm0.05}$	$\underline{10.16_{\pm0.07}}$	12.67$_{\pm0.00}$	11.12$_{\pm0.01}$	13.11$_{\pm0.00}$	11.85$_{\pm0.01}$	14.34$_{\pm0.00}$	15.75$_{\pm0.28}$	13.26$_{\pm0.00}$	11.66$_{\pm0.00}$	**9.48$_{\pm0.03}$**
RMSE	3	17.36$_{\pm0.11}$	16.92$_{\pm0.07}$	18.29$_{\pm0.18}$	17.96$_{\pm0.15}$	33.60$_{\pm1.18}$	16.59$_{\pm0.07}$	23.55$_{\pm0.72}$	16.33$_{\pm0.09}$	16.61$_{\pm0.11}$	17.14$_{\pm0.48}$	16.84$_{\pm0.20}$	22.45$_{\pm3.39}$	16.58$_{\pm0.06}$	16.46$_{\pm0.06}$	17.64$_{\pm0.10}$	18.03$_{\pm0.22}$	16.69$_{\pm0.08}$	17.33$_{\pm0.06}$	$\underline{15.54_{\pm0.09}}$	17.14$_{\pm0.00}$	16.31$_{\pm0.01}$	17.78$_{\pm0.00}$	16.98$_{\pm0.00}$	19.54$_{\pm0.00}$	21.76$_{\pm0.54}$	18.39$_{\pm0.00}$	16.77$_{\pm0.01}$	**14.55$_{\pm0.03}$**
	6	18.03$_{\pm0.07}$	18.31$_{\pm0.07}$	19.20$_{\pm0.16}$	19.12$_{\pm0.18}$	34.52$_{\pm1.07}$	17.84$_{\pm0.06}$	25.94$_{\pm0.75}$	17.49$_{\pm0.13}$	17.87$_{\pm0.12}$	18.33$_{\pm0.43}$	18.07$_{\pm0.18}$	23.58$_{\pm3.68}$	17.79$_{\pm0.05}$	17.71$_{\pm0.06}$	18.43$_{\pm0.07}$	18.81$_{\pm0.16}$	17.86$_{\pm0.07}$	19.09$_{\pm0.05}$	$\underline{16.02_{\pm0.10}}$	19.78$_{\pm0.00}$	18.19$_{\pm0.01}$	20.61$_{\pm0.00}$	19.44$_{\pm0.01}$	22.68$_{\pm0.00}$	24.70$_{\pm0.40}$	20.68$_{\pm0.00}$	18.85$_{\pm0.01}$	**15.11$_{\pm0.04}$**
	12	20.16$_{\pm0.08}$	21.07$_{\pm0.10}$	21.40$_{\pm0.15}$	22.14$_{\pm0.27}$	36.72$_{\pm1.19}$	20.30$_{\pm0.12}$	30.76$_{\pm0.61}$	19.68$_{\pm0.22}$	20.32$_{\pm0.22}$	20.68$_{\pm0.38}$	20.57$_{\pm0.27}$	25.93$_{\pm3.97}$	20.19$_{\pm0.09}$	20.17$_{\pm0.12}$	20.58$_{\pm0.06}$	20.61$_{\pm0.16}$	20.20$_{\pm0.08}$	22.73$_{\pm0.08}$	$\underline{16.89_{\pm0.13}}$	26.27$_{\pm0.00}$	22.43$_{\pm0.04}$	26.26$_{\pm0.00}$	24.56$_{\pm0.02}$	28.35$_{\pm0.00}$	30.31$_{\pm0.41}$	25.89$_{\pm0.00}$	23.50$_{\pm0.02}$	**15.97$_{\pm0.06}$**
	Avg	18.34$_{\pm0.09}$	18.52$_{\pm0.08}$	19.43$_{\pm0.16}$	19.49$_{\pm0.18}$	34.77$_{\pm1.06}$	18.03$_{\pm0.07}$	26.35$_{\pm0.67}$	17.63$_{\pm0.13}$	18.04$_{\pm0.14}$	18.50$_{\pm0.44}$	18.26$_{\pm0.21}$	23.77$_{\pm3.62}$	17.97$_{\pm0.04}$	17.89$_{\pm0.07}$	18.69$_{\pm0.08}$	19.00$_{\pm0.17}$	18.04$_{\pm0.07}$	19.40$_{\pm0.06}$	$\underline{16.07_{\pm0.10}}$	20.54$_{\pm0.00}$	18.61$_{\pm0.02}$	21.07$_{\pm0.00}$	19.92$_{\pm0.01}$	23.12$_{\pm0.00}$	25.07$_{\pm0.41}$	21.14$_{\pm0.00}$	19.31$_{\pm0.00}$	**15.12$_{\pm0.04}$**
MAPE (%)	3	26.28$_{\pm0.86}$	23.31$_{\pm0.62}$	25.81$_{\pm0.68}$	25.60$_{\pm0.88}$	114.78$_{\pm6.27}$	23.06$_{\pm0.42}$	27.15$_{\pm0.88}$	22.28$_{\pm0.43}$	23.18$_{\pm0.88}$	23.88$_{\pm0.92}$	23.69$_{\pm0.81}$	59.98$_{\pm37.51}$	23.90$_{\pm0.94}$	22.97$_{\pm0.42}$	23.40$_{\pm0.55}$	33.91$_{\pm0.52}$	23.53$_{\pm0.44}$	23.50$_{\pm0.68}$	$\underline{22.26_{\pm0.46}}$	21.96$_{\pm0.00}$	20.12$_{\pm0.01}$	23.23$_{\pm0.00}$	20.62$_{\pm0.02}$	22.62$_{\pm0.00}$	23.96$_{\pm0.28}$	22.69$_{\pm0.00}$	21.26$_{\pm0.01}$	**20.93$_{\pm0.13}$**
	6	26.59$_{\pm0.75}$	24.92$_{\pm0.62}$	26.46$_{\pm0.61}$	26.73$_{\pm0.88}$	115.48$_{\pm6.90}$	24.15$_{\pm0.42}$	28.67$_{\pm0.81}$	23.24$_{\pm0.42}$	24.29$_{\pm0.95}$	24.84$_{\pm0.87}$	24.82$_{\pm0.74}$	60.44$_{\pm37.27}$	24.93$_{\pm0.92}$	24.05$_{\pm0.42}$	24.20$_{\pm0.51}$	35.01$_{\pm0.59}$	24.65$_{\pm0.36}$	25.54$_{\pm0.69}$	$\underline{22.67_{\pm0.37}}$	24.72$_{\pm0.00}$	21.39$_{\pm0.01}$	26.15$_{\pm0.00}$	22.15$_{\pm0.01}$	25.53$_{\pm0.00}$	27.63$_{\pm0.14}$	25.33$_{\pm0.00}$	22.97$_{\pm0.01}$	**21.71$_{\pm0.15}$**
	12	28.62$_{\pm0.63}$	28.72$_{\pm0.60}$	28.84$_{\pm0.49}$	30.08$_{\pm0.80}$	118.51$_{\pm8.78}$	26.77$_{\pm0.42}$	32.35$_{\pm0.78}$	25.61$_{\pm0.55}$	26.89$_{\pm1.00}$	27.37$_{\pm0.73}$	27.67$_{\pm0.80}$	63.79$_{\pm37.30}$	27.61$_{\pm0.75}$	26.67$_{\pm0.42}$	26.50$_{\pm0.51}$	38.11$_{\pm0.99}$	27.28$_{\pm0.38}$	30.73$_{\pm0.73}$	$\underline{23.86_{\pm0.49}}$	32.01$_{\pm0.00}$	24.30$_{\pm0.03}$	32.54$_{\pm0.00}$	25.58$_{\pm0.01}$	31.30$_{\pm0.00}$	33.24$_{\pm0.06}$	31.29$_{\pm0.00}$	26.70$_{\pm0.01}$	**23.13$_{\pm0.25}$**
	Avg	26.98$_{\pm0.74}$	25.31$_{\pm0.60}$	26.83$_{\pm0.59}$	27.19$_{\pm0.86}$	116.10$_{\pm7.29}$	24.42$_{\pm0.41}$	29.06$_{\pm0.76}$	23.49$_{\pm0.44}$	24.56$_{\pm0.93}$	25.13$_{\pm0.85}$	25.11$_{\pm0.78}$	61.09$_{\pm37.33}$	25.25$_{\pm0.88}$	24.33$_{\pm0.41}$	24.51$_{\pm0.52}$	35.42$_{\pm0.64}$	24.92$_{\pm0.37}$	26.15$_{\pm0.70}$	$\underline{22.83_{\pm0.44}}$	25.64$_{\pm0.00}$	21.67$_{\pm0.02}$	26.75$_{\pm0.00}$	22.50$_{\pm0.02}$	25.96$_{\pm0.00}$	27.92$_{\pm0.16}$	25.93$_{\pm0.00}$	23.30$_{\pm0.01}$	**21.77$_{\pm0.17}$**
PEMS06																													
MAE	3	14.29$_{\pm0.34}$	14.25$_{\pm0.15}$	14.75$_{\pm0.46}$	14.91$_{\pm0.58}$	78.18$_{\pm1.49}$	13.39$_{\pm0.36}$	26.65$_{\pm0.40}$	13.10$_{\pm0.23}$	13.47$_{\pm0.33}$	17.23$_{\pm3.32}$	13.86$_{\pm0.26}$	19.38$_{\pm4.55}$	13.53$_{\pm0.45}$	13.28$_{\pm0.36}$	14.65$_{\pm0.27}$	15.70$_{\pm0.34}$	14.19$_{\pm0.45}$	14.86$_{\pm0.15}$	$\underline{12.73_{\pm0.61}}$	13.03$_{\pm0.00}$	12.20$_{\pm0.01}$	13.60$_{\pm0.00}$	12.46$_{\pm0.01}$	14.70$_{\pm0.00}$	17.57$_{\pm0.43}$	14.03$_{\pm0.00}$	12.68$_{\pm0.01}$	**11.06$_{\pm0.20}$**
	6	14.60$_{\pm0.30}$	15.36$_{\pm0.17}$	15.33$_{\pm0.46}$	15.92$_{\pm0.53}$	78.77$_{\pm1.35}$	14.24$_{\pm0.34}$	27.53$_{\pm0.65}$	13.91$_{\pm0.21}$	14.33$_{\pm0.33}$	17.90$_{\pm3.21}$	14.75$_{\pm0.23}$	20.83$_{\pm4.24}$	14.30$_{\pm0.46}$	14.13$_{\pm0.34}$	15.20$_{\pm0.21}$	16.52$_{\pm0.33}$	15.02$_{\pm0.42}$	16.16$_{\pm0.13}$	$\underline{13.09_{\pm0.61}}$	15.11$_{\pm0.00}$	13.47$_{\pm0.01}$	15.79$_{\pm0.00}$	14.02$_{\pm0.01}$	17.33$_{\pm0.00}$	19.73$_{\pm0.46}$	16.01$_{\pm0.00}$	14.08$_{\pm0.01}$	**11.49$_{\pm0.20}$**
	12	16.11$_{\pm0.30}$	17.62$_{\pm0.22}$	16.90$_{\pm0.44}$	18.24$_{\pm0.51}$	79.36$_{\pm1.36}$	15.99$_{\pm0.31}$	29.23$_{\pm1.05}$	15.54$_{\pm0.26}$	16.09$_{\pm0.35}$	19.37$_{\pm2.99}$	16.67$_{\pm0.21}$	23.88$_{\pm3.70}$	15.99$_{\pm0.52}$	15.88$_{\pm0.32}$	16.77$_{\pm0.20}$	18.34$_{\pm0.40}$	16.83$_{\pm0.37}$	18.94$_{\pm0.11}$	$\underline{13.78_{\pm0.56}}$	19.88$_{\pm0.00}$	16.06$_{\pm0.01}$	20.01$_{\pm0.00}$	17.21$_{\pm0.01}$	22.01$_{\pm0.00}$	24.25$_{\pm0.31}$	20.28$_{\pm0.00}$	16.91$_{\pm0.01}$	**12.21$_{\pm0.21}$**
	Avg	14.87$_{\pm0.31}$	15.52$_{\pm0.16}$	15.52$_{\pm0.46}$	16.16$_{\pm0.54}$	78.66$_{\pm1.39}$	14.37$_{\pm0.34}$	27.64$_{\pm0.65}$	14.02$_{\pm0.21}$	14.46$_{\pm0.34}$	18.02$_{\pm3.19}$	14.90$_{\pm0.24}$	21.10$_{\pm4.19}$	14.44$_{\pm0.46}$	14.25$_{\pm0.34}$	15.39$_{\pm0.23}$	16.70$_{\pm0.35}$	15.18$_{\pm0.41}$	16.40$_{\pm0.13}$	$\underline{13.13_{\pm0.60}}$	15.59$_{\pm0.00}$	13.66$_{\pm0.01}$	16.08$_{\pm0.00}$	14.28$_{\pm0.01}$	17.67$_{\pm0.00}$	20.04$_{\pm0.40}$	16.33$_{\pm0.00}$	14.28$_{\pm0.01}$	**11.51$_{\pm0.20}$**
RMSE	3	22.13$_{\pm0.48}$	21.74$_{\pm0.27}$	23.99$_{\pm0.57}$	22.66$_{\pm0.70}$	96.67$_{\pm2.18}$	20.57$_{\pm0.42}$	36.24$_{\pm0.70}$	20.00$_{\pm0.18}$	20.57$_{\pm0.42}$	27.60$_{\pm6.52}$	21.19$_{\pm0.35}$	28.19$_{\pm5.72}$	20.82$_{\pm0.45}$	20.39$_{\pm0.41}$	23.12$_{\pm0.45}$	23.42$_{\pm0.45}$	21.56$_{\pm0.48}$	22.68$_{\pm0.15}$	$\underline{19.15_{\pm0.71}}$	20.74$_{\pm0.00}$	19.77$_{\pm0.05}$	21.65$_{\pm0.00}$	20.03$_{\pm0.02}$	23.69$_{\pm0.00}$	27.96$_{\pm0.62}$	22.24$_{\pm0.00}$	20.34$_{\pm0.02}$	**17.11$_{\pm0.30}$**
	6	22.84$_{\pm0.42}$	23.84$_{\pm0.29}$	25.08$_{\pm0.56}$	24.49$_{\pm0.62}$	98.00$_{\pm2.04}$	22.23$_{\pm0.39}$	38.11$_{\pm1.19}$	21.54$_{\pm0.29}$	22.27$_{\pm0.46}$	28.90$_{\pm6.09}$	22.98$_{\pm0.33}$	30.83$_{\pm5.09}$	22.35$_{\pm0.47}$	22.06$_{\pm0.39}$	24.03$_{\pm0.42}$	24.72$_{\pm0.43}$	23.16$_{\pm0.48}$	25.24$_{\pm0.14}$	$\underline{19.84_{\pm0.70}}$	24.87$_{\pm0.00}$	22.65$_{\pm0.03}$	25.77$_{\pm0.00}$	23.64$_{\pm0.01}$	28.35$_{\pm0.00}$	32.03$_{\pm0.80}$	25.81$_{\pm0.00}$	23.30$_{\pm0.04}$	**17.87$_{\pm0.29}$**
	12	25.62$_{\pm0.41}$	27.71$_{\pm0.34}$	27.79$_{\pm0.53}$	28.42$_{\pm0.62}$	100.02$_{\pm1.93}$	25.34$_{\pm0.36}$	41.47$_{\pm2.03}$	24.41$_{\pm0.45}$	25.39$_{\pm0.54}$	31.36$_{\pm5.41}$	26.49$_{\pm0.34}$	36.07$_{\pm4.02}$	25.35$_{\pm0.62}$	25.18$_{\pm0.37}$	26.62$_{\pm0.35}$	27.54$_{\pm0.47}$	26.36$_{\pm0.43}$	30.13$_{\pm0.17}$	$\underline{21.03_{\pm0.61}}$	33.64$_{\pm0.00}$	28.54$_{\pm0.08}$	33.36$_{\pm0.00}$	30.70$_{\pm0.03}$	36.75$_{\pm0.00}$	44.91$_{\pm8.74}$	33.00$_{\pm0.00}$	29.32$_{\pm0.08}$	**19.08$_{\pm0.27}$**
	Avg	23.29$_{\pm0.43}$	24.04$_{\pm0.28}$	25.38$_{\pm0.56}$	24.84$_{\pm0.65}$	97.96$_{\pm2.03}$	22.39$_{\pm0.39}$	38.29$_{\pm1.20}$	21.68$_{\pm0.28}$	22.41$_{\pm0.47}$	29.03$_{\pm6.07}$	23.19$_{\pm0.34}$	31.23$_{\pm4.99}$	22.53$_{\pm0.49}$	22.21$_{\pm0.38}$	24.34$_{\pm0.40}$	24.97$_{\pm0.45}$	23.39$_{\pm0.46}$	25.55$_{\pm0.15}$	$\underline{19.88_{\pm0.68}}$	25.65$_{\pm0.00}$	23.12$_{\pm0.04}$	26.22$_{\pm0.00}$	24.21$_{\pm0.01}$	28.85$_{\pm0.00}$	32.90$_{\pm1.12}$	26.23$_{\pm0.00}$	23.75$_{\pm0.04}$	**17.87$_{\pm0.29}$**
MAPE (%)	3	24.10$_{\pm0.81}$	21.76$_{\pm0.38}$	23.21$_{\pm0.43}$	23.31$_{\pm0.78}$	79.39$_{\pm1.35}$	21.38$_{\pm0.30}$	31.34$_{\pm0.34}$	20.89$_{\pm0.38}$	21.45$_{\pm0.51}$	25.89$_{\pm5.52}$	23.18$_{\pm0.54}$	34.95$_{\pm16.21}$	21.52$_{\pm0.65}$	21.25$_{\pm0.30}$	21.75$_{\pm0.39}$	30.95$_{\pm1.10}$	22.76$_{\pm0.99}$	22.34$_{\pm0.37}$	$\underline{19.81_{\pm0.84}}$	19.36$_{\pm0.00}$	17.99$_{\pm0.04}$	20.16$_{\pm0.00}$	18.36$_{\pm0.01}$	19.80$_{\pm0.00}$	21.95$_{\pm0.35}$	19.90$_{\pm0.00}$	18.89$_{\pm0.01}$	**18.56$_{\pm0.32}$**
	6	24.33$_{\pm0.75}$	23.07$_{\pm0.47}$	23.76$_{\pm0.39}$	24.42$_{\pm0.81}$	79.52$_{\pm1.43}$	22.31$_{\pm0.32}$	32.02$_{\pm0.39}$	21.73$_{\pm0.47}$	22.24$_{\pm0.49}$	26.69$_{\pm5.49}$	23.91$_{\pm0.56}$	36.41$_{\pm15.97}$	22.36$_{\pm0.64}$	22.19$_{\pm0.32}$	22.48$_{\pm0.39}$	32.23$_{\pm1.07}$	23.77$_{\pm0.94}$	24.02$_{\pm0.37}$	$\underline{20.23_{\pm0.87}}$	21.61$_{\pm0.00}$	19.19$_{\pm0.04}$	22.53$_{\pm0.00}$	19.79$_{\pm0.01}$	22.19$_{\pm0.00}$	24.49$_{\pm0.32}$	22.05$_{\pm0.00}$	20.42$_{\pm0.01}$	**19.15$_{\pm0.35}$**
	12	25.99$_{\pm0.73}$	26.28$_{\pm0.63}$	25.75$_{\pm0.39}$	27.32$_{\pm0.86}$	79.07$_{\pm1.37}$	24.56$_{\pm0.38}$	33.58$_{\pm0.89}$	23.69$_{\pm0.83}$	24.14$_{\pm0.51}$	28.62$_{\pm5.53}$	25.98$_{\pm0.56}$	39.24$_{\pm15.64}$	24.53$_{\pm0.73}$	24.43$_{\pm0.38}$	24.45$_{\pm0.40}$	35.38$_{\pm1.06}$	26.08$_{\pm0.96}$	28.21$_{\pm0.42}$	$\underline{21.03_{\pm0.84}}$	27.27$_{\pm0.00}$	21.83$_{\pm0.05}$	27.51$_{\pm0.00}$	22.86$_{\pm0.01}$	26.87$_{\pm0.00}$	28.81$_{\pm0.24}$	26.63$_{\pm0.00}$	23.73$_{\pm0.02}$	**20.20$_{\pm0.39}$**
	Avg	24.66$_{\pm0.77}$	23.41$_{\pm0.46}$	24.06$_{\pm0.40}$	24.76$_{\pm0.81}$	79.40$_{\pm1.33}$	22.54$_{\pm0.33}$	32.16$_{\pm0.38}$	21.93$_{\pm0.52}$	22.43$_{\pm0.49}$	26.88$_{\pm5.50}$	24.14$_{\pm0.58}$	36.63$_{\pm16.01}$	22.60$_{\pm0.66}$	22.42$_{\pm0.33}$	22.72$_{\pm0.38}$	32.58$_{\pm1.07}$	24.00$_{\pm0.96}$	24.49$_{\pm0.37}$	$\underline{20.28_{\pm0.85}}$	22.27$_{\pm0.00}$	19.42$_{\pm0.04}$	22.96$_{\pm0.00}$	20.06$_{\pm0.01}$	22.54$_{\pm0.00}$	24.70$_{\pm0.29}$	22.46$_{\pm0.00}$	20.71$_{\pm0.01}$	**19.18$_{\pm0.35}$**
PEMS07																													
MAE	3	18.59$_{\pm0.29}$	17.59$_{\pm0.06}$	19.66$_{\pm0.44}$	18.38$_{\pm1.08}$	40.22$_{\pm2.16}$	16.53$_{\pm0.25}$	21.47$_{\pm0.50}$	16.22$_{\pm0.28}$	16.12$_{\pm0.30}$	17.90$_{\pm1.53}$	15.62$_{\pm1.26}$	17.21$_{\pm0.25}$	16.37$_{\pm0.15}$	16.24$_{\pm0.26}$	18.02$_{\pm0.25}$	23.09$_{\pm0.91}$	17.17$_{\pm0.11}$	17.63$_{\pm0.04}$	$\underline{14.67_{\pm0.34}}$	15.19$_{\pm0.00}$	14.38$_{\pm0.02}$	15.97$_{\pm0.00}$	14.96$_{\pm0.03}$	18.20$_{\pm0.00}$	20.28$_{\pm0.24}$	17.22$_{\pm0.00}$	14.98$_{\pm0.02}$	**12.73$_{\pm0.04}$**
	6	18.93$_{\pm0.27}$	19.22$_{\pm0.05}$	20.25$_{\pm0.40}$	19.21$_{\pm0.96}$	40.32$_{\pm1.94}$	17.83$_{\pm0.27}$	22.58$_{\pm0.58}$	17.53$_{\pm0.32}$	17.40$_{\pm0.31}$	19.09$_{\pm1.46}$	16.93$_{\pm1.25}$	18.14$_{\pm0.23}$	17.64$_{\pm0.15}$	17.55$_{\pm0.27}$	18.94$_{\pm0.17}$	24.77$_{\pm1.12}$	18.44$_{\pm0.08}$	19.49$_{\pm0.01}$	$\underline{15.29_{\pm0.29}}$	18.03$_{\pm0.00}$	16.36$_{\pm0.03}$	19.10$_{\pm0.00}$	17.51$_{\pm0.02}$	21.91$_{\pm0.00}$	23.40$_{\pm0.16}$	20.19$_{\pm0.00}$	17.24$_{\pm0.01}$	**13.47$_{\pm0.06}$**
	12	21.19$_{\pm0.27}$	22.81$_{\pm0.08}$	22.49$_{\pm0.38}$	21.54$_{\pm0.96}$	40.46$_{\pm2.10}$	20.40$_{\pm0.30}$	24.70$_{\pm0.74}$	19.94$_{\pm0.36}$	19.79$_{\pm0.36}$	21.32$_{\pm1.38}$	19.34$_{\pm1.26}$	19.99$_{\pm0.38}$	20.09$_{\pm0.22}$	20.14$_{\pm0.30}$	21.55$_{\pm0.15}$	28.54$_{\pm1.56}$	21.02$_{\pm0.10}$	23.49$_{\pm0.03}$	$\underline{16.32_{\pm0.21}}$	24.47$_{\pm0.00}$	20.20$_{\pm0.06}$	25.20$_{\pm0.00}$	22.33$_{\pm0.02}$	28.49$_{\pm0.00}$	29.63$_{\pm0.12}$	26.07$_{\pm0.00}$	21.56$_{\pm0.01}$	**14.53$_{\pm0.09}$**
	Avg	19.39$_{\pm0.28}$	19.56$_{\pm0.05}$	20.62$_{\pm0.41}$	19.51$_{\pm0.99}$	40.30$_{\pm2.05}$	18.02$_{\pm0.27}$	22.72$_{\pm0.59}$	17.67$_{\pm0.31}$	17.54$_{\pm0.31}$	19.23$_{\pm1.47}$	17.07$_{\pm1.26}$	18.26$_{\pm0.27}$	17.81$_{\pm0.16}$	17.74$_{\pm0.27}$	19.27$_{\pm0.19}$	25.17$_{\pm1.15}$	18.64$_{\pm0.09}$	19.85$_{\pm0.01}$	$\underline{15.32_{\pm0.28}}$	18.67$_{\pm0.00}$	16.63$_{\pm0.04}$	19.55$_{\pm0.00}$	17.86$_{\pm0.02}$	22.38$_{\pm0.00}$	23.89$_{\pm0.16}$	20.58$_{\pm0.00}$	17.53$_{\pm0.01}$	**13.45$_{\pm0.06}$**
RMSE	3	30.83$_{\pm0.32}$	29.38$_{\pm0.14}$	36.00$_{\pm0.80}$	30.15$_{\pm1.49}$	52.97$_{\pm2.13}$	27.82$_{\pm0.20}$	33.25$_{\pm0.58}$	27.41$_{\pm0.29}$	27.28$_{\pm0.25}$	29.96$_{\pm2.69}$	26.60$_{\pm1.78}$	28.38$_{\pm0.35}$	27.62$_{\pm0.15}$	27.46$_{\pm0.21}$	29.94$_{\pm0.28}$	36.09$_{\pm1.22}$	28.42$_{\pm0.19}$	29.63$_{\pm0.04}$	$\underline{25.56_{\pm0.32}}$	27.55$_{\pm0.00}$	26.41$_{\pm0.05}$	28.60$_{\pm0.00}$	27.46$_{\pm0.05}$	32.69$_{\pm0.00}$	36.91$_{\pm0.74}$	30.43$_{\pm0.00}$	27.18$_{\pm0.07}$	**23.14$_{\pm0.07}$**
	6	31.89$_{\pm0.28}$	32.54$_{\pm0.12}$	37.21$_{\pm0.75}$	31.78$_{\pm1.30}$	54.08$_{\pm1.79}$	30.46$_{\pm0.22}$	35.59$_{\pm0.67}$	29.94$_{\pm0.33}$	29.84$_{\pm0.28}$	32.30$_{\pm2.45}$	29.14$_{\pm1.82}$	30.27$_{\pm0.30}$	30.13$_{\pm0.17}$	30.10$_{\pm0.22}$	31.73$_{\pm0.17}$	38.85$_{\pm1.49}$	30.90$_{\pm0.16}$	33.33$_{\pm0.06}$	$\underline{26.92_{\pm0.27}}$	32.91$_{\pm0.00}$	30.53$_{\pm0.10}$	34.44$_{\pm0.00}$	32.80$_{\pm0.03}$	39.61$_{\pm0.00}$	42.56$_{\pm0.17}$	35.69$_{\pm0.00}$	31.79$_{\pm0.06}$	**24.64$_{\pm0.09}$**
	12	36.27$_{\pm0.30}$	39.02$_{\pm0.14}$	41.01$_{\pm0.68}$	35.84$_{\pm1.30}$	55.87$_{\pm1.97}$	35.29$_{\pm0.28}$	39.86$_{\pm0.86}$	34.34$_{\pm0.39}$	34.29$_{\pm0.38}$	36.50$_{\pm2.15}$	33.55$_{\pm1.93}$	33.61$_{\pm0.60}$	34.67$_{\pm0.25}$	34.96$_{\pm0.28}$	36.28$_{\pm0.20}$	45.39$_{\pm2.19}$	35.56$_{\pm0.17}$	40.75$_{\pm0.12}$	$\underline{28.97_{\pm0.18}}$	45.31$_{\pm0.00}$	38.69$_{\pm0.25}$	46.03$_{\pm0.00}$	42.92$_{\pm0.05}$	52.40$_{\pm0.00}$	59.72$_{\pm6.83}$	46.50$_{\pm0.00}$	40.60$_{\pm0.03}$	**26.64$_{\pm0.14}$**
	Avg	32.62$_{\pm0.30}$	33.05$_{\pm0.13}$	37.75$_{\pm0.75}$	32.22$_{\pm1.35}$	54.10$_{\pm1.93}$	30.73$_{\pm0.23}$	35.82$_{\pm0.68}$	30.13$_{\pm0.32}$	30.04$_{\pm0.29}$	32.52$_{\pm2.46}$	29.33$_{\pm1.84}$	30.41$_{\pm0.37}$	30.37$_{\pm0.18}$	30.37$_{\pm0.23}$	32.22$_{\pm0.21}$	39.57$_{\pm1.56}$	31.19$_{\pm0.17}$	33.90$_{\pm0.07}$	$\underline{26.93_{\pm0.26}}$	34.18$_{\pm0.00}$	31.15$_{\pm0.13}$	35.34$_{\pm0.00}$	33.57$_{\pm0.03}$	40.57$_{\pm0.00}$	44.62$_{\pm1.28}$	36.42$_{\pm0.00}$	32.42$_{\pm0.05}$	**24.56$_{\pm0.09}$**
MAPE (%)	3	30.48$_{\pm1.24}$	27.98$_{\pm0.33}$	31.59$_{\pm1.03}$	30.73$_{\pm2.59}$	50.26$_{\pm2.43}$	26.51$_{\pm1.30}$	34.13$_{\pm1.18}$	25.57$_{\pm1.32}$	25.51$_{\pm2.27}$	26.75$_{\pm1.59}$	24.19$_{\pm3.12}$	27.34$_{\pm0.96}$	25.71$_{\pm0.67}$	26.27$_{\pm1.27}$	32.44$_{\pm0.94}$	55.33$_{\pm2.34}$	29.41$_{\pm1.13}$	27.44$_{\pm0.37}$	$\underline{20.97_{\pm0.86}}$	19.33$_{\pm0.00}$	18.12$_{\pm0.03}$	20.30$_{\pm0.00}$	18.88$_{\pm0.03}$	20.43$_{\pm0.00}$	21.13$_{\pm0.06}$	20.34$_{\pm0.00}$	18.90$_{\pm0.03}$	**18.98$_{\pm0.18}$**
	6	30.62$_{\pm1.27}$	29.47$_{\pm0.24}$	31.86$_{\pm1.01}$	31.46$_{\pm2.35}$	49.24$_{\pm2.48}$	27.39$_{\pm1.34}$	34.81$_{\pm1.48}$	26.44$_{\pm1.29}$	26.43$_{\pm2.20}$	27.64$_{\pm1.71}$	25.38$_{\pm3.00}$	28.43$_{\pm0.54}$	26.65$_{\pm0.63}$	27.16$_{\pm1.32}$	33.59$_{\pm0.86}$	58.00$_{\pm2.92}$	30.46$_{\pm1.12}$	29.34$_{\pm0.46}$	$\underline{21.31_{\pm0.65}}$	21.88$_{\pm0.00}$	19.74$_{\pm0.04}$	23.14$_{\pm0.00}$	20.91$_{\pm0.02}$	23.33$_{\pm0.00}$	24.52$_{\pm0.11}$	23.12$_{\pm0.00}$	20.85$_{\pm0.03}$	**19.80$_{\pm0.18}$**
	12	32.67$_{\pm1.31}$	33.46$_{\pm0.27}$	34.54$_{\pm1.08}$	33.85$_{\pm2.10}$	47.91$_{\pm3.13}$	29.62$_{\pm1.46}$	36.82$_{\pm2.23}$	28.63$_{\pm1.18}$	28.65$_{\pm2.11}$	29.78$_{\pm1.91}$	28.11$_{\pm3.10}$	31.10$_{\pm1.22}$	28.92$_{\pm0.73}$	29.40$_{\pm1.43}$	37.31$_{\pm0.74}$	64.27$_{\pm4.24}$	33.08$_{\pm1.16}$	34.30$_{\pm0.78}$	$\underline{22.17_{\pm0.41}}$	28.40$_{\pm0.00}$	23.16$_{\pm0.04}$	29.16$_{\pm0.00}$	24.96$_{\pm0.02}$	28.97$_{\pm0.00}$	29.96$_{\pm0.09}$	28.69$_{\pm0.00}$	24.87$_{\pm0.05}$	**21.22$_{\pm0.26}$**
	Avg	31.09$_{\pm1.27}$	29.95$_{\pm0.26}$	32.43$_{\pm1.02}$	31.81$_{\pm2.36}$	49.24$_{\pm2.54}$	27.64$_{\pm1.35}$	35.08$_{\pm1.56}$	26.68$_{\pm1.27}$	26.67$_{\pm2.19}$	27.85$_{\pm1.72}$	25.66$_{\pm3.04}$	28.73$_{\pm0.39}$	26.88$_{\pm0.66}$	27.40$_{\pm1.33}$	34.13$_{\pm0.84}$	58.70$_{\pm3.03}$	30.76$_{\pm1.13}$	29.94$_{\pm0.50}$	$\underline{21.39_{\pm0.65}}$	22.66$_{\pm0.00}$	20.04$_{\pm0.03}$	23.68$_{\pm0.00}$	21.24$_{\pm0.02}$	23.73$_{\pm0.00}$	24.80$_{\pm0.09}$	23.57$_{\pm0.00}$	21.17$_{\pm0.03}$	**19.85$_{\pm0.19}$**

All sensors. In Figure 10, EvoST has lower error than Frozen in all 10×3×12 = 360 combinations of network, metric, and step, and lower error than Frozen+RLS in 343. The 17 exceptions are MAPE at steps 1–7 and 9–12 on PEMS05 and at steps 4–6 on PEMS04, RMSE at the first step on PEMS04 and PEMS05, and MAPE at the last step on TFNSW. Averaged equally over networks and steps, EvoST lowers MAE, RMSE, and MAPE by 8.9%, 7.2%, and 6.3% relative to Frozen and by 1.0%, 0.8%, and 1.2% relative to Frozen+RLS.

A.4 Annual Benchmark Protocol and Controls

Protocol. Identity fitting in the annual benchmark uses at most 16 epochs and patience 3 for each candidate learning rate, and samples every second window for both fitting and holdout (every fourth above 1,600 sensors). On TFNSW in 2026, the training split has only 1,704 windows and the validation split 552. The identity fit still holds out the final 512 window starts, which leaves 596 fitting and 256 holdout windows after sampling; the calibrators are initialized on all 1,704 training windows, and their settings are selected on all 552 validation windows. On PEMS06 in 2005, the fitting windows contain no valid target. For this year, the five seeds keep the checkpoint identities, and the calibrators take the standard deviation of the year's training inputs as their shared scale and start from zero residual coefficients; validation only selects the calibration settings, and the test state is initialized again and absorbs only labels released under the same rule. Refresh therefore coincides with Frozen, and EvoST with Frozen+RLS, in this year. The lowest candidate learning rate was selected in 1,100 of the other 1,160 runs (337 of 350 on PEMS03, PEMS04, and TFNSW). The selected

Table 8: Per-horizon results, part 2/2 (PEMS08, PEMS10, PEMS11, PEMS12, and TFNSW; mean±SD; for EvoST, the sample SD over five seeds).

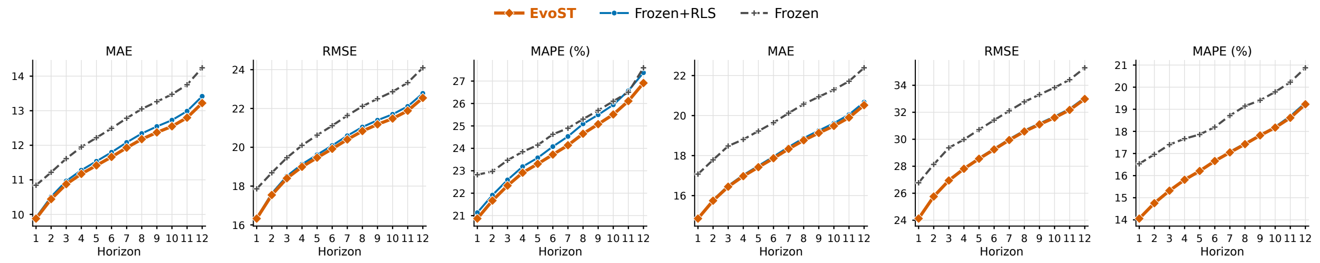
		Static STGNN Backbones				Naïve Schemes				Evolving Graph				Retrieval TTC		Static Forecasting Backbones					TSFMs								Ours
Metric	Len	DCRNN	ASTGNN	TGCN	GWN	Pretrain	Retrain	OL-NN	OL-AN	TrafStm	PECPM	STKEC	EAC	STRAP	ST-TTC	STID	STNorm	iTrans	DLinear	STAE	Chro2-Z	Chro2-F	TimesF-Z	TimesF-T	Moira1-Z	Moira1-T	Moira2-Z	Moira2-T	EvoST
PEMS08																													
MAE	3	$14.86_{\pm0.14}$	$13.63_{\pm0.05}$	$15.01_{\pm0.19}$	$14.62_{\pm0.19}$	$52.44_{\pm2.51}$	$13.13_{\pm0.06}$	$16.14_{\pm0.04}$	$12.76_{\pm0.09}$	$12.85_{\pm0.12}$	$14.58_{\pm0.83}$	$13.32_{\pm0.44}$	$13.86_{\pm0.28}$	$12.98_{\pm0.04}$	$12.95_{\pm0.06}$	$14.23_{\pm0.08}$	$17.19_{\pm0.35}$	$13.42_{\pm0.09}$	$13.79_{\pm0.05}$	$11.99_{\pm0.04}$	$13.26_{\pm0.00}$	$12.46_{\pm0.01}$	$13.94_{\pm0.00}$	$12.77_{\pm0.02}$	$15.65_{\pm0.00}$	$17.49_{\pm0.13}$	$14.90_{\pm0.00}$	$13.02_{\pm0.02}$	**$11.01_{\pm0.03}$**
	6	$15.17_{\pm0.13}$	$14.97_{\pm0.05}$	$15.60_{\pm0.16}$	$15.39_{\pm0.16}$	$52.24_{\pm2.61}$	$14.18_{\pm0.07}$	$17.16_{\pm0.06}$	$13.80_{\pm0.10}$	$13.91_{\pm0.12}$	$15.47_{\pm0.78}$	$14.31_{\pm0.33}$	$14.66_{\pm0.28}$	$14.02_{\pm0.03}$	$14.00_{\pm0.06}$	$14.83_{\pm0.06}$	$18.07_{\pm0.30}$	$14.44_{\pm0.08}$	$15.28_{\pm0.04}$	$12.47_{\pm0.03}$	$15.62_{\pm0.00}$	$14.00_{\pm0.02}$	$16.49_{\pm0.00}$	$14.62_{\pm0.01}$	$18.77_{\pm0.00}$	$19.98_{\pm0.13}$	$17.30_{\pm0.00}$	$14.75_{\pm0.02}$	**$11.54_{\pm0.04}$**
	12	$16.88_{\pm0.13}$	$17.84_{\pm0.05}$	$17.43_{\pm0.16}$	$17.61_{\pm0.20}$	$52.07_{\pm3.05}$	$16.27_{\pm0.09}$	$19.13_{\pm0.11}$	$15.75_{\pm0.11}$	$15.90_{\pm0.17}$	$17.19_{\pm0.67}$	$16.32_{\pm0.36}$	$16.25_{\pm0.27}$	$16.00_{\pm0.03}$	$16.10_{\pm0.08}$	$16.83_{\pm0.06}$	$20.27_{\pm0.37}$	$16.48_{\pm0.09}$	$18.53_{\pm0.04}$	$13.32_{\pm0.03}$	$21.09_{\pm0.00}$	$17.01_{\pm0.04}$	$21.51_{\pm0.00}$	$18.18_{\pm0.01}$	$24.43_{\pm0.00}$	$25.47_{\pm0.26}$	$22.15_{\pm0.00}$	$18.10_{\pm0.03}$	**$12.37_{\pm0.06}$**
	Avg	$15.50_{\pm0.13}$	$15.22_{\pm0.05}$	$15.86_{\pm0.17}$	$15.69_{\pm0.16}$	$52.25_{\pm2.71}$	$14.34_{\pm0.07}$	$17.30_{\pm0.05}$	$13.92_{\pm0.10}$	$14.04_{\pm0.13}$	$15.58_{\pm0.77}$	$14.47_{\pm0.38}$	$14.76_{\pm0.28}$	$14.15_{\pm0.03}$	$14.16_{\pm0.07}$	$15.12_{\pm0.07}$	$18.33_{\pm0.31}$	$14.60_{\pm0.08}$	$15.58_{\pm0.04}$	$12.52_{\pm0.03}$	$16.19_{\pm0.00}$	$14.22_{\pm0.02}$	$16.87_{\pm0.00}$	$14.90_{\pm0.01}$	$19.21_{\pm0.00}$	$20.52_{\pm0.16}$	$17.64_{\pm0.00}$	$14.99_{\pm0.02}$	**$11.55_{\pm0.04}$**
RMSE	3	$23.73_{\pm0.20}$	$22.36_{\pm0.08}$	$26.25_{\pm0.43}$	$23.28_{\pm0.29}$	$61.63_{\pm2.94}$	$21.49_{\pm0.07}$	$24.62_{\pm0.06}$	$21.01_{\pm0.12}$	$21.09_{\pm0.13}$	$23.27_{\pm1.20}$	$21.64_{\pm0.51}$	$21.99_{\pm0.33}$	$21.29_{\pm0.07}$	$21.26_{\pm0.05}$	$24.03_{\pm0.21}$	$26.14_{\pm0.40}$	$21.79_{\pm0.12}$	$22.74_{\pm0.04}$	$19.83_{\pm0.09}$	$22.30_{\pm0.00}$	$21.40_{\pm0.03}$	$23.20_{\pm0.00}$	$21.87_{\pm0.03}$	$26.33_{\pm0.00}$	$29.66_{\pm0.56}$	$24.72_{\pm0.00}$	$22.10_{\pm0.05}$	**$18.54_{\pm0.07}$**
	6	$24.70_{\pm0.18}$	$24.88_{\pm0.10}$	$27.48_{\pm0.38}$	$24.83_{\pm0.26}$	$62.11_{\pm3.02}$	$23.63_{\pm0.09}$	$26.65_{\pm0.12}$	$23.05_{\pm0.15}$	$23.19_{\pm0.16}$	$25.08_{\pm1.05}$	$23.66_{\pm0.40}$	$23.65_{\pm0.34}$	$23.35_{\pm0.04}$	$23.41_{\pm0.08}$	$25.24_{\pm0.16}$	$27.58_{\pm0.31}$	$23.77_{\pm0.13}$	$25.65_{\pm0.03}$	$20.82_{\pm0.08}$	$26.57_{\pm0.00}$	$24.61_{\pm0.06}$	$27.72_{\pm0.00}$	$25.66_{\pm0.02}$	$31.97_{\pm0.00}$	$33.94_{\pm0.36}$	$28.76_{\pm0.00}$	$25.56_{\pm0.05}$	**$19.59_{\pm0.09}$**
	12	$28.02_{\pm0.18}$	$29.80_{\pm0.13}$	$30.67_{\pm0.37}$	$28.68_{\pm0.40}$	$63.24_{\pm3.50}$	$27.52_{\pm0.13}$	$30.31_{\pm0.23}$	$26.53_{\pm0.20}$	$26.81_{\pm0.29}$	$28.33_{\pm0.80}$	$27.41_{\pm0.62}$	$26.49_{\pm0.45}$	$27.00_{\pm0.05}$	$27.31_{\pm0.12}$	$28.72_{\pm0.14}$	$31.18_{\pm0.47}$	$27.42_{\pm0.17}$	$31.37_{\pm0.05}$	$22.44_{\pm0.10}$	$36.33_{\pm0.00}$	$30.88_{\pm0.15}$	$36.53_{\pm0.00}$	$32.74_{\pm0.02}$	$41.51_{\pm0.00}$	$47.49_{\pm1.38}$	$37.04_{\pm0.00}$	$31.97_{\pm0.08}$	**$21.10_{\pm0.13}$**
	Avg	$25.20_{\pm0.19}$	$25.23_{\pm0.10}$	$27.85_{\pm0.40}$	$25.27_{\pm0.29}$	$62.21_{\pm3.13}$	$23.85_{\pm0.09}$	$26.86_{\pm0.13}$	$23.20_{\pm0.15}$	$23.35_{\pm0.18}$	$25.25_{\pm1.04}$	$23.89_{\pm0.50}$	$23.76_{\pm0.37}$	$23.54_{\pm0.04}$	$23.63_{\pm0.08}$	$25.69_{\pm0.17}$	$28.00_{\pm0.33}$	$23.99_{\pm0.13}$	$26.07_{\pm0.04}$	$20.88_{\pm0.09}$	$27.58_{\pm0.00}$	$25.08_{\pm0.08}$	$28.38_{\pm0.00}$	$26.18_{\pm0.02}$	$32.55_{\pm0.00}$	$35.46_{\pm0.39}$	$29.31_{\pm0.00}$	$25.97_{\pm0.06}$	**$19.57_{\pm0.09}$**
MAPE (%)	3	$22.82_{\pm0.42}$	$19.07_{\pm0.41}$	$21.25_{\pm0.69}$	$20.76_{\pm0.73}$	$57.52_{\pm1.13}$	$19.25_{\pm0.27}$	$22.79_{\pm0.31}$	$18.05_{\pm0.22}$	$19.62_{\pm1.41}$	$21.16_{\pm1.57}$	$20.30_{\pm1.59}$	$26.19_{\pm1.99}$	$18.63_{\pm0.35}$	$19.13_{\pm0.27}$	$19.38_{\pm0.76}$	$41.36_{\pm1.88}$	$20.82_{\pm1.06}$	$18.96_{\pm0.43}$	$17.16_{\pm0.26}$	$17.01_{\pm0.00}$	$15.85_{\pm0.02}$	$17.93_{\pm0.00}$	$16.36_{\pm0.01}$	$18.02_{\pm0.00}$	$18.90_{\pm0.04}$	$17.86_{\pm0.00}$	$16.69_{\pm0.04}$	**$16.21_{\pm0.04}$**
	6	$22.94_{\pm0.42}$	$20.47_{\pm0.38}$	$21.71_{\pm0.71}$	$21.51_{\pm0.73}$	$56.94_{\pm1.54}$	$20.33_{\pm0.24}$	$23.83_{\pm0.33}$	$19.02_{\pm0.21}$	$20.70_{\pm1.34}$	$22.07_{\pm1.65}$	$21.30_{\pm1.42}$	$26.35_{\pm1.97}$	$19.69_{\pm0.40}$	$20.20_{\pm0.23}$	$19.98_{\pm0.75}$	$42.87_{\pm1.97}$	$21.81_{\pm1.00}$	$20.66_{\pm0.40}$	$17.53_{\pm0.25}$	$19.33_{\pm0.00}$	$17.22_{\pm0.03}$	$20.46_{\pm0.00}$	$17.97_{\pm0.00}$	$20.63_{\pm0.00}$	$21.71_{\pm0.05}$	$20.31_{\pm0.00}$	$18.33_{\pm0.04}$	**$16.85_{\pm0.07}$**
	12	$24.58_{\pm0.45}$	$23.92_{\pm0.38}$	$24.00_{\pm0.73}$	$24.07_{\pm0.76}$	$56.54_{\pm2.39}$	$22.74_{\pm0.27}$	$26.26_{\pm0.49}$	$21.37_{\pm0.32}$	$23.18_{\pm1.24}$	$24.14_{\pm1.73}$	$23.62_{\pm0.94}$	$27.80_{\pm2.10}$	$22.19_{\pm0.54}$	$22.61_{\pm0.27}$	$22.15_{\pm0.79}$	$46.66_{\pm2.26}$	$24.16_{\pm0.88}$	$24.95_{\pm0.43}$	$18.49_{\pm0.24}$	$25.32_{\pm0.00}$	$20.17_{\pm0.04}$	$25.89_{\pm0.00}$	$21.27_{\pm0.00}$	$25.69_{\pm0.00}$	$26.68_{\pm0.07}$	$25.19_{\pm0.00}$	$21.82_{\pm0.04}$	**$17.99_{\pm0.10}$**
	Avg	$23.31_{\pm0.41}$	$20.85_{\pm0.39}$	$22.13_{\pm0.71}$	$21.90_{\pm0.74}$	$57.09_{\pm1.58}$	$20.58_{\pm0.25}$	$24.09_{\pm0.35}$	$19.28_{\pm0.22}$	$20.96_{\pm1.33}$	$22.26_{\pm1.64}$	$21.55_{\pm1.34}$	$26.61_{\pm1.73}$	$19.96_{\pm0.41}$	$20.45_{\pm0.25}$	$20.33_{\pm0.76}$	$43.33_{\pm2.01}$	$22.07_{\pm0.99}$	$21.16_{\pm0.42}$	$17.66_{\pm0.23}$	$20.06_{\pm0.00}$	$17.49_{\pm0.03}$	$20.96_{\pm0.00}$	$18.26_{\pm0.00}$	$21.00_{\pm0.00}$	$22.04_{\pm0.05}$	$20.70_{\pm0.00}$	$18.63_{\pm0.04}$	**$16.89_{\pm0.07}$**
PEMS10																													
MAE	3	$11.81_{\pm0.19}$	$11.46_{\pm0.05}$	$11.82_{\pm0.09}$	$11.87_{\pm0.11}$	$81.21_{\pm3.23}$	$11.09_{\pm0.04}$	$11.49_{\pm0.20}$	$10.86_{\pm0.08}$	$11.14_{\pm0.06}$	$11.90_{\pm1.24}$	$11.15_{\pm0.08}$	$12.84_{\pm1.70}$	$11.05_{\pm0.08}$	$11.01_{\pm0.04}$	$11.85_{\pm0.09}$	$13.13_{\pm0.29}$	$11.13_{\pm0.07}$	$11.73_{\pm0.04}$	$10.09_{\pm0.03}$	$11.46_{\pm0.00}$	$10.79_{\pm0.01}$	$11.82_{\pm0.00}$	$11.04_{\pm0.01}$	$13.03_{\pm0.00}$	$14.44_{\pm0.12}$	$12.56_{\pm0.00}$	$11.24_{\pm0.00}$	**$9.51_{\pm0.03}$**
	6	$12.06_{\pm0.14}$	$12.32_{\pm0.05}$	$12.29_{\pm0.08}$	$12.38_{\pm0.11}$	$81.43_{\pm3.27}$	$11.74_{\pm0.04}$	$12.23_{\pm0.23}$	$11.50_{\pm0.10}$	$11.82_{\pm0.08}$	$12.51_{\pm1.18}$	$11.81_{\pm0.07}$	$13.48_{\pm1.73}$	$11.68_{\pm0.09}$	$11.65_{\pm0.04}$	$12.36_{\pm0.07}$	$13.62_{\pm0.32}$	$11.74_{\pm0.08}$	$12.69_{\pm0.04}$	$10.32_{\pm0.03}$	$12.93_{\pm0.00}$	$11.74_{\pm0.01}$	$13.38_{\pm0.00}$	$12.10_{\pm0.01}$	$15.01_{\pm0.00}$	$16.15_{\pm0.13}$	$13.93_{\pm0.00}$	$12.30_{\pm0.00}$	**$9.75_{\pm0.03}$**
	12	$13.19_{\pm0.15}$	$14.05_{\pm0.05}$	$13.48_{\pm0.08}$	$13.70_{\pm0.14}$	$81.53_{\pm3.26}$	$13.06_{\pm0.06}$	$13.70_{\pm0.29}$	$12.76_{\pm0.12}$	$13.18_{\pm0.13}$	$13.72_{\pm1.04}$	$13.14_{\pm0.08}$	$14.87_{\pm1.91}$	$12.97_{\pm0.10}$	$12.97_{\pm0.05}$	$13.73_{\pm0.09}$	$14.81_{\pm0.35}$	$12.95_{\pm0.07}$	$14.67_{\pm0.04}$	$10.73_{\pm0.04}$	$16.37_{\pm0.00}$	$13.51_{\pm0.01}$	$16.49_{\pm0.00}$	$14.17_{\pm0.01}$	$18.64_{\pm0.00}$	$19.70_{\pm0.22}$	$16.97_{\pm0.00}$	$14.30_{\pm0.01}$	**$10.14_{\pm0.03}$**
	Avg	$12.26_{\pm0.16}$	$12.45_{\pm0.05}$	$12.43_{\pm0.08}$	$12.54_{\pm0.11}$	$81.39_{\pm3.16}$	$11.85_{\pm0.04}$	$12.35_{\pm0.23}$	$11.60_{\pm0.09}$	$11.93_{\pm0.08}$	$12.61_{\pm1.16}$	$11.92_{\pm0.07}$	$13.60_{\pm1.76}$	$11.79_{\pm0.09}$	$11.76_{\pm0.04}$	$12.53_{\pm0.08}$	$13.75_{\pm0.32}$	$11.83_{\pm0.07}$	$12.86_{\pm0.04}$	$10.34_{\pm0.03}$	$13.30_{\pm0.00}$	$11.85_{\pm0.01}$	$13.63_{\pm0.00}$	$12.27_{\pm0.01}$	$15.31_{\pm0.00}$	$16.45_{\pm0.14}$	$14.18_{\pm0.00}$	$12.43_{\pm0.01}$	**$9.76_{\pm0.03}$**
RMSE	3	$19.17_{\pm0.29}$	$19.03_{\pm0.09}$	$20.39_{\pm0.16}$	$19.21_{\pm0.12}$	$83.86_{\pm3.27}$	$18.21_{\pm0.02}$	$18.96_{\pm0.38}$	$17.84_{\pm0.14}$	$18.31_{\pm0.08}$	$20.00_{\pm2.91}$	$18.35_{\pm0.11}$	$19.71_{\pm1.41}$	$18.17_{\pm0.10}$	$18.08_{\pm0.02}$	$19.33_{\pm0.09}$	$20.78_{\pm0.37}$	$18.33_{\pm0.09}$	$19.35_{\pm0.05}$	$16.69_{\pm0.07}$	$19.29_{\pm0.00}$	$18.45_{\pm0.03}$	$19.75_{\pm0.00}$	$18.67_{\pm0.01}$	$21.83_{\pm0.00}$	$23.87_{\pm0.10}$	$20.99_{\pm0.00}$	$19.00_{\pm0.02}$	**$15.82_{\pm0.05}$**
	6	$19.79_{\pm0.24}$	$20.59_{\pm0.06}$	$21.24_{\pm0.14}$	$20.20_{\pm0.12}$	$84.50_{\pm3.37}$	$19.49_{\pm0.04}$	$20.39_{\pm0.45}$	$19.06_{\pm0.18}$	$19.58_{\pm0.14}$	$21.13_{\pm2.71}$	$19.63_{\pm0.10}$	$20.94_{\pm1.41}$	$19.40_{\pm0.10}$	$19.35_{\pm0.04}$	$20.30_{\pm0.07}$	$21.60_{\pm0.41}$	$19.49_{\pm0.10}$	$21.11_{\pm0.05}$	$17.17_{\pm0.05}$	$21.87_{\pm0.00}$	$20.39_{\pm0.04}$	$22.48_{\pm0.00}$	$20.80_{\pm0.01}$	$25.46_{\pm0.00}$	$28.17_{\pm2.25}$	$23.25_{\pm0.00}$	$21.08_{\pm0.02}$	**$16.29_{\pm0.04}$**
	12	$21.96_{\pm0.25}$	$23.58_{\pm0.05}$	$23.35_{\pm0.13}$	$22.52_{\pm0.20}$	$85.41_{\pm3.41}$	$21.94_{\pm0.07}$	$23.10_{\pm0.55}$	$21.39_{\pm0.21}$	$22.03_{\pm0.24}$	$23.31_{\pm2.32}$	$22.08_{\pm0.12}$	$23.29_{\pm1.57}$	$21.80_{\pm0.14}$	$21.80_{\pm0.07}$	$22.59_{\pm0.11}$	$23.56_{\pm0.44}$	$21.66_{\pm0.11}$	$24.54_{\pm0.05}$	$17.96_{\pm0.05}$	$27.96_{\pm0.00}$	$24.06_{\pm0.05}$	$27.93_{\pm0.00}$	$24.87_{\pm0.01}$	$32.24_{\pm0.00}$	$33.91_{\pm1.29}$	$28.30_{\pm0.00}$	$24.93_{\pm0.04}$	**$17.02_{\pm0.04}$**
	Avg	$20.13_{\pm0.26}$	$20.80_{\pm0.06}$	$21.49_{\pm0.14}$	$20.45_{\pm0.13}$	$84.52_{\pm3.24}$	$19.66_{\pm0.03}$	$20.58_{\pm0.45}$	$19.22_{\pm0.17}$	$19.77_{\pm0.14}$	$21.29_{\pm2.68}$	$19.80_{\pm0.11}$	$21.08_{\pm1.43}$	$19.58_{\pm0.10}$	$19.53_{\pm0.04}$	$20.54_{\pm0.09}$	$21.81_{\pm0.40}$	$19.63_{\pm0.10}$	$21.36_{\pm0.05}$	$17.20_{\pm0.05}$	$22.54_{\pm0.00}$	$20.64_{\pm0.04}$	$22.92_{\pm0.00}$	$21.11_{\pm0.01}$	$25.99_{\pm0.00}$	$28.09_{\pm1.21}$	$23.64_{\pm0.00}$	$21.32_{\pm0.03}$	**$16.30_{\pm0.04}$**
MAPE (%)	3	$30.74_{\pm0.34}$	$28.38_{\pm0.32}$	$29.73_{\pm0.29}$	$29.49_{\pm0.74}$	$756.76_{\pm29.47}$	$29.11_{\pm0.29}$	$30.04_{\pm0.94}$	$28.70_{\pm0.37}$	$30.52_{\pm0.75}$	$29.20_{\pm1.71}$	$29.69_{\pm0.29}$	$58.30_{\pm38.83}$	$29.13_{\pm0.55}$	$28.95_{\pm0.28}$	$31.70_{\pm0.31}$	$40.22_{\pm1.74}$	$29.02_{\pm0.61}$	$28.77_{\pm0.41}$	$27.39_{\pm0.42}$	$27.81_{\pm0.00}$	$26.36_{\pm0.07}$	$28.80_{\pm0.00}$	$27.03_{\pm0.03}$	$28.16_{\pm0.00}$	$29.19_{\pm0.16}$	$28.44_{\pm0.00}$	$27.36_{\pm0.03}$	**$26.98_{\pm0.12}$**
	6	$31.29_{\pm0.33}$	$29.84_{\pm0.32}$	$30.52_{\pm0.23}$	$30.31_{\pm0.69}$	$755.22_{\pm29.03}$	$30.00_{\pm0.23}$	$31.38_{\pm0.98}$	$29.47_{\pm0.34}$	$31.73_{\pm0.67}$	$30.11_{\pm1.70}$	$30.78_{\pm0.50}$	$58.55_{\pm40.02}$	$30.04_{\pm0.59}$	$29.84_{\pm0.22}$	$32.71_{\pm0.23}$	$41.21_{\pm1.76}$	$30.02_{\pm0.71}$	$30.57_{\pm0.38}$	$27.74_{\pm0.45}$	$29.81_{\pm0.00}$	$27.60_{\pm0.07}$	$30.98_{\pm0.00}$	$28.35_{\pm0.03}$	$30.54_{\pm0.00}$	$31.69_{\pm0.10}$	$30.28_{\pm0.00}$	$28.91_{\pm0.03}$	**$27.51_{\pm0.11}$**
	12	$33.45_{\pm0.41}$	$33.35_{\pm0.34}$	$32.93_{\pm0.18}$	$32.78_{\pm0.66}$	$752.24_{\pm26.19}$	$32.42_{\pm0.18}$	$34.68_{\pm1.59}$	$31.77_{\pm0.42}$	$35.13_{\pm0.79}$	$32.53_{\pm1.65}$	$33.28_{\pm0.51}$	$62.45_{\pm42.54}$	$32.43_{\pm0.61}$	$32.25_{\pm0.17}$	$35.68_{\pm0.25}$	$43.96_{\pm1.88}$	$32.40_{\pm0.94}$	$34.98_{\pm0.32}$	$28.55_{\pm0.44}$	$35.14_{\pm0.00}$	$30.28_{\pm0.06}$	$35.86_{\pm0.00}$	$31.18_{\pm0.03}$	$35.72_{\pm0.00}$	$36.23_{\pm0.13}$	$34.51_{\pm0.00}$	$32.29_{\pm0.03}$	**$28.49_{\pm0.14}$**
	Avg	$31.65_{\pm0.35}$	$30.22_{\pm0.32}$	$30.86_{\pm0.22}$	$30.65_{\pm0.69}$	$755.06_{\pm28.01}$	$30.30_{\pm0.22}$	$31.79_{\pm1.11}$	$29.80_{\pm0.34}$	$32.18_{\pm0.70}$	$30.41_{\pm1.69}$	$31.06_{\pm0.42}$	$59.52_{\pm40.27}$	$30.34_{\pm0.58}$	$30.14_{\pm0.22}$	$33.11_{\pm0.25}$	$41.57_{\pm1.79}$	$30.28_{\pm0.74}$	$31.07_{\pm0.38}$	$27.83_{\pm0.45}$	$30.51_{\pm0.00}$	$27.84_{\pm0.07}$	$31.47_{\pm0.00}$	$28.62_{\pm0.03}$	$30.97_{\pm0.00}$	$32.02_{\pm0.11}$	$30.73_{\pm0.00}$	$29.22_{\pm0.02}$	**$27.56_{\pm0.12}$**
PEMS11																													
MAE	3	$18.69_{\pm0.36}$	$17.88_{\pm0.20}$	$20.19_{\pm0.31}$	$18.90_{\pm0.41}$	$83.38_{\pm3.81}$	$17.03_{\pm0.13}$	$19.74_{\pm1.56}$	$16.62_{\pm0.13}$	$16.93_{\pm0.27}$	$24.48_{\pm2.64}$	$17.24_{\pm0.28}$	$23.71_{\pm3.81}$	$16.91_{\pm0.12}$	$16.76_{\pm0.12}$	$18.57_{\pm0.32}$	$20.18_{\pm0.43}$	$17.54_{\pm0.29}$	$18.32_{\pm0.15}$	$16.33_{\pm0.16}$	$15.80_{\pm0.00}$	$14.70_{\pm0.01}$	$16.84_{\pm0.00}$	$15.25_{\pm0.02}$	$18.69_{\pm0.00}$	$20.45_{\pm0.14}$	$17.40_{\pm0.00}$	$15.09_{\pm0.00}$	**$12.97_{\pm0.07}$**
	6	$19.19_{\pm0.28}$	$19.63_{\pm0.17}$	$20.93_{\pm0.31}$	$19.91_{\pm0.39}$	$85.10_{\pm3.52}$	$18.50_{\pm0.12}$	$21.22_{\pm1.61}$	$18.07_{\pm0.12}$	$18.43_{\pm0.34}$	$25.70_{\pm2.54}$	$18.77_{\pm0.21}$	$25.31_{\pm4.21}$	$18.36_{\pm0.15}$	$18.26_{\pm0.12}$	$19.61_{\pm0.32}$	$21.46_{\pm0.36}$	$18.98_{\pm0.29}$	$20.20_{\pm0.11}$	$16.94_{\pm0.14}$	$19.21_{\pm0.00}$	$16.84_{\pm0.02}$	$20.39_{\pm0.00}$	$17.88_{\pm0.02}$	$22.74_{\pm0.00}$	$24.00_{\pm0.10}$	$20.82_{\pm0.00}$	$17.34_{\pm0.01}$	**$13.69_{\pm0.09}$**
	12	$21.83_{\pm0.24}$	$23.48_{\pm0.12}$	$23.55_{\pm0.32}$	$22.77_{\pm0.55}$	$88.77_{\pm3.23}$	$21.61_{\pm0.15}$	$24.45_{\pm1.70}$	$20.92_{\pm0.13}$	$21.44_{\pm0.47}$	$28.19_{\pm2.36}$	$21.86_{\pm0.16}$	$29.22_{\pm4.63}$	$21.32_{\pm0.24}$	$21.38_{\pm0.15}$	$22.64_{\pm0.34}$	$24.40_{\pm0.67}$	$21.99_{\pm0.27}$	$24.43_{\pm0.09}$	$18.28_{\pm0.12}$	$27.09_{\pm0.00}$	$21.20_{\pm0.04}$	$27.29_{\pm0.00}$	$23.15_{\pm0.03}$	$29.56_{\pm0.00}$	$30.42_{\pm0.16}$	$27.38_{\pm0.00}$	$21.94_{\pm0.03}$	**$14.84_{\pm0.13}$**
	Avg	$19.69_{\pm0.30}$	$20.00_{\pm0.16}$	$21.35_{\pm0.31}$	$20.29_{\pm0.44}$	$85.43_{\pm3.52}$	$18.77_{\pm0.13}$	$21.54_{\pm1.59}$	$18.29_{\pm0.13}$	$18.67_{\pm0.35}$	$25.89_{\pm2.53}$	$19.02_{\pm0.23}$	$25.76_{\pm4.14}$	$18.60_{\pm0.15}$	$18.52_{\pm0.13}$	$20.02_{\pm0.32}$	$21.77_{\pm0.34}$	$19.24_{\pm0.28}$	$20.62_{\pm0.12}$	$17.05_{\pm0.12}$	$20.04_{\pm0.00}$	$17.19_{\pm0.02}$	$20.91_{\pm0.00}$	$18.32_{\pm0.02}$	$23.16_{\pm0.00}$	$24.37_{\pm0.09}$	$21.24_{\pm0.00}$	$17.71_{\pm0.01}$	**$13.70_{\pm0.09}$**
RMSE	3	$31.04_{\pm0.45}$	$29.94_{\pm0.35}$	$37.69_{\pm0.55}$	$31.36_{\pm0.74}$	$112.94_{\pm3.71}$	$28.37_{\pm0.17}$	$32.65_{\pm2.94}$	$27.81_{\pm0.14}$	$28.11_{\pm0.36}$	$42.15_{\pm5.33}$	$28.52_{\pm0.27}$	$37.06_{\pm5.06}$	$28.19_{\pm0.16}$	$27.97_{\pm0.16}$	$30.85_{\pm0.38}$	$32.56_{\pm0.56}$	$29.15_{\pm0.31}$	$30.60_{\pm0.15}$	$27.05_{\pm0.36}$	$28.35_{\pm0.00}$	$26.78_{\pm0.04}$	$30.08_{\pm0.00}$	$27.62_{\pm0.04}$	$33.70_{\pm0.00}$	$36.68_{\pm0.41}$	$31.28_{\pm0.00}$	$27.08_{\pm0.01}$	**$23.77_{\pm0.08}$**
	6	$32.77_{\pm0.30}$	$33.71_{\pm0.26}$	$39.44_{\pm0.53}$	$33.73_{\pm0.69}$	$116.32_{\pm3.41}$	$31.76_{\pm0.14}$	$35.91_{\pm2.87}$	$31.05_{\pm0.11}$	$31.50_{\pm0.44}$	$44.47_{\pm4.94}$	$31.98_{\pm0.20}$	$40.37_{\pm5.70}$	$31.47_{\pm0.20}$	$31.38_{\pm0.13}$	$33.20_{\pm0.37}$	$35.12_{\pm0.52}$	$32.37_{\pm0.31}$	$34.80_{\pm0.12}$	$28.65_{\pm0.25}$	$35.45_{\pm0.00}$	$31.71_{\pm0.06}$	$37.19_{\pm0.00}$	$33.61_{\pm0.08}$	$41.63_{\pm0.00}$	$43.90_{\pm0.27}$	$37.82_{\pm0.00}$	$32.06_{\pm0.04}$	**$25.24_{\pm0.12}$**
	12	$38.65_{\pm0.27}$	$41.33_{\pm0.17}$	$44.47_{\pm0.51}$	$39.49_{\pm1.05}$	$123.41_{\pm3.10}$	$38.27_{\pm0.17}$	$42.48_{\pm2.88}$	$36.98_{\pm0.16}$	$37.79_{\pm0.69}$	$49.10_{\pm4.27}$	$38.39_{\pm0.19}$	$47.69_{\pm6.19}$	$37.68_{\pm0.32}$	$37.94_{\pm0.17}$	$39.19_{\pm0.37}$	$40.62_{\pm1.15}$	$38.55_{\pm0.30}$	$43.44_{\pm0.13}$	$31.70_{\pm0.19}$	$51.41_{\pm0.00}$	$41.40_{\pm0.19}$	$50.75_{\pm0.00}$	$44.93_{\pm0.10}$	$54.25_{\pm0.00}$	$56.11_{\pm0.11}$	$50.35_{\pm0.00}$	$41.93_{\pm0.10}$	**$27.58_{\pm0.18}$**
	Avg	$33.66_{\pm0.34}$	$34.32_{\pm0.27}$	$40.11_{\pm0.53}$	$34.35_{\pm0.80}$	$116.95_{\pm3.38}$	$32.21_{\pm0.15}$	$36.45_{\pm2.86}$	$31.40_{\pm0.13}$	$31.90_{\pm0.47}$	$44.81_{\pm4.91}$	$32.38_{\pm0.23}$	$41.08_{\pm5.52}$	$31.88_{\pm0.20}$	$31.83_{\pm0.15}$	$33.89_{\pm0.36}$	$35.62_{\pm0.53}$	$32.79_{\pm0.31}$	$35.52_{\pm0.13}$	$28.82_{\pm0.20}$	$37.05_{\pm0.00}$	$32.41_{\pm0.08}$	$38.14_{\pm0.00}$	$34.43_{\pm0.07}$	$42.24_{\pm0.00}$	$44.37_{\pm0.19}$	$38.56_{\pm0.00}$	$32.78_{\pm0.04}$	**$25.24_{\pm0.12}$**
MAPE (%)	3	$30.78_{\pm0.93}$	$24.39_{\pm0.34}$	$29.45_{\pm0.64}$	$26.32_{\pm0.78}$	$77.32_{\pm15.78}$	$26.11_{\pm0.80}$	$28.92_{\pm0.90}$	$23.82_{\pm0.70}$	$26.53_{\pm1.24}$	$35.21_{\pm6.88}$	$28.98_{\pm1.96}$	$45.25_{\pm14.22}$	$27.10_{\pm0.83}$	$25.97_{\pm0.79}$	$30.64_{\pm1.81}$	$45.43_{\pm2.56}$	$29.58_{\pm2.86}$	$24.98_{\pm0.73}$	$23.94_{\pm0.63}$	$21.66_{\pm0.00}$	$20.03_{\pm0.04}$	$22.80_{\pm0.00}$	$20.81_{\pm0.03}$	$22.44_{\pm0.00}$	$23.02_{\pm0.01}$	$22.43_{\pm0.00}$	$20.95_{\pm0.01}$	**$21.22_{\pm0.13}$**
	6	$30.74_{\pm0.87}$	$26.12_{\pm0.37}$	$29.65_{\pm0.33}$	$27.14_{\pm0.80}$	$80.97_{\pm18.12}$	$27.25_{\pm0.78}$	$30.36_{\pm1.20}$	$25.01_{\pm0.79}$	$27.76_{\pm1.43}$	$36.53_{\pm6.71}$	$30.24_{\pm1.63}$	$46.71_{\pm12.65}$	$28.41_{\pm1.01}$	$27.11_{\pm0.78}$	$31.58_{\pm1.76}$	$47.41_{\pm2.67}$	$30.58_{\pm2.79}$	$26.94_{\pm0.61}$	$24.36_{\pm0.69}$	$24.49_{\pm0.00}$	$21.62_{\pm0.04}$	$25.85_{\pm0.00}$	$22.74_{\pm0.04}$	$25.41_{\pm0.00}$	$26.42_{\pm0.04}$	$25.28_{\pm0.00}$	$22.88_{\pm0.02}$	**$22.16_{\pm0.17}$**
	12	$32.84_{\pm0.78}$	$30.75_{\pm0.53}$	$32.76_{\pm0.27}$	$29.87_{\pm0.87}$	$89.84_{\pm23.97}$	$30.48_{\pm0.87}$	$34.14_{\pm1.85}$	$28.09_{\pm0.80}$	$31.40_{\pm2.51}$	$39.89_{\pm6.43}$	$33.90_{\pm1.15}$	$51.27_{\pm11.76}$	$31.92_{\pm1.30}$	$30.33_{\pm0.87}$	$35.10_{\pm1.52}$	$52.32_{\pm3.03}$	$33.38_{\pm2.70}$	$32.38_{\pm0.59}$	$25.71_{\pm0.78}$	$31.70_{\pm0.00}$	$25.05_{\pm0.03}$	$32.22_{\pm0.00}$	$26.79_{\pm0.04}$	$30.99_{\pm0.00}$	$31.71_{\pm0.03}$	$31.06_{\pm0.00}$	$26.96_{\pm0.05}$	**$23.93_{\pm0.23}$**
	Avg	$31.28_{\pm0.86}$	$26.69_{\pm0.39}$	$30.37_{\pm0.32}$	$27.55_{\pm0.81}$	$82.12_{\pm18.72}$	$27.67_{\pm0.81}$	$30.83_{\pm1.25}$	$25.39_{\pm0.75}$	$28.29_{\pm1.63}$	$36.90_{\pm6.66}$	$30.78_{\pm1.62}$	$47.34_{\pm12.72}$	$28.85_{\pm1.00}$	$27.53_{\pm0.80}$	$32.17_{\pm1.75}$	$48.00_{\pm2.68}$	$30.95_{\pm2.77}$	$27.66_{\pm0.61}$	$24.54_{\pm0.69}$	$25.36_{\pm0.00}$	$21.93_{\pm0.04}$	$26.41_{\pm0.00}$	$23.11_{\pm0.03}$	$25.78_{\pm0.00}$	$26.67_{\pm0.03}$	$25.78_{\pm0.00}$	$23.23_{\pm0.03}$	**$22.27_{\pm0.17}$**
PEMS12																													
MAE	3	$15.43_{\pm0.22}$	$14.23_{\pm0.12}$	$16.26_{\pm0.09}$	$15.16_{\pm0.28}$	$15.17_{\pm0.18}$	$13.59_{\pm0.05}$	$16.25_{\pm0.47}$	$13.24_{\pm0.05}$	$13.42_{\pm0.06}$	$14.97_{\pm1.63}$	$13.42_{\pm0.10}$	$13.96_{\pm0.25}$	$13.59_{\pm0.15}$	$13.41_{\pm0.05}$	$15.00_{\pm0.09}$	$17.97_{\pm0.53}$	$13.92_{\pm0.13}$	$14.31_{\pm0.02}$	$12.39_{\pm0.11}$	$13.51_{\pm0.00}$	$12.73_{\pm0.00}$	$14.31_{\pm0.00}$	$13.24_{\pm0.03}$	$15.91_{\pm0.00}$	$17.32_{\pm0.62}$	$14.88_{\pm0.00}$	$13.21_{\pm0.01}$	**$11.20_{\pm0.05}$**
	6	$15.76_{\pm0.16}$	$15.67_{\pm0.09}$	$16.80_{\pm0.08}$	$16.11_{\pm0.25}$	$16.28_{\pm0.20}$	$14.75_{\pm0.05}$	$17.78_{\pm0.56}$	$14.40_{\pm0.06}$	$14.60_{\pm0.06}$	$16.02_{\pm1.50}$	$14.57_{\pm0.08}$	$14.84_{\pm0.28}$	$14.72_{\pm0.14}$	$14.57_{\pm0.05}$	$15.76_{\pm0.07}$	$19.17_{\pm0.41}$	$15.06_{\pm0.10}$	$15.90_{\pm0.02}$	$12.87_{\pm0.10}$	$16.05_{\pm0.00}$	$14.41_{\pm0.01}$	$17.06_{\pm0.00}$	$15.37_{\pm0.03}$	$19.07_{\pm0.00}$	$20.17_{\pm0.58}$	$17.51_{\pm0.00}$	$15.06_{\pm0.01}$	**$11.76_{\pm0.06}$**
	12	$17.77_{\pm0.14}$	$18.87_{\pm0.10}$	$18.83_{\pm0.06}$	$18.46_{\pm0.20}$	$18.34_{\pm0.30}$	$17.02_{\pm0.06}$	$20.91_{\pm0.77}$	$16.58_{\pm0.09}$	$16.81_{\pm0.13}$	$18.11_{\pm1.30}$	$16.75_{\pm0.10}$	$16.63_{\pm0.44}$	$16.95_{\pm0.14}$	$16.87_{\pm0.07}$	$18.07_{\pm0.06}$	$22.03_{\pm0.30}$	$17.48_{\pm0.08}$	$19.37_{\pm0.03}$	$13.70_{\pm0.12}$	$21.94_{\pm0.00}$	$17.73_{\pm0.04}$	$22.48_{\pm0.00}$	$19.51_{\pm0.03}$	$24.73_{\pm0.00}$	$25.32_{\pm0.63}$	$22.80_{\pm0.00}$	$18.67_{\pm0.01}$	**$12.59_{\pm0.09}$**
	Avg	$16.16_{\pm0.17}$	$15.98_{\pm0.10}$	$17.13_{\pm0.08}$	$16.38_{\pm0.23}$	$16.41_{\pm0.21}$	$14.91_{\pm0.05}$	$18.04_{\pm0.57}$	$14.54_{\pm0.06}$	$14.74_{\pm0.07}$	$16.18_{\pm1.49}$	$14.71_{\pm0.08}$	$14.98_{\pm0.30}$	$14.88_{\pm0.14}$	$14.74_{\pm0.05}$	$16.07_{\pm0.07}$	$19.49_{\pm0.39}$	$15.27_{\pm0.11}$	$16.23_{\pm0.02}$	$12.91_{\pm0.11}$	$16.67_{\pm0.00}$	$14.66_{\pm0.01}$	$17.48_{\pm0.00}$	$15.69_{\pm0.02}$	$19.49_{\pm0.00}$	$20.49_{\pm0.59}$	$17.90_{\pm0.00}$	$15.32_{\pm0.01}$	**$11.76_{\pm0.06}$**
RMSE	3	$26.80_{\pm0.31}$	$25.07_{\pm0.18}$	$31.84_{\pm0.21}$	$26.13_{\pm0.45}$	$25.43_{\pm0.16}$	$23.87_{\pm0.08}$	$28.37_{\pm0.92}$	$23.41_{\pm0.03}$	$23.68_{\pm0.09}$	$26.16_{\pm3.54}$	$23.69_{\pm0.15}$	$24.14_{\pm0.31}$	$23.87_{\pm0.18}$	$23.58_{\pm0.05}$	$26.16_{\pm0.22}$	$29.50_{\pm0.83}$	$24.26_{\pm0.18}$	$25.36_{\pm0.03}$	$22.03_{\pm0.12}$	$24.46_{\pm0.00}$	$23.44_{\pm0.02}$	$25.67_{\pm0.00}$	$24.31_{\pm0.05}$	$29.08_{\pm0.00}$	$32.10_{\pm1.13}$	$26.66_{\pm0.00}$	$24.10_{\pm0.03}$	**$20.25_{\pm0.07}$**
	6	$27.76_{\pm0.23}$	$28.00_{\pm0.16}$	$32.92_{\pm0.19}$	$28.03_{\pm0.39}$	$27.77_{\pm0.22}$	$26.29_{\pm0.06}$	$31.54_{\pm1.11}$	$25.78_{\pm0.06}$	$26.12_{\pm0.09}$	$28.34_{\pm3.19}$	$26.08_{\pm0.12}$	$26.01_{\pm0.36}$	$26.23_{\pm0.17}$	$26.03_{\pm0.04}$	$27.64_{\pm0.19}$	$31.61_{\pm0.57}$	$26.59_{\pm0.14}$	$28.70_{\pm0.05}$	$23.09_{\pm0.11}$	$29.57_{\pm0.00}$	$27.08_{\pm0.04}$	$31.07_{\pm0.00}$	$29.01_{\pm0.05}$	$35.26_{\pm0.00}$	$37.46_{\pm1.02}$	$31.59_{\pm0.00}$	$28.07_{\pm0.03}$	**$21.42_{\pm0.10}$**
	12	$31.86_{\pm0.20}$	$34.05_{\pm0.23}$	$36.42_{\pm0.17}$	$32.38_{\pm0.27}$	$31.84_{\pm0.41}$	$30.73_{\pm0.07}$	$37.79_{\pm1.53}$	$29.93_{\pm0.15}$	$30.40_{\pm0.22}$	$32.39_{\pm2.68}$	$30.26_{\pm0.19}$	$29.33_{\pm0.71}$	$30.58_{\pm0.21}$	$30.50_{\pm0.08}$	$31.73_{\pm0.20}$	$36.73_{\pm0.45}$	$31.16_{\pm0.13}$	$35.51_{\pm0.08}$	$24.77_{\pm0.15}$	$41.53_{\pm0.00}$	$34.49_{\pm0.19}$	$41.80_{\pm0.00}$	$38.08_{\pm0.07}$	$46.15_{\pm0.00}$	$47.62_{\pm0.88}$	$41.70_{\pm0.00}$	$35.73_{\pm0.01}$	**$22.99_{\pm0.14}$**
	Avg	$28.46_{\pm0.25}$	$28.49_{\pm0.18}$	$33.43_{\pm0.19}$	$28.47_{\pm0.35}$	$27.96_{\pm0.24}$	$26.54_{\pm0.06}$	$32.01_{\pm1.14}$	$25.97_{\pm0.07}$	$26.32_{\pm0.10}$	$28.57_{\pm3.19}$	$26.28_{\pm0.12}$	$26.18_{\pm0.40}$	$26.48_{\pm0.17}$	$26.28_{\pm0.05}$	$28.14_{\pm0.19}$	$32.19_{\pm0.51}$	$26.92_{\pm0.15}$	$29.24_{\pm0.05}$	$23.13_{\pm0.12}$	$30.85_{\pm0.00}$	$27.68_{\pm0.07}$	$31.92_{\pm0.00}$	$29.74_{\pm0.05}$	$36.00_{\pm0.00}$	$38.04_{\pm1.02}$	$32.32_{\pm0.00}$	$28.62_{\pm0.02}$	**$21.36_{\pm0.10}$**
MAPE (%)	3	$30.24_{\pm0.93}$	$24.84_{\pm0.51}$	$29.71_{\pm0.50}$	$25.72_{\pm0.40}$	$27.23_{\pm0.80}$	$25.35_{\pm0.64}$	$27.74_{\pm0.49}$	$24.09_{\pm0.35}$	$25.52_{\pm0.76}$	$28.04_{\pm1.67}$	$24.99_{\pm0.83}$	$28.88_{\pm3.23}$	$25.59_{\pm1.28}$	$25.24_{\pm0.63}$	$31.60_{\pm0.74}$	$55.15_{\pm2.04}$	$27.92_{\pm1.57}$	$24.78_{\pm0.36}$	$23.40_{\pm0.78}$	$22.43_{\pm0.00}$	$20.71_{\pm0.03}$	$23.78_{\pm0.00}$	$21.67_{\pm0.02}$	$23.72_{\pm0.00}$	$24.09_{\pm0.26}$	$23.33_{\pm0.00}$	$21.79_{\pm0.02}$	**$21.76_{\pm0.17}$**
	6	$30.30_{\pm0.77}$	$26.69_{\pm0.51}$	$29.80_{\pm0.42}$	$26.82_{\pm0.39}$	$28.24_{\pm0.93}$	$26.56_{\pm0.55}$	$29.07_{\pm0.52}$	$25.13_{\pm0.23}$	$26.46_{\pm0.63}$	$29.04_{\pm1.40}$	$26.32_{\pm0.61}$	$29.31_{\pm2.63}$	$26.69_{\pm1.16}$	$26.45_{\pm0.53}$	$33.05_{\pm0.86}$	$58.10_{\pm2.24}$	$28.96_{\pm1.38}$	$27.01_{\pm0.44}$	$23.81_{\pm0.72}$	$25.58_{\pm0.00}$	$22.46_{\pm0.04}$	$27.22_{\pm0.00}$	$23.88_{\pm0.03}$	$27.23_{\pm0.00}$	$28.45_{\pm0.18}$	$26.61_{\pm0.00}$	$23.97_{\pm0.02}$	**$22.67_{\pm0.20}$**
	12	$32.77_{\pm0.67}$	$31.71_{\pm0.65}$	$33.40_{\pm0.39}$	$30.26_{\pm0.53}$	$30.60_{\pm1.19}$	$29.49_{\pm0.46}$	$32.30_{\pm0.73}$	$27.98_{\pm0.40}$	$29.56_{\pm0.75}$	$31.53_{\pm0.96}$	$29.50_{\pm0.47}$	$31.58_{\pm2.32}$	$29.76_{\pm1.26}$	$29.39_{\pm0.45}$	$37.84_{\pm1.06}$	$65.26_{\pm2.85}$	$31.77_{\pm1.26}$	$32.90_{\pm0.72}$	$24.90_{\pm0.70}$	$33.94_{\pm0.00}$	$26.38_{\pm0.07}$	$34.72_{\pm0.00}$	$28.54_{\pm0.03}$	$34.24_{\pm0.00}$	$35.19_{\pm0.14}$	$33.62_{\pm0.00}$	$28.62_{\pm0.03}$	**$24.25_{\pm0.27}$**
	Avg	$30.90_{\pm0.78}$	$27.32_{\pm0.53}$	$30.68_{\pm0.38}$	$27.32_{\pm0.41}$	$28.48_{\pm0.81}$	$26.88_{\pm0.52}$	$29.42_{\pm0.55}$	$25.48_{\pm0.28}$	$26.88_{\pm0.62}$	$29.30_{\pm1.38}$	$26.66_{\pm0.60}$	$29.71_{\pm2.65}$	$27.09_{\pm1.22}$	$26.77_{\pm0.51}$	$33.77_{\pm0.88}$	$58.93_{\pm2.28}$	$29.32_{\pm1.41}$	$27.74_{\pm0.47}$	$23.94_{\pm0.74}$	$26.63_{\pm0.00}$	$22.84_{\pm0.04}$	$27.93_{\pm0.00}$	$24.31_{\pm0.02}$	$27.76_{\pm0.00}$	$28.81_{\pm0.19}$	$27.26_{\pm0.00}$	$24.38_{\pm0.02}$	**$22.73_{\pm0.21}$**
TFNSW																													
MAE	3	$168.71_{\pm15.14}$	$122.66_{\pm1.26}$	$195.29_{\pm1.44}$	$143.12_{\pm2.43}$	$257.73_{\pm11.31}$	$120.27_{\pm0.46}$	$148.65_{\pm3.87}$	$114.38_{\pm1.09}$	$126.95_{\pm2.36}$	$160.08_{\pm23.60}$	$129.61_{\pm0.97}$	$124.47_{\pm2.51}$	$118.46_{\pm1.42}$	$129.39_{\pm0.47}$	$146.99_{\pm8.64}$	$132.28_{\pm2.29}$	$127.68_{\pm0.63}$	$155.32_{\pm0.84}$	$88.89_{\pm1.68}$	$160.32_{\pm0.00}$	$128.86_{\pm0.38}$	$190.66_{\pm0.00}$	$165.69_{\pm0.60}$	$255.65_{\pm0.00}$	$288.08_{\pm1.27}$	$204.29_{\pm0.00}$	$114.89_{\pm0.23}$	**$58.23_{\pm0.27}$**
	6	$181.43_{\pm14.38}$	$145.13_{\pm0.79}$	$211.00_{\pm1.58}$	$175.19_{\pm3.28}$	$261.85_{\pm11.96}$	$142.34_{\pm0.66}$	$180.04_{\pm3.54}$	$134.84_{\pm1.26}$	$150.05_{\pm2.33}$	$177.30_{\pm21.89}$	$152.15_{\pm1.41}$	$139.81_{\pm1.76}$	$139.74_{\pm2.06}$	$147.98_{\pm0.77}$	$151.13_{\pm5.35}$	$150.92_{\pm2.57}$	$148.02_{\pm0.71}$	$191.54_{\pm1.27}$	$95.46_{\pm1.73}$	$245.53_{\pm0.00}$	$171.51_{\pm0.85}$	$282.31_{\pm0.00}$	$234.82_{\pm0.32}$	$324.03_{\pm0.00}$	$362.17_{\pm0.58}$	$301.81_{\pm0.00}$	$151.95_{\pm0.30}$	**$65.33_{\pm0.37}$**
	12	$191.66_{\pm13.98}$	$158.22_{\pm0.48}$	$219.72_{\pm1.46}$	$203.53_{\pm4.54}$	$269.70_{\pm12.03}$	$156.34_{\pm0.69}$	$199.28_{\pm4.63}$	$148.00_{\pm1.26}$	$164.32_{\pm2.37}$	$189.11_{\pm20.37}$	$167.11_{\pm1.75}$	$154.35_{\pm1.24}$	$153.70_{\pm2.03}$	$160.78_{\pm0.84}$	$162.81_{\pm4.96}$	$170.66_{\pm2.26}$	$161.09_{\pm1.47}$	$205.55_{\pm1.20}$	$107.50_{\pm1.60}$	$288.66_{\pm0.00}$	$187.39_{\pm0.77}$	$345.79_{\pm0.00}$	$256.73_{\pm0.39}$	$359.26_{\pm0.00}$	$371.62_{\pm0.68}$	$399.37_{\pm0.00}$	$167.54_{\pm0.34}$	**$76.02_{\pm0.41}$**
	Avg	$179.87_{\pm14.50}$	$140.15_{\pm0.82}$	$208.33_{\pm1.48}$	$170.89_{\pm3.29}$	$262.63_{\pm10.87}$	$138.04_{\pm0.62}$	$174.36_{\pm3.31}$	$130.61_{\pm0.96}$	$145.42_{\pm2.41}$	$174.19_{\pm22.07}$	$147.86_{\pm1.05}$	$137.90_{\pm1.42}$	$135.57_{\pm1.80}$	$144.41_{\pm0.72}$	$153.63_{\pm6.20}$	$148.78_{\pm2.33}$	$144.04_{\pm0.82}$	$182.20_{\pm1.03}$	$96.12_{\pm1.67}$	$227.74_{\pm0.00}$	$160.20_{\pm0.68}$	$266.77_{\pm0.00}$	$215.22_{\pm0.33}$	$306.03_{\pm0.00}$	$334.59_{\pm0.89}$	$294.05_{\pm0.00}$	$142.36_{\pm0.28}$	**$65.16_{\pm0.34}$**
RMSE	3	$280.25_{\pm20.45}$	$217.63_{\pm1.09}$	$371.81_{\pm3.45}$	$262.49_{\pm4.16}$	$370.25_{\pm14.96}$	$215.03_{\pm0.78}$	$269.38_{\pm4.49}$	$205.28_{\pm2.18}$	$224.54_{\pm4.17}$	$292.97_{\pm49.14}$	$224.34_{\pm1.36}$	$219.33_{\pm3.97}$	$211.67_{\pm3.60}$	$229.39_{\pm0.90}$	$291.53_{\pm13.80}$	$241.64_{\pm3.36}$	$226.79_{\pm1.65}$	$277.14_{\pm1.12}$	$171.62_{\pm2.96}$	$327.55_{\pm0.00}$	$268.64_{\pm1.59}$	$369.29_{\pm0.00}$	$336.90_{\pm0.43}$	$462.40_{\pm0.00}$	$561.94_{\pm2.60}$	$417.01_{\pm0.00}$	$252.28_{\pm0.75}$	**$120.73_{\pm0.43}$**
	6	$308.03_{\pm18.86}$	$256.45_{\pm1.03}$	$394.80_{\pm3.35}$	$326.67_{\pm6.57}$	$384.60_{\pm12.76}$	$257.56_{\pm1.26}$	$333.40_{\pm6.29}$	$244.87_{\pm2.33}$	$270.67_{\pm4.71}$	$326.33_{\pm44.92}$	$271.09_{\pm2.95}$	$250.85_{\pm2.25}$	$253.32_{\pm4.77}$	$265.94_{\pm1.21}$	$303.05_{\pm8.85}$	$280.63_{\pm3.49}$	$266.95_{\pm2.18}$	$335.83_{\pm1.27}$	$189.36_{\pm3.02}$	$473.73_{\pm0.00}$	$348.32_{\pm2.49}$	$519.48_{\pm0.00}$	$458.77_{\pm1.05}$	$570.16_{\pm0.00}$	$649.78_{\pm1.17}$	$565.48_{\pm0.00}$	$326.33_{\pm0.66}$	**$137.47_{\pm0.67}$**
	12	$327.96_{\pm18.20}$	$282.09_{\pm0.90}$	$407.82_{\pm2.97}$	$370.26_{\pm8.00}$	$402.93_{\pm11.34}$	$283.78_{\pm1.17}$	$365.29_{\pm9.55}$	$269.48_{\pm2.24}$	$296.76_{\pm4.74}$	$346.80_{\pm41.06}$	$297.74_{\pm2.86}$	$277.73_{\pm1.60}$	$279.24_{\pm4.41}$	$290.96_{\pm1.24}$	$330.87_{\pm7.88}$	$318.01_{\pm2.67}$	$290.09_{\pm2.68}$	$358.40_{\pm1.12}$	$219.12_{\pm2.63}$	$546.05_{\pm0.00}$	$372.36_{\pm2.11}$	$608.15_{\pm0.00}$	$493.26_{\pm0.47}$	$648.32_{\pm0.00}$	$652.88_{\pm0.85}$	$692.51_{\pm0.00}$	$352.26_{\pm0.80}$	**$164.90_{\pm0.69}$**
	Avg	$303.16_{\pm19.20}$	$247.22_{\pm1.02}$	$390.75_{\pm3.25}$	$313.61_{\pm5.86}$	$383.38_{\pm12.29}$	$247.58_{\pm1.00}$	$318.17_{\pm5.87}$	$235.35_{\pm1.95}$	$259.40_{\pm4.67}$	$318.61_{\pm45.67}$	$259.82_{\pm2.21}$	$245.35_{\pm1.93}$	$243.47_{\pm4.15}$	$257.96_{\pm1.08}$	$307.19_{\pm10.11}$	$275.25_{\pm3.16}$	$257.27_{\pm2.16}$	$318.13_{\pm1.10}$	$190.33_{\pm2.83}$	$440.18_{\pm0.00}$	$323.47_{\pm2.06}$	$487.06_{\pm0.00}$	$420.65_{\pm0.52}$	$544.51_{\pm0.00}$	$609.59_{\pm1.67}$	$545.09_{\pm0.00}$	$303.78_{\pm0.69}$	**$137.77_{\pm0.59}$**
MAPE (%)	3	$203.98_{\pm42.96}$	$101.54_{\pm2.40}$	$167.79_{\pm4.05}$	$109.87_{\pm6.56}$	$419.02_{\pm18.83}$	$93.75_{\pm3.87}$	$121.57_{\pm11.39}$	$87.14_{\pm5.01}$	$94.78_{\pm7.81}$	$94.19_{\pm7.03}$	$134.83_{\pm7.19}$	$109.08_{\pm12.15}$	$93.03_{\pm3.88}$	$99.28_{\pm3.48}$	$112.69_{\pm14.09}$	$162.98_{\pm5.57}$	$98.77_{\pm5.85}$	$154.16_{\pm5.94}$	$59.94_{\pm3.07}$	$60.65_{\pm0.00}$	$54.11_{\pm0.13}$	$84.30_{\pm0.00}$	$73.71_{\pm1.57}$	$103.01_{\pm0.00}$	$54.54_{\pm0.87}$	$68.13_{\pm0.00}$	$43.39_{\pm0.01}$	**$44.61_{\pm0.29}$**
	6	$210.50_{\pm43.26}$	$130.32_{\pm3.10}$	$176.75_{\pm3.86}$	$137.59_{\pm9.85}$	$427.56_{\pm35.35}$	$104.87_{\pm3.92}$	$131.86_{\pm9.09}$	$102.09_{\pm3.95}$	$108.40_{\pm8.04}$	$108.65_{\pm5.53}$	$147.27_{\pm10.71}$	$122.40_{\pm16.15}$	$105.70_{\pm3.72}$	$106.79_{\pm3.16}$	$110.90_{\pm12.12}$	$183.80_{\pm8.11}$	$114.90_{\pm9.27}$	$194.19_{\pm6.66}$	$64.60_{\pm2.35}$	$117.43_{\pm0.00}$	$82.43_{\pm0.32}$	$163.00_{\pm0.00}$	$128.42_{\pm0.85}$	$164.37_{\pm0.00}$	$164.10_{\pm1.19}$	$131.57_{\pm0.00}$	$65.49_{\pm0.27}$	**$52.86_{\pm0.35}$**
	12	$221.79_{\pm41.52}$	$146.31_{\pm4.51}$	$188.49_{\pm3.75}$	$174.63_{\pm17.60}$	$438.74_{\pm48.59}$	$115.98_{\pm3.12}$	$137.54_{\pm6.98}$	$114.71_{\pm2.98}$	$120.47_{\pm9.27}$	$119.62_{\pm3.65}$	$162.82_{\pm10.53}$	$132.49_{\pm17.32}$	$117.09_{\pm2.11}$	$115.50_{\pm2.33}$	$111.46_{\pm8.46}$	$206.11_{\pm5.57}$	$130.80_{\pm6.23}$	$218.83_{\pm4.82}$	$76.98_{\pm1.65}$	$182.97_{\pm0.00}$	$99.59_{\pm0.44}$	$242.48_{\pm0.00}$	$171.51_{\pm0.37}$	$207.96_{\pm0.00}$	$206.18_{\pm1.08}$	$226.64_{\pm0.00}$	$80.90_{\pm0.33}$	**$64.93_{\pm0.39}$**
	Avg	$211.72_{\pm42.72}$	$125.28_{\pm3.02}$	$177.95_{\pm3.89}$	$138.49_{\pm10.57}$	$430.83_{\pm35.95}$	$104.00_{\pm3.59}$	$129.82_{\pm8.24}$	$99.93_{\pm3.54}$	$106.92_{\pm8.01}$	$106.24_{\pm4.44}$	$148.01_{\pm7.69}$	$119.37_{\pm14.85}$	$104.21_{\pm3.09}$	$106.33_{\pm3.01}$	$112.25_{\pm11.68}$	$182.00_{\pm6.10}$	$113.97_{\pm6.78}$	$187.17_{\pm5.55}$	$66.43_{\pm2.35}$	$120.67_{\pm0.00}$	$78.26_{\pm0.28}$	$160.45_{\pm0.00}$	$124.25_{\pm0.83}$	$155.80_{\pm0.00}$	$144.74_{\pm1.00}$	$137.97_{\pm0.00}$	$62.69_{\pm0.22}$	**$52.89_{\pm0.29}$**

Table 9: Matched controls of the annual benchmark on all ten networks. Avg errors of the same per-year STAEFormer checkpoints without adaptation (Frozen), with the calibrator only (Frozen+RLS), with refreshed identities only (Refresh), and with both (EvoST); mean ± sample standard deviation over five seeds. MAPE is in percent.

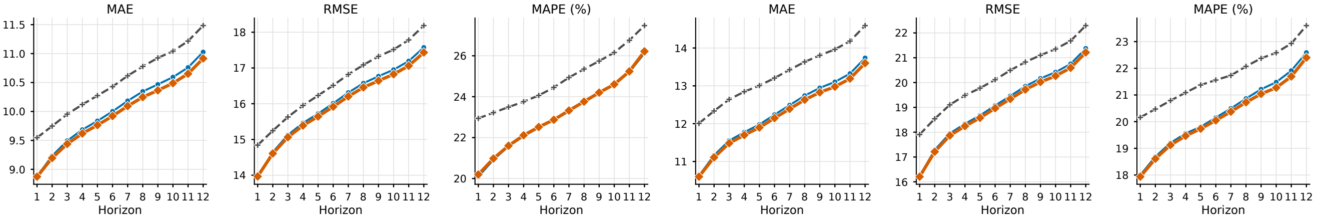
	Frozen			Frozen+RLS			Refresh			EvoST		
Network	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
PEMS03	$11.76_{\pm0.15}$	$19.72_{\pm0.19}$	$23.75_{\pm0.73}$	$11.01_{\pm0.04}$	$18.58_{\pm0.06}$	$22.82_{\pm0.26}$	$11.26_{\pm0.09}$	$19.24_{\pm0.15}$	$21.66_{\pm0.24}$	$10.92_{\pm0.05}$	$18.47_{\pm0.07}$	$22.52_{\pm0.21}$
PEMS04	$18.57_{\pm0.91}$	$29.49_{\pm1.11}$	$17.55_{\pm1.20}$	$16.63_{\pm0.15}$	$27.17_{\pm0.18}$	$15.53_{\pm0.20}$	$18.18_{\pm1.22}$	$29.32_{\pm1.47}$	$16.88_{\pm1.86}$	$16.58_{\pm0.21}$	$27.14_{\pm0.26}$	$15.52_{\pm0.23}$
PEMS05	$10.05_{\pm0.04}$	$15.79_{\pm0.04}$	$23.81_{\pm0.50}$	$9.53_{\pm0.03}$	$15.17_{\pm0.05}$	$21.72_{\pm0.15}$	$9.68_{\pm0.05}$	$15.49_{\pm0.06}$	$21.90_{\pm0.41}$	$9.48_{\pm0.03}$	$15.12_{\pm0.04}$	$21.77_{\pm0.17}$
PEMS06	$12.70_{\pm0.50}$	$19.21_{\pm0.63}$	$20.95_{\pm0.89}$	$11.57_{\pm0.20}$	$17.96_{\pm0.29}$	$19.28_{\pm0.29}$	$12.35_{\pm0.49}$	$18.87_{\pm0.64}$	$19.59_{\pm0.58}$	$11.51_{\pm0.20}$	$17.87_{\pm0.29}$	$19.18_{\pm0.35}$
PEMS07	$14.62_{\pm0.31}$	$25.98_{\pm0.30}$	$21.39_{\pm0.67}$	$13.56_{\pm0.07}$	$24.65_{\pm0.09}$	$20.29_{\pm0.33}$	$13.96_{\pm0.11}$	$25.47_{\pm0.13}$	$19.30_{\pm0.16}$	$13.45_{\pm0.06}$	$24.56_{\pm0.09}$	$19.85_{\pm0.19}$
PEMS08	$12.35_{\pm0.07}$	$20.58_{\pm0.10}$	$18.08_{\pm0.28}$	$11.63_{\pm0.04}$	$19.66_{\pm0.08}$	$17.24_{\pm0.10}$	$11.88_{\pm0.04}$	$20.17_{\pm0.11}$	$16.54_{\pm0.11}$	$11.55_{\pm0.04}$	$19.57_{\pm0.09}$	$16.89_{\pm0.07}$
PEMS10	$10.19_{\pm0.06}$	$16.94_{\pm0.07}$	$28.49_{\pm0.53}$	$9.82_{\pm0.03}$	$16.34_{\pm0.04}$	$27.81_{\pm0.15}$	$9.83_{\pm0.04}$	$16.78_{\pm0.05}$	$26.64_{\pm0.12}$	$9.76_{\pm0.03}$	$16.30_{\pm0.04}$	$27.56_{\pm0.12}$
PEMS11	$16.04_{\pm0.46}$	$27.38_{\pm0.45}$	$25.05_{\pm0.97}$	$14.00_{\pm0.08}$	$25.89_{\pm0.12}$	$22.81_{\pm0.23}$	$14.87_{\pm0.36}$	$25.78_{\pm0.40}$	$22.24_{\pm0.43}$	$13.70_{\pm0.09}$	$25.24_{\pm0.12}$	$22.27_{\pm0.17}$
PEMS12	$12.54_{\pm0.14}$	$22.42_{\pm0.20}$	$24.80_{\pm0.84}$	$11.85_{\pm0.06}$	$21.45_{\pm0.10}$	$23.32_{\pm0.27}$	$12.10_{\pm0.09}$	$22.08_{\pm0.16}$	$21.87_{\pm0.20}$	$11.76_{\pm0.06}$	$21.36_{\pm0.10}$	$22.73_{\pm0.21}$
TFNSW	$83.54_{\pm1.68}$	$176.72_{\pm3.38}$	$59.00_{\pm0.34}$	$65.54_{\pm0.35}$	$138.29_{\pm0.53}$	$53.33_{\pm0.21}$	$81.93_{\pm1.37}$	$176.64_{\pm3.43}$	$57.26_{\pm0.98}$	$65.16_{\pm0.34}$	$137.77_{\pm0.59}$	$52.89_{\pm0.29}$

candidate ran all 16 epochs in 82 runs, 47 of them on TFNSW, and its best epoch was the 16th in 30, 19 of them on TFNSW. The 1,165 runs took 100.3 H100 GPU-hours in total, including candidate fits, calibration selection, and diagnostic replays.

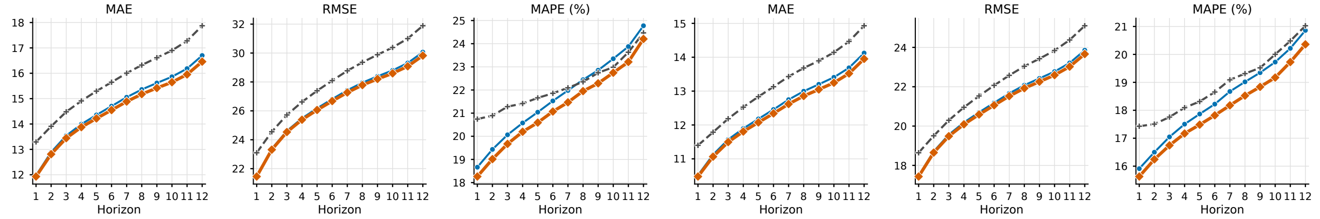
Matched controls. The four streams in Figure 11 share the same per-year checkpoints, training-split input normalization, windows, and seeds. Table 9 lists their absolute errors. Averaged equally over the ten networks and relative to Frozen, EvoST lowers Avg-MAE, Avg-RMSE, and Avg-MAPE by 9.4%, 7.4%, and 8.1%, Frozen+RLS by 8.7%, 6.9%, and 6.9%, and Refresh by 3.7%, 1.9%, and 7.8%. Relative



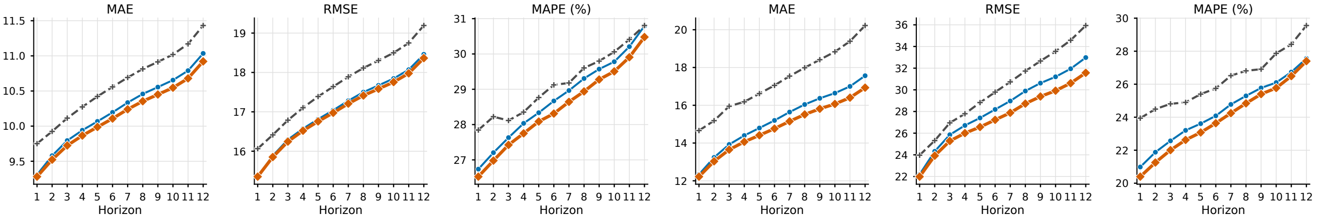
(a) PEMS03 (b) PEMS04



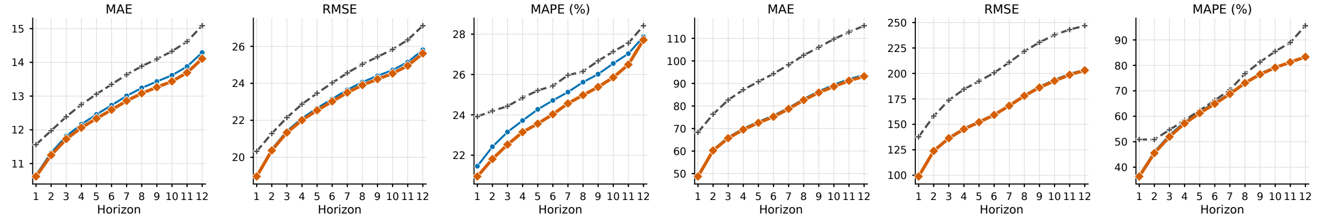
(c) PEMS05 (d) PEMS06



(e) PEMS07 (f) PEMS08



(g) PEMS10 (h) PEMS11



(i) PEMS12 (j) TFNSW

Figure 10: All-sensor per-step errors of EvoST and its matched controls on all ten networks: the same checkpoints with the calibrator only (Frozen+RLS) and without adaptation (Frozen), over the same years and five seeds. Lower is better.

to Frozen+RLS, EvoST lowers the three errors by 0.8%, 0.6%, and 1.2%. Its Avg-MAE is lower on every network, by 0.3% on PEMS04 to 2.1% on PEMS11, with lower cross-year means in all five paired seeds except on PEMS04 (three). On PEMS04, its Avg-RMSE and Avg-MAPE are lower in only two of the five seeds; on PEMS05, its Avg-MAPE is 0.2% higher. Calibration lowers MAE and RMSE for both the frozen and the refreshed model on every network, and MAPE for the frozen model on every network. For the refreshed model, it raises MAPE on six networks, PEMS03, PEMS07, PEMS08, PEMS10, PEMS11, and PEMS12, and lowers it on the other four.

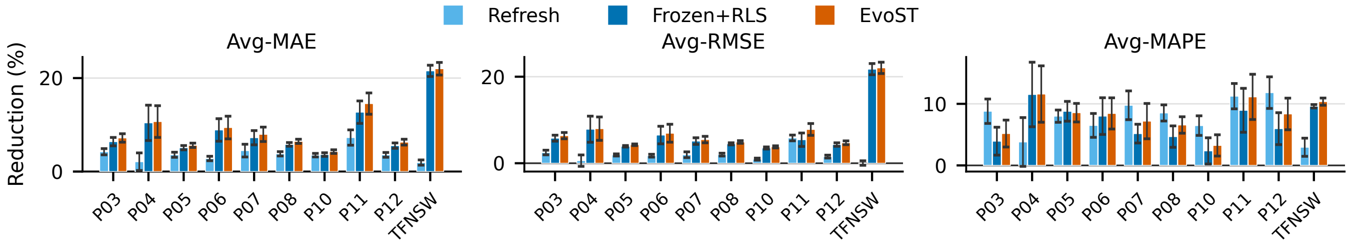


Figure 11: Matched controls of the annual benchmark on all ten networks. Reduction in Avg-MAE, Avg-RMSE, and Avg-MAPE relative to the same per-year STAEFormer checkpoints without adaptation (Frozen), with refreshed identities only (Refresh), with the calibrator only (Frozen+RLS), and with both (EvoST). Bars are reductions of the five-seed means; whiskers give the sample standard deviation of the per-seed reductions.

A.5 EvoST Implementation and Feedback Protocol

Identity parameters and fitting. The identity blocks have shapes $12 \times N_y \times 80$ for STAEFormer, $N_y \times 32$ for STID, and two $N_y \times 10$ tables for Graph WaveNet. The main STAEFormer maintenance comparison uses the latest 2,048 training windows, reserving the final quarter for early stopping and selecting the identity learning rate from $\{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$. Arrival experiments use a fixed rate of 3×10^{-3}. These fits use at most 16 epochs and patience 3.

Maintenance controls. The fitted variants shown in Figure 4 use the same recent windows and holdout as identity refresh and select their learning rates on that holdout. Head and Prompt select from $\{3{\times}10^{-3}, 10^{-2}, 3{\times}10^{-2}\}$ and Shared from $\{10^{-4}, 3{\times}10^{-4}, 10^{-3}\}$. Full FT selects the rate of the shared weights from $\{3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}\}$ and uses the identity rate selected for EvoST in the same run; EvoST+Prompt keeps that identity rate and selects the prompt rate. These fits also use at most 16 epochs and patience 3. Scratch starts from random initialization with target-year training statistics, selects from $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$, and uses at most 100 epochs with patience 5. The holdout-selected Scratch fits ran for at most 41 epochs. The lowest rate of the grid was selected in 27 of 36 runs for EvoST, 20 for Full FT, 21 for Shared, and 23 for Scratch. Lower-bound selections were common across several variants, so the comparisons are conditional on the reported grids and fitting budgets. The prompt is a rank-6 EAC-style input prompt $\mathbf{UV}$ with $\mathbf{U} \in \mathbb{R}^{N_y \times 6}$ and $\mathbf{V} \in \mathbb{R}^{6 \times 12}$, that is, $6N_y + 72$ trainable values (median 8,856, against a median of 1,405,440 identity values). It is not a reproduction of the full EAC procedure.

Fitting cost. Figure 12 shows the median fitting costs of the main maintenance comparison, measured on H100 GPUs. Identity refresh updates 1.41M values (960 per sensor), takes 11.5 s per epoch and 259 s for its three-rate search, and peaks at 10.4 GiB; full fine-tuning takes 14.0 s, 327 s, and 11.4 GiB. Seconds per epoch first average training-epoch times within each candidate, then average the three candidate means; holdout evaluation is excluded. Search times sum each arm's three candidate fits and exclude calibration; the Full FT entry also excludes the preceding identity-rate search. Memory is the peak process allocation observed during fitting, including resident states and caches. All entries are medians over 36 runs.

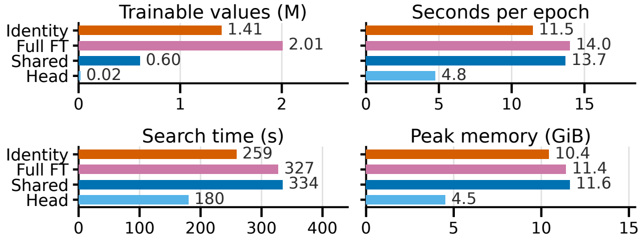


Figure 12: Median fitting costs in the primary STAEFormer maintenance experiment (36 H100 runs): identity refresh (Identity), full fine-tuning (Full FT), shared weights (Shared), and output head (Head). Search time sums three learning-rate candidates. Times exclude calibration; memory is the peak process allocation during fitting.

Ablation and sensitivity figure. In Figure 5(d, e), colored curves average within each network and then over the ten networks; grey lines are networks. In (b), (c), (f) (twelve settings each) and (g), ratios are geometric means over settings of seed-averaged test H12 MAE relative to the target-year checkpoint, both calibrated or both not. In (f), the target-year reference is calibrated at the same budget. Whiskers are 95% district-bootstrap intervals (none for dashed curves). Percentages in (g) are the share of the gap recovered; (f) and (g) fix the identity learning rate at 3×10^{-3}.

Case study. Figure 7 uses evaluation seed 51 and the 2015 checkpoint. Panel (a) uses the original maintenance evaluation and includes all 2,171 sensors with valid test targets (249 have none); panel (b) records a separate run of the same protocol with the same seed and fixed sensors. All forecasts are calibrated; (a) uses the MAE over steps 1–12 and (b) step 12. The p50 and p90 sensors have the nearest-rank median and 90th-percentile relative MAE reductions among the 1,126 known sensors with at least 95% valid test targets, and the unseen sensor has the highest validity, ties broken by sensor ID; these choices were fixed before the run of (b). The day shown is the first complete weekday by 60-minute target time. Differences in Section 6.9 are given in the order p50, p90, unseen.

History budgets. With a budget of b fitting windows, the latest b training windows both refresh the identities and initialize the calibrators; the final quarter is the identity holdout, and identity fitting uses every second window (every fourth above 1,600 sensors). Calibration settings are selected on an additional prefix of $b/2$ validation windows; the calibrators are then initialized again and the validation and test portions are replayed. The identities stay fixed while the calibrators keep absorbing released labels, so the budget limits the history available at the start, not the labels used during the stream. All budgets use at most 16 epochs and patience 3 with the same number of windows per epoch; the total number of steps depends on early stopping.

Checkpoint age. The checkpoint-age study uses the six main-panel districts with five-year-old checkpoints in 2015, 2020, and 2025; ten-year-old checkpoints in 2020 and 2025; and fifteen-year-old checkpoints in 2025 (18, 12, and 6 settings).

Backbone and age extensions. For the checkpoint-age, STID, and time-input experiments, identities are refitted with a fixed learning rate of 3×10^{-3}, at most 16 epochs, and patience 3. Graph WaveNet selects its rate from $\{10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}\}$ on the same fitting holdout, with the same budget. STID and Graph WaveNet are the benchmark implementations, retrained per year with checkpoint seeds 42–44; Graph WaveNet is the benchmark's compact two-layer version, which averages the given and the learned adjacency. On the nine PeMS districts in 2025, the uncalibrated test H12 MAE of the retrained models is 0.961 (STID) and 1.006 (Graph WaveNet) times that of the benchmark's own checkpoints. With calibration, the same-year STID and Graph WaveNet have 26.9% and 22.9% higher error than STAEFormer on the sixteen settings of Table 2, so the ratios there measure reuse within each backbone, not an accuracy ranking. The time-input models add time-of-day and day-of-week embeddings, initialized from $\mathcal{N}(0, 0.02^2)$ by a rule fixed on development data; their width is 152 against 104 without them, and the time embeddings are shared parameters that stay frozen during refresh.

Retraining checks. The original check of the Graph WaveNet retraining flagged 11 of 90 models whose training-mode validation error was not below that of the last-value forecast; an evaluation-mode check found all 90 below it (largest ratio 0.976). Excluding the six affected settings, as the original rule prescribed, gives 1.098 for the old checkpoint and 1.005 after refresh (95% recovery), against 1.091 and 1.011 with all sixteen. One time-input model diverged (PEMS06/2025, checkpoint seed 43); we keep its best checkpoint (epoch 2 of 8). Excluding PEMS06/2025 afterwards gives 15.8% ageing and 1.027 after refresh (83% recovery), against 14.0% and 1.014 with all twelve settings.

Arrival history and donors. The arrival experiment provides 64 post-arrival fitting windows before the stream. Matching uses finite raw readings, including zeros. A donor must have observations on at least 20% of the available times and share finite observations with the incoming sensor on at least 20%. If fewer than k donors qualify, all k are sampled without replacement from the established sensors using a seed fixed for the run. At zero-based stream index $t_\star = 1{,}024$, with $H = 12$, records $0, \dots, 1{,}012$ are eligible. These 1,013 records and the last 11 eligible post-arrival prestream windows supply the $W = 1{,}024$ refitting windows. The refit follows that step's RLS update and precedes its prediction.

Arrival refits. The new-only refit fits the new identities and rebuilds only their calibrators; the all-identity refit fits every identity and rebuilds every calibrator. For the established-only backbone reference, identities are refreshed before the stream; calibration hyperparameters are still selected using all-node validation error, including last-value forecasts for the new nodes. The established column reports only the established-node errors. The all-identity refit was specified after the new-only refit left an established-sensor gap and was evaluated on the same region-year settings. Before this evaluation, we required the upper 95% district-bootstrap bound to be at most 1.01 for established sensors relative to the established-only backbone reference and for new sensors relative to the new-only refit. The bounds were 1.009 and 1.002, respectively. Without output calibration, the new-sensor MAE ratio to the uncalibrated target-year model is 0.994 (95% CI 0.973–1.012). Rebuilding the new-sensor calibrators on cross-fitted forecasts did not help (ratio 1.004 to the new-only refit).

Calibration and evaluation. We use $L = H = 12$, giving $D = 29$, and update RLS every $\delta = 64$ windows. The history slope is fitted against centered sample indices. The recursion uses float64 arithmetic and symmetrizes $\mathbf{P}_n$ after each update. Except in the history-budget study, where the selection prefix is $b/2$ windows, for each branch of the maintenance experiments $\rho \in \{1, 100\}$ and $\lambda \in \{1, 0.999, 0.995, 0.99\}$ are selected on the first 1,024 validation windows. With these choices fixed, the calibrators are initialized again and validation and test windows are replayed chronologically; reported performance uses the test portion. For the annual benchmark, selection uses up to the first 1,024 validation windows. After selection, new calibrators are fitted on the training tail and replay only the test portion; validation-updated states are discarded. PEMS06 in 2005, whose training tail has no valid target, follows the fallback in Appendix A.4.

We compute the shared scale s as the standard deviation of valid targets in the initial training tail. The arrival experiments compute this statistic before applying the node-specific arrival masks. The arrival restriction applies to donor observations and sensor-specific fitting records. The shared scale uses the full initial tail and stays fixed.